Biased Surveys☆

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Abstract

We find empirical evidence that surveys of professional forecasters are biased by strategic incentives. First, we find that individual forecasts overreact to idiosyncratic information but underreact to common information. We show this is consistent with a model of strategic diversification incentives in forecast reporting where forecasters want to optimally "stand out" from the crowd, and thus report forecasts that exaggerate the agents' true beliefs. Second, we show that no such biases are present in forecasts data that is not subject to strategic incentives. We also test further comparative statics that also confirm the strategic incentive model. Overall, we conclude that strategic reporting biases the inference an econometrician can draw on the true underlying expectations formation process, and the precision and heterogeneity in agents' information sets, and lastly we show how to correct for this.

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1. Introduction

Expectations play a pivotal role in macroeconomic models. Recently, the academic literature has leveraged surveys of professional forecasters data to directly estimate economic agents' expectations, and this line of research has found a bevy of apparent violations of the full information rational expectations (FIRE) paradigm at the heart of standard macroeconomic models. On the one hand, Coibion and Gorodnichenko (2015) document that aggregate ("consensus") forecasts, defined as the mean across survey participants, are "sticky" in a way that implies that the forecasters are operating with imperfect noisy information. On the other hand, Bordalo et al. (2020) document that individual-level forecasts "overreact" to new information, indicating a violation of rational expectations.

The established methodology in the literature generally assumes that forecast surveys provide a direct and unbiased measure of the respondents' true beliefs and expectations. However, an older, theoretical literature (Laster et al., 1999; Ottaviani and Sørensen, 2006), argues that professional forecasters, who compete with one another for customers, might strategically shade their survey responses away from their true expectations because of reputational concerns. For example, there are several long-standing, prestigious and highly visible "Best Forecaster Awards", such as the Lawrence R. Klein Best BlueChip Forecaster Award which has been running since the 1980s. These awards are coveted by professional forecasters, as evidenced by the fact that when one of their team members wins such an award, economic consulting and forecasting firms publicize it widely.[3] But since only one forecaster can win – the most accurate over the time period the award covers – these salient awards and related publicity create a winner-take-all environment.

Ottaviani and Sørensen (2006) show that formally this gives rise to *strategic diversification* incentives in responding to the surveys, meaning that reported forecasts will optimally

[3] See for example press releases such as: https://news.comerica.com/news-releases?item=108196 and https://www.eiu.com/n/eiu-most-awarded-economic-forecaster/. In addition, firms also often upkeep a list of all recent awards they have received in a prominent place on their websites, e.g. https://www.spglobal.com/marketintelligence/en/mi/products/economics-country-risk-accolades.html.

"overreact", because this help forecasters "stand out" from the crowd. We generalize their framework to a setup with both common and idiosyncratic signals, and show that this more general strategic environment can indeed generate the same type of "overreaction" in forecasts that has been recently documented by Bordalo et al. (2020). Moreover, the strategic incentives also bias the estimated information stickiness or rigidity in the sense of Coibion and Gorodnichenko (2012). Thus, understanding whether surveys are an unbiased proxy of the professional forecasters beliefs or rather are affected by strategic incentives can deeply influence an econometrician's inference about expectations based on survey data.

We use our model's key implications to devise a novel, direct empirical test for the presence of strategic incentives. The intuition is that if strategic forces are empirically relevant, then forecast errors will react in a systematically different way to "public" information that is commonly observed by all survey participants, such as the latest GDP data release or the consensus forecast in the last survey round, as compared to forecaster-specific information, like an update in the forecaster's own model.

Specifically, to "stand out" forecasters want to be right when everyone else is wrong. Thus, when agents have access to two types of noisy signals – a signal with idiosyncratic noise and a common noisy signal that is the same for everyone – the reported forecasts optimally overweight idiosyncratic signals at the expense of underweighting common signals. This embellishes individual agents' forecasts in a way that helps them stand out and differentiate from one another, which is the optimal response in this strategic setting.

We design an empirical methodology that directly tests this implication, by augmenting the Bordalo et al. (2020) regression of individual forecast errors on individual forecast revisions with a proxy for a commonly observed, public signal. To implement our test, we use data from both the Survey of Professional Forecasters (SPF), the most commonly used forecast survey dataset, and the BlueChip survey.[4] As a first step in our analysis, we docu-

[4]The SPF is anonymous, but its respondents also participate in many other, non-anonymous surveys. As we discuss below and also cite direct evidence to this effect, it appears that the SPF respondents provide the same forecast to the SPF as they do to the other, non-anonymous outlets they participate in. Thus any strategic considerations that they face in their public interactions are also present in their SPF responses.

ment that commonly observed noisy signals are indeed a quantitatively important feature of the information sets of professional forecasters. We do so by exploiting the fact that, in the presence of public signals, the methodology in Coibion and Gorodnichenko (2015) displays an omitted variable bias which attenuates their estimated coefficient (as also noted in their Appendix).[5] We correct for this bias, and find that it is quite significant in practice (attenuating the estimated degree of information rigidity by 50%), which indicates that there is indeed a meaningful common component in the information set of forecasters.[6]

In turn, our main empirical analysis tests whether common, publicly observed signals are indeed underweighted in reported forecasts, as predicted by the strategic diversification hypothesis. We use the consensus forecast from the previous round of the survey as our main proxy for a commonly observed signal. The lagged consensus is "new" information for the forecasters at time t, because the results of the survey are announced to everyone only after they submit their $t-1$ responses, and before they submit their quarter t forecasts. Moreover, the consensus is not only commonly available to all agents, but objectively it is a fairly precise forecast, and is thus a salient piece of information.

Our key finding is that the individual forecasts in both the SPF and BlueChip *underweight* the lagged consensus forecast significantly, while they *overweight* the rest of their information set. We stress that this finding is not consistent with a large class of behavioral models that have been proposed to address FIRE deviations documented by Bordalo et al. (2020), such as diagnostic expectations or over-extrapolation, since those models imply over-reaction to all new information. Instead, our finding that commonly observed signals are *underweighted* is a characteristic and defining feature of forecasts made when agents face strategic diversification incentives in their response to the survey.

We also present the results of several further tests that support the strategic diversification hypothesis. First, we run the same regression on the forecasts published in the

[5]This bias does not impact the key conclusions of Coibion and Gorodnichenko (2015), because the bias works against their main hypothesis. Hence, their estimates and conclusions are conservative.

[6]To do so, we follow an empirical methodology inspired by Goldstein (2023).

Federal Reserve's Greenbook, which forecasts the same set of variables as the SPF. However, the Federal Reserve's forecasts are made for internal monetary policy purposes and are only released with a 5 year lag. Thus, they are arguably not subject to the same strategic forces that arise among the professional forecasters, and as such we would expect that the Fed's Greenbook forecasts do not exhibit the same predictable errors and biases. And indeed we find no evidence of under or overreaction to any kind of information in the Greenbook. This striking contrast, where the professional forecasts display clear deviations from rationality while the Fed's forecasts do not, is indicative of strategic incentives in the professionals' survey.

In a different test, we leverage the fact that the lagged consensus is not just a generic commonly observed signal, but is also a direct estimate of everyone else's recent beliefs. As such, strategic diversification incentives imply that agents would optimally underweight the lagged consensus more than alternative common signals that do not also directly speak to the forecasts of others. We show that this is indeed true in the data, using as an alternative common signal proxy the past realization of the variable being forecasted (e.g. lagged realized inflation). And lastly, we also test a key comparative static of the model and show that in the cross-section, for variables where we estimate that the common component of information sets is relatively more precise, the observed forecasts also display stronger evidence of underreaction to the common information – again as predicted by the model.

Lastly, we estimate a quantitative version of the model and use it to recover the true underlying beliefs of the agents. This structural analysis showcases that strategic incentives also greatly influence estimates of the degree of information imperfection and dispersion (e.g. Coibion and Gorodnichenko, 2015). Using our model to back out the true underlying individual beliefs, we find that the degree of information rigidity is 30% higher than previously estimated, while the true dispersion in beliefs is 20-80% lower than in the raw data. Furthermore, we also show that the strategic incentives increase the precision of the consensus forecast by 30-100%, as compared to the precision of the true beliefs.

Overall, we conclude that strategic incentives are a likely rational explanation of a number of “puzzles” in survey data, suggesting professional forecasters do not deviate from rationality. However, the strategic incentives bias econometric inference, and our results imply that standard survey-based estimates of information imperfection are likely to be significantly understated, while survey-based measures of disagreement are overstated.

Related literature. This paper relates to three strands of the literature. First, are the papers using surveys of professional forecasters to test the full information hypothesis. A key finding in this literature is consensus underreaction, meaning a positive relation between consensus forecast errors and consensus forecast revisions, which can be used to proxy for the precision of the information sets underlying the forecasts (Crowe, 2010; Coibion and Gorodnichenko, 2012, 2015). Our main contribution relative to this literature is twofold. First, we document that strategic incentives are likely making reported forecasts a biased proxy of the true beliefs of agents, which among other things, biases the estimated information precision upwards. Second, we use a structural model to estimate the actual information rigidity of the true underlying beliefs, and find it is significantly lower.

Another strand of the literature uses surveys to test rational expectations. In particular, Bordalo et al. (2020) document overreaction in individual forecasts, in the sense of a negative correlation between individual forecast errors and forecast revisions. As forecast errors should not be predictable if the forecast is rational, this could be evidence of a behavioral bias in expectations formation. Instead, we show that this evidence can be explained by a departure from truthful revelation as a result of strategic incentives, even when agents have rational expectations.

Moreover, we provide direct evidence of the strategic incentives hypothesis, as captured by our novel finding that reported forecasts underreact to public information. Related to this finding, in a recent paper Broer and Kohlhas (2022) also use a public signal proxy to build on existing tests of RE, but they find mixed results in terms of under and over-reaction. In contrast to them, we use an empirical methodology that differs in two ways: first we focus on multivariate regressions, and second, when proxying for the common information we isolate the *surprise* component in public signals and use that as our key regressor. A novel feature of our empirical methodology is that it allows us to directly map our main regression coefficient of interest to the structural weight put on the public information in the updated forecast. Furthermore, it also leads to estimates that consistently point to *underreaction* to public information across all forecasted variables in the SPF and the Blue Chip, in line with our strategic diversification model.[7] Lastly, our notion of underreaction is conceptually different than that used in Kučinskas and Peters (2022), as we measure deviations from rational expectations, while they measure deviations from the joint hypothesis of FIRE.[8]

A third group of papers analyzes, theoretically, the strategic incentives in forecasters behavior (see Marinovic et al. 2013 for a review). A first major reference, Laster et al. (1999), develop a model of strategic diversification incentives based on the key idea that some of the forecasters' customers simply follow only the forecaster that happened to be most accurate last period. Ottaviani and Sørensen (2006) present several different models of competition between forecasters, to demonstrate that depending on the broader economic environment, it is theoretically possible to generate both strategic coordination and strategic diversification forces among forecasters. More recently, Baley and Turén (2023) study how strategic incentives interacted with fixed costs to updating forecasts can help explain the lumpiness in forecast revisions in the Bloomberg survey. Our main contributions relative to this literature is to first provide direct empirical evidence that common information is underweighted, a key implications of a broad class of strategic incentives theories, and second we propose and structurally estimate a quantitative, fully dynamic version of the model and use it recover the underlying true expectations.

[7]Moreover, as we discuss in Section 3.4, our strategic diversification framework is also consistent with some, although not all, formulations of the univariate analysis.

[8]Born et al. (2023) study how firms responds differently to new information about macroeconomic variables versus information about their own firm-specific variables, and find that firms underweight the former but overweight the latter. Instead, we only consider information about the macroeconomy, but differentiate between public and idiosyncratic information, as further detailed below.

2. Data

Forecasts. We use data on forecasts from the Survey of Professional Forecasters (SPF), the most widely used survey in the academic literature, and also from the Blue Chip survey. One important distinction between the two surveys is that the SPF is anonymous, while the Blue Chip is not. It could be argued that strategic considerations do not apply to anonymous surveys like the SPF, however in practice it appears that forecasters (who typically participate in both anonymous and non-anonymous surveys typically) submit the same forecast to both types of surveys, likely due to convenience (Marinovic et al., 2013).[9] This means that strategic considerations would influence the results of anonymous surveys as well, and indeed consistent with this view, our estimates and results are virtually the same across both the SPF and the Blue Chip.

The SPF is quarterly, and each quarter around 40 professional forecasters contribute forecasts for outcomes in the current quarter and the next four quarters. The forecasted variables span both macroeconomic and financial quantities, including GDP, price indices, consumption, investment, unemployment, government consumption, and yields on government bonds and corporate bonds. The Blue Chip has a very similar structure, with respondents providing forecasts for almost exactly the same set of variables over the next four quarters, however the Blue Chip survey is collected monthly. To create a quarterly data set of the same structure as the SPF, we only keep the Blue Chip forecasts made in the third month of each quarter (following the methodology of Bordalo et al. (2020), henceforth BGMS).

While most forecasts are of the level of a given variable, we follow BGMS and Coibion and Gorodnichenko (2015) (henceforth CG) in transforming the GDP, price indices, consumption, investment and government consumption series into growth rates h periods ahead, while we keep the forecasts in level for unemployment and financial variables.

[9] We find direct evidence of this behavior in a supplement to the European Central Bank's Survey of Professional (European Central Bank, 2014), which asked forecasters about this and found that indeed the great majority submits the same forecast to all surveys they participate in. For full details, please see Appendix C.

Lastly, both the SPF and Blue Chip are known to suffer from outliers, hence as in BGMS, we winsorize the data by removing forecasts that are more than 5 interquartile ranges away from the median of each horizon in each quarter. We also only keep individual forecasters for which we have a time series of at least 10 quarters with responses from this forecaster. Appendix A provides a detailed description of variable construction, and Tables A.1 and A.2 present key summary statistics for our data.

Forecasted variables. Macroeconomic data is released quarterly, and is also subsequently revised. At the time of the survey (second month of a given quarter t), the most recent information forecasters have is about the initial release of the previous quarter's macroeconomic variables. Thus, following Bordalo et al. (2020), we construct implied forecasted growth rates between the forecast for quarter $t+h$ and the *initial* release of the macroeconomic variable for quarter $t-1$, using the Philadelphia Fed's Real-Time Data Set for Macroeconomics. Financial variables are not revised, so we use historical data from the Federal Reserve Bank of St. Louis.

We label the realizations of forecasted variables as x_t, and denote the time t forecast by forecaster i as $\widetilde{E}_t^{(i)}[x_{t+h}]$. We use the tilde notation to be explicit about the fact that these forecasts are not necessarily the rational expectation, that is $\widetilde{E}_t^{(i)} \neq \mathbb{E}_t^{(i)}$ where $\mathbb{E}_t^{(i)}$ is the rational expectations operator conditional on the time t information of agent i.

It is useful to also define notation for the implied forecast error

$$fe_{t+h,t}^{(i)} = x_{t+h} - \widetilde{E}_t^{(i)}[x_{t+h}]$$

and also the time t revision in the forecast of x_{t+h}

$$fr_{t+h,t}^{(i)} = \widetilde{E}_t^{(i)}[x_{t+h}] - \widetilde{E}_{t-1}^{(i)}[x_{t+h}]$$

Following the literature, we also define the *consensus* forecast as the average forecast across all participating forecasters i:

$$\overline{E}_t[x_{t+h}] = \int_i \widetilde{E}_t^{(i)}[x_{t+h}]di,$$

and the consensus forecast error and revision, $\overline{fe}_{t+h,t} = \int_i fe_{t+h,t}^{(i)}di$ and $\overline{fr}_{t+h,t} = \int_i fr_{t+h,t}^{(i)}di$.

2.1. Motivational Evidence

The paper is motivated by two famous stylized facts in the survey data. First, the "underreaction" to new information in the *average* forecast, documented by CG. Second, the "overreaction" to new information documented at the individual level by BGMS.

Conceptual framework To help structure the discussion, and provide a formal definition of "under-" and "over-" reaction, we organize our analysis in the context of the following general framework of beliefs updating which allows for both dispersed and imperfect information. The framework generalizes the settings typically considered by the prior literature, such that the specific settings of BGMS and CG are special cases of our framework.

At time t agents provide a forecast of the h period ahead realization of a random variable x_{t+h} (e.g. GDP growth rate, inflation and etc.). We impose very limited restrictions on the time series process for x_t, assuming only that it is a stationary, linear Gaussian process and thus has a Wold representation with Gaussian innovations, that is

$$x_t = \mu + \sum_{k=0}^{\infty} \psi_k \varepsilon_{t-k}^x \tag{1}$$

where ε_{t-k}^x are i.i.d. standard normal variables and $\sum_{k=0}^{\infty} \psi_k^2 < \infty$.

We allow for the possibility that the information sets underlying the forecasts $\widetilde{E}_t^{(i)}(x_{t+h})$ contain both components with idiosyncratic errors and also components with a common error term that is correlated across forecasters. In our conceptual framework, we model this as agents having access to the following two noisy, but informative signals

$$\begin{aligned} g_t &= x_{t+h} + e_t \\ s_t^{(i)} &= x_{t+h} + \eta_t^{(i)} \end{aligned} \tag{2}$$

where $\eta_t^{(i)} \sim N(0, \tau^{-1})$ is i.i.d. across time and across agents, while $e_t \sim N(0, \nu^{-1})$ is i.i.d. only across time, but common across agents.[10] For simplicity we assume the idiosyncratic noise has the same variance τ^{-1} across all agents, but this is not necessary. For expositional purposes, we will refer to the signal subject to idiosyncratic errors as the "private" signal, but we are not making an explicit assumptions that forecasters possess some fundamentally private or "inside" information – the source of the idiosyncratic error could also be subjective perception noise. This signal is simply meant to capture the part of the information set that makes forecaster i different from everyone else.

We do not impose any ex-ante assumptions about whether the forecasts $\widetilde{E}_t^{(i)}[x_{t+h}]$ agents come up with based on this information are statistically optimal (i.e. rational). The only assumption we make is that each forecast is a convex combination of the previous period's forecast of x_{t+h}, that is $\widetilde{E}_{t-1}^{(i)}[x_{t+h}]$, and the two new signals g_t and s_t^i:

$$\widetilde{E}_t^{(i)}[x_{t+h}] = (1 - G_1 - G_2)\widetilde{E}_{t-1}^{(i)}[x_{t+h}] + G_1 s_t^{(i)} + G_2 g_t \tag{3}$$

where G_1 is the weight agents put on the idiosyncratic signal and G_2 is the weight put on the common signal. Rational expectations, denoted by $\mathbb{E}_t^{(i)}[x_{t+h}]$, are a special case of our general representation in (3), with the following weights: $G_1^{RE} = \frac{\tau}{\tau+\nu+1/\Sigma^{RE}}$ and $G_2^{RE} = \frac{\nu}{\tau+\nu+1/\Sigma^{RE}}$, where $\Sigma^{RE} \equiv var(x_{t+h} - \mathbb{E}_{t-1}^{(i)}[x_{t+h}])$.

Our conceptual framework is more general than the settings considered in the previous literature for three reasons. First, we do not assume that expectations are either optimal

[10] We model the new information arriving each period in the form of news about the future, i.e. x_{t+h}, as opposed to information about current realizations x_t. We make this assumption for expositional convenience, but all results carry through if the signals are centered on x_t instead. Moreover, Goldstein and Gorodnichenko (2022) find that this type of "forward" information structure is indeed the most empirically relevant.

or suboptimal in any particular way; we only assume they are linear and keep the weights put on the different pieces of information available to the agents unrestricted. Second, we do not need to assume any particular time series process for x_t (e.g. the literature has mainly focused on AR(1) processes), just that it is stationary (a weak assumption satisfied by the data). And third, we allow for the agents beliefs to be driven by informative signals with both idiosyncratic and common errors – the previous literature has overwhelmingly focused on information structures where agents only have signals with idiosyncratic errors.

2.2. Underreaction in consensus forecasts

In a seminal contribution, Coibion and Gorodnichenko (2015) consider whether consensus forecast errors are predictable by consensus forecast revisions, estimating

$$\overline{fe}_{t+h,t} = \alpha + \beta_{CG}\overline{fr}_{t+h,t} + err_t \tag{4}$$

This is not a regression that tests forecast rationality, since the average forecast revision $\overline{fr}_{t+h,t}$ is not available to the forecasters when they respond to the time t survey. Hence, the fact that $\overline{fe}_{t+h,t}$ is predictable by $\overline{fr}_{t+h,t}$ does not reject rationality. Still, the CG finding of $\beta_{CG} > 0$ is often referred to as an "under-reaction" of the average forecast, since it intuitively indicates that the average forecast revision does not adjust fully to all of the new information (across all agents) arriving at time t.

Formally, the regression coefficient β_{CG} is informative about the weight put on new information in the forecasts $\widetilde{E}_t^{(i)}[x_{t+h}]$. In the context of our conceptual framework in (3), this weight is given by $G \equiv G_1 + G_2$. A finding of $\beta_{CG} > 0$ is indicative of $G < 1$, which implies that the available signals are likely noisy, as they get a weight of less than one.

As our conceptual framework makes explicit, there are two potential types of noise that could be present in the forecasters' information sets – idiosyncratic noise, i.e. $\eta_t^{(i)}$, and noise that is common to all forecasters, i.e. e_t. Both of these noise terms affect the estimated G that can be inferred from the CG regression coefficient. To gain some intuition, notice that re-arranging the LHS of (3), and averaging across forecasters we get:

$$\overline{fe}_{t+h,t} = \frac{1-G}{G}\overline{fr}_{t+h,t} - \frac{G_2}{G}e_t \tag{5}$$

Since the consensus forecast revision $\overline{fr}_{t+h,t}$ is positively correlated with the time t common signal error e_t, the regression coefficient β_{CG} is biased downward from $\frac{1-G}{G}$.

The focus of Coibion and Gorodnichenko (2015) is to estimate whether information is noisy at all, hence in their benchmark specification they worked under the simpler information structure that there are no common signals, which is equivalent to $\nu^{-1} = 0$ and implies $\beta_{CG} = \frac{1-G}{G}$. In that case $\frac{1}{1+\beta_{CG}} = G$ is a direct measure of the weight put on new information G. Nevertheless, the Appendix to CG itself acknowledges that if there is a common noise component like e_t, this will bias β_{CG} downward, so that $\beta_{CG} < \frac{1-G}{G}$, which would in turn bias their estimate of G upward (implying less noise than there really is).

One can account for both the idiosyncratic and common noise terms and estimate G robustly, by using a different approach. Following Goldstein (2023), we re-arrange (3) to express the difference between *individual* and consensus forecast revisions as:

$$(fr^i_{t+h,t}) - (\overline{fr}_{t+h,t}) = G(\overline{E}_{t-1}[x_{t+h}] - \widetilde{E}^{(i)}_{t-1}[x_{t+h}]) + G_1\eta^i_t \tag{6}$$

Standard OLS estimation of this regression provides an unbiased measure of the elasticity of forecasts to new information, G, since η^i_t is uncorrelated with $\widetilde{E}^i_{t-1}[x_{t+h}]$ and $\overline{E}_{t-1}[x_{t+h}]$.

Figure 1, panel (a) reports the estimate of G from regression (6) in the SPF and compares them with the estimated gains from the original CG specification, which we label $G_{CG} \equiv \frac{1}{1+\beta_{CG}}$. We discuss the estimates based on the SPF data with $h = 3$ in the main text, as these are the most widely used data (and horizon) in the literature so we make them our benchmark case. In Appendix E.1 we reports detailed tables of the estimates for both $h = 2$ and $h = 3$, and also all of the same specifications for the Blue Chip survey data. In all cases, results are very similar across both horizons and surveys.

The key takeaway from Figure 1 is that the robustly estimated gain G we obtain is consistently lower than that implied by the standard CG regression, across all forecasted variables, suggesting that the bias due to common noise terms indeed plays a significant role in the data. The differences are quite substantial – the average information gain based on the CG methodology is $G_{CG} = 0.75$, while our robust estimation yields $G = 0.5$ on average, so the implied "information stickiness" $1 - G$ doubles when we account for common error components. Lastly, our estimated gains are also remarkably stable across variables and have small standard errors, while the original CG regression leads to much noisier estimates. All of this leads us to conclude that the underlying information sets forecasters use indeed contain a substantial common error component.

2.3. Overreaction in individual forecasts

In another seminal paper, Bordalo et al. (2020) directly test rational expectations by regressing forecast errors on forecast revisions at the level of *individual* forecasters

$$fe^{(i)}_{t+h,t} = \alpha_i + \beta_{BGMS} fr^{(i)}_{t+h,t} + err^{(i)}_t \tag{7}$$

If the forecasts were rational, it should not be possible to predict forecast errors using forecast revisions, as the forecast revisions are clearly part of the forecasters' time t information sets. Thus, rational expectations predicts $\beta_{BGMS} = 0$, but to the contrary, Bordalo et al. (2020) find that for many macroeconomic series $\beta_{BGMS} < 0$. Intuitively, this implies that individual forecasts *overreact* to the arrival of new information, since the time t forecast revision was "too much", and ends up overshooting the actual realization x_{t+h}.

We replicate their results by estimating (7) as a panel with individual forecaster fixed effects, and plot the estimated β_{BGMS} in Figure 1, panel (b). Including fixed effects is important because the panel specification otherwise exploits cross-sectional variation in addition to the time series variation, which is more information than available to the forecasters in real time (i.e. they do not know the other respondents' contemporaneously submitted forecasts). In the main text we again focus on the SPF data with horizon $h = 3$, and leave the detailed tables of estimates at different horizons for both the SPF and Blue Chip in Appendix E.2. Results are again very similar across horizons and surveys.

3. Strategic Incentives in Survey of Forecasts

The findings of under-reaction in average forecasts and the simultaneous over-reaction in individual forecasts have been important and influential in the recent macro literature, informing both the calibration of information frictions in models and also disciplining theories of behavioral expectations formation. However, that literature has generally interpreted the predictability of forecast errors based on the key assumption that the survey forecasts are *unbiased* measures of the respondents true beliefs and expectations.

This assumption is contrary to an older, theoretical literature, like Laster et al. (1999) and Ottaviani and Sørensen (2006), which emphasizes that the main business goal of professional forecasters is to compete with one another for customers, and that they participate in surveys primarily as a way to grow their reputation and attract more customers. One important reputation building channel is winning one of the prestigious "Best Forecaster Awards" the surveys themselves publicize, such as the Lawrence R. Klein Best BlueChip Forecaster Award which has been running since the 1980s. Many other surveys of professional forecasters, such as those conducted by FocusEconomics, Consensus Economics, and Bloomberg, also publicly recognize a select number of forecasters based on their relative accuracy. These awards are coveted by professional forecasters and notably, only the top few forecasters (and usually just one) – the most accurate over the time period the award covers – are publicized as the "winner" of such awards. Hence, the reputational payoff for being the best is significantly higher than finishing in any other position in terms of relative accuracy, since none of the "losers" are ranked or publicized anyways.

Motivated by these observations, Ottaviani and Sørensen (2006) formalize the competitive incentives forecasters face in answering surveys in the form of a model where survey respondents participate in a prize-winning forecasting contest, where only the most accurate agent (in a given period) is awarded the prize. Ottaviani and Sørensen (2006) show that this microfounded model generates *strategic diversification* incentives in responding to the survey, implying that forecasters will optimally seek to "stand out" from the crowd and report forecasts that are shaded away from what they expect everyone else to report. This maximizes their chances of being the "winner" in any given period, even if, on average, their forecasts will exhibit higher forecast error variance.

In this section, we present a tractable version of a model with strategic incentives based on the beauty-contest framework of Morris and Shin (2002), and also generalized relative to Ottaviani and Sørensen (2006) to include both common and idiosyncratic sources of information. In Appendix N we show that the equilibrium conditions and key implications of the beauty-contest framework we present here are the same as a formal winner-take-all prize winning contest, hence we can think of a winner-take-all game as one type of microfoundation of the more reduced form beauty-contest framework we analyze here.

We use the model to derive two key results. First, we show that the reporting bias generated by the strategic incentives delivers simultaneously both of the stylized facts that the recent literature has emphasized – (i) the BGMS over-reaction in individual forecasts and (ii) the CG under-reaction in the consensus forecast. Second, we show that our strategic framework makes the further characteristic prediction that while reported forecasts overreact to idiosyncratic information, they *underreact to common information*.

Our main contribution is to test and extensively confirm this novel and differentiating implication in the data. Overall, our results suggest that the well-documented "biases" in forecasting surveys are likely symptoms of the strategic incentives respondents face.

3.1. Analytical Model of Strategic Incentives

For the analytical results of this subsection we make the simplifying assumption that x_t is an i.i.d., white noise process, which helps with analytics since the solution to the strategic interaction game necessitates finding a fixed point. Nevertheless, this simplification is only assumed for the analytical results, and we design our empirical tests and derive their properties in Section 3.2 for the general case of x_t being any stationary process. Moreover, in Section 4 we also present a quantitative version of the model where we assume that x_t follows an AR(1) process that is estimated from the actual data.

Agents are rational and have access to the two signals, g_t and $s_t^{(i)}$, as defined in (2). In the spirit of Morris and Shin (2002), the forecasters' problem is to submit a forecast $\widetilde{E}_t^{(i)}[x_{t+h}]$ that minimizes the weighted average of the expected squared forecast error and the expected distance from the average forecast:

$$\min_{\widetilde{E}_t^{(i)}[x_{t+h}]} \mathbb{E}\left[(\widetilde{E}_t^{(i)}[x_{t+h}] - x_{t+h})^2 - \lambda(\widetilde{E}_t^{(i)}[x_{t+h}] - \overline{E}_t[x_{t+h}])^2 \middle| g_t, s_t^{(i)} \right] \tag{8}$$

The strategic interactions incentives are parameterized by $\lambda \in (-1, 1)$, where $\lambda > 0$ indicates a game of strategic diversification – i.e. a game where agents want to stand-out from the crowd, as in a prize-winning contest (see Appendix N). In that case, agents optimally report forecasts that balance accuracy and the incentive to be different from the average forecast across the other agents. This can be seen in the first order conditions:

$$\widetilde{E}_t^{(i)}[x_{t+h}] = \frac{1}{1-\lambda}\mathbb{E}(x_{t+h}\,|g_t, s_t^{(i)}) - \frac{\lambda}{1-\lambda}\mathbb{E}(\overline{E}_t[x_{t+h}]\,|g_t, s_t^{(i)}) \tag{9}$$

If $\lambda = 0$, there are no strategic incentives and agents report their true rational beliefs. Otherwise, when $\lambda > 0$ agents optimally shade their reported forecasts *away* from their expectation of the average forecast $\overline{E}_t[x_{t+h}]$.

In the special case under consideration here, where x_t is i.i.d., the rational expectation of x_{t+h} is given by

$$\mathbb{E}[x_{t+h}\,|g_t, s_t^{(i)}] = \mu + G_1^{RE}(s_t^{(i)} - \mu) + G_2^{RE}(g_t - \mu) \tag{10}$$

with $G_1^{RE} = \frac{\tau}{\tau+\nu+\chi}$, $G_2^{RE} = \frac{\nu}{\tau+\nu+\chi}$, where $\chi = 1/Var(x_t)$. And to solve for the optimal reported forecast, $\widetilde{E}_t^{(i)}[x_{t+h}]$, we guess a similar linear form

$$\widetilde{E}_t^{(i)}[x_{t+h}] = \mu + G_1(s_t^{(i)} - \mu) + G_2(g_t - \mu) \tag{11}$$

We refer to (10) as the *true* beliefs and to (11) as the *reported* forecast. Solving the equilibrium fixed point jointly defined by (9) and (11), we obtain

$$G_1 = \frac{G_1^{RE}}{(1-\lambda)+\lambda G_1^{RE}}$$

$$G_2 = \frac{(1-\lambda)G_2^{RE}}{(1-\lambda)+\lambda G_1^{RE}}$$

Thus, when agents face strategic *diversification* incentives ($\lambda > 0$), and hence want to stand out from the crowd, they overweight their idiosyncratic signals (that is $G_1 > G_1^{RE}$), at the expense of underweighting the common (public) signal ($G_2 < G_2^{RE}$). In other words, dividing the above two equations by the respective optimal weights G_1^{RE} and G_2^{RE} we have

$$\frac{G_1}{G_1^{RE}} > \frac{G_2}{G_2^{RE}} \iff \lambda > 0 \tag{12}$$

Now let us relate to the CG and BGMS regressions we estimated in the previous section. Formally we have the following Proposition (all proofs are in the Appendix).

Proposition 1. *In the strategic incentives game in (8), the resulting forecasts imply*

$$\beta_{BGMS} = \frac{-\lambda\tau\chi}{([(1-\lambda)\nu+\tau]^2+(1-\lambda)^2\nu\chi)} < 0 \qquad \iff \lambda > 0$$
$$\beta_{CG} = \frac{(1-\lambda)\tau\chi}{([(1-\lambda)\nu+\tau]^2+[(1-\lambda)^2\nu+\tau]\chi)} > 0 \qquad \iff \lambda < 1$$

The first result in the Proposition states that in the presence of strategic diversification incentives ($\lambda > 0$) reported forecasts are over-reacting to time t information, in the sense of BGMS. The key intuition is that the new information at time t, i.e. the signals g_t and $s_t^{(i)}$, is partly idiosyncratic, hence the model implies that those new signals as a whole get overweighted relative to the prior, μ, which is common across agents. And we show in Section 4, this result carries through to dynamic versions of the model as well, with the intuition being that the time t priors of agents are weighted average of past signals, and hence less dispersed than the new time t idiosyncratic signals, giving us again $\beta_{BGMS} < 0$.

The second result of the Proposition states that the model is also consistent with the CG finding of $\beta_{CG} > 0$. The reason is two-fold. First, because there is dispersed information, the average belief is sticky and the CG regression coefficient would be positive if reported forecasts $\widetilde{E}^{(i)}_{t-1}[x_{t+h}]$ were rational, for the same reasons as in Coibion and Gorodnichenko (2015). Second, while with $\lambda > 0$ the reported forecasts overreact to new information, as long as the strategic diversification incentives are not extreme ($\lambda < 1$), these two effects balance out so that the model implies $\beta_{CG} > 0$. Still, it remains the case that $G < G^{RE}$ for $\lambda > 0$, hence even if one correctly estimates the information gain G from the data, they would infer a smaller degree of information rigidity than there really is.

Thus, a model of strategic diversification incentives is consistent with both the under-reaction in aggregate forecasts, and the over-reaction in individual forecasts, and also biases the econometrician's inference on the underlying information sets and beliefs.[11]

In addition to matching these influential stylized facts, the strategic diversification model also makes the further, differentiating prediction that forecasts should specifically overweight new *idiosyncratic* information, at the expense of new common information (see equation (12)). Intuitively, forecasters are willing to sacrifice some accuracy in order to stand out from the crowd, and they do that by tilting their forecast towards idiosyncratic and away from common information. Simple models of behavioral overreaction often used to explain the BGMS overreaction results, such as diagnostic expectations for example, do not imply such differential under- and overreaction to different kinds of signals.

In the next section, we test this implication directly. But before we turn to that, we briefly note that this characteristic relative overweighting of idiosyncratic signals also implies that the consensus forecast $\overline{E}_t(x_{t+h})$ is in fact *more accurate* than the average of the true rational individual beliefs (i.e. $\overline{\mathbb{E}}_t[x_{t+h}]$). This counter-intuitive finding is due to the fact that idiosyncratic errors, which are overweighted in the suboptimal reported forecasts,

[11] In a concurrent paper, Baley and Turén (2023) study forecast lumpiness and its interaction with strategic incentives. Given their focus on lumpiness, they take a reduced form approach to modeling strategic interactions and assume agents treat the consensus as an exogenous variable rather than taking into account it is a function of others' actions (as in our beauty-contest equilibrium). Their analysis is complementary but distinct to ours, taking a reduced form approach to strategic forces and instead focusing on lumpiness.

wash out when we aggregate to the consensus forecast $\overline{E}_t[x_{t+h}]$. So while the individual forecasts are suboptimally volatile, this washes out in the aggregate and leaves the consensus forecast surprisingly accurate.[12] There is also a similar result due to behavioral overconfidence in Broer and Kohlhas (2022), but here it arises as a result of strategic incentives. Formally, we find the following relationship between the mean squared errors of the consensus forecast with and without strategic incentives biasing reported forecasts.

Proposition 2. *In the strategic incentives game in (8) with $\lambda \in (0, 1)$,*

$$Var(x_{t+h} - \overline{E}_t(x_{t+h})) = \frac{(1-\lambda)^2}{(1-\lambda+\lambda G_2^{RE})^2} Var(x_{t+h} - \overline{\mathbb{E}}_t(x_{t+h})) < Var(x_{t+h} - \overline{\mathbb{E}}_t(x_{t+h}))$$

This interesting implication is also consistent with the puzzling empirical finding that while individual forecasts display predictable errors, the consensus forecasts are surprisingly accurate, and virtually impossible to beat (e.g. Kohlhas and Robertson (2022)).

3.2. Under-reaction to public information in the data

We propose a new empirical methodology that allows us to directly test for the characteristic implication of underweighting of common signals relative to idiosyncratic ones. The methodology works under the very mild assumptions that x_t follows an arbitrary stationary process and that the reported forecasts follow the linear recursive structure in (3).

We observe that manipulating (3) we can express individual forecast errors as:

$$fe^{(i)}_{t+h,t} = \frac{1-G_1}{G_1}(\widetilde{E}^{(i)}_t[x_{t+h}] - \widetilde{E}^{(i)}_{t-1}[x_{t+h}]) - \frac{G_2}{G_1}(g_t - \widetilde{E}^{(i)}_{t-1}[x_{t+h}]) - \eta^{(i)}_t \tag{13}$$

and then use an empirical proxy for the public signal g_t to estimate the regression

$$fe^{(i)}_{t+h,t} = \alpha + \beta_1 fr^{(i)}_{t+h,t} + \beta_2 pi^{(i)}_{t+h,t} + err^{(i)}_t \tag{14}$$

where $pi^{(i)}_{t+h,t} = g_t - \widetilde{E}^{(i)}_{t-1}[x_{t+h}]$ is the deviation of the public signal from agent i's previous

[12] See also Lichtendahl et al. (2013) for a related result in the specific context of "winner-take-all" games.

period forecast. Loosely speaking, this is the individual-level "surprise" in the public signal, the same way it appears on the RHS of the forecast error equation (13).

If the reported forecasts are rational then the forecast errors should not predictable and hence $\beta_1 = \beta_2 = 0$. However, when forecasts deviate from optimal the β_1 coefficient speaks to deviations in regards to the weight put on the idiosyncratic signal and β_2 is informative about the differential under- and over-weighting of the common signal. Formally, we can derive the following Proposition regarding these regression coefficients.

Proposition 3. *If agents forecasts follow (3), then for any stationary process x_t*

$$\beta_1 < 0 \iff G_1 > \frac{G_1^{RE}}{G^{RE} + \frac{\Sigma^{RE}}{\tilde{\Sigma}}(1 - G^{RE})}$$

$$\beta_2 > 0 \iff \frac{G_1}{G_1^{RE}} > \frac{G_2}{G_2^{RE}}$$

where $\widetilde{\Sigma} \equiv var(x_t - \widetilde{E}_{t-1}^{(i)}[x_t])$ *and* $\Sigma^{RE} \equiv var(x_{t+h} - \mathbb{E}_{t-1}^{(i)}[x_{t+h}])$.

Intuitively, $\beta_2 > 0$ implies that the common signal, g_t, is underweighted relative to the idiosyncratic signal, $\eta_t^{(i)}$, and more generally the null hypothesis $\beta_2 = 0$ directly tests the key characteristic implication of the strategic diversification model we derived in equation (12). On the other hand, $\beta_1 < 0$ implies an overreaction to the idiosyncratic signal – since $G^{RE} + \frac{\Sigma^{RE}}{\tilde{\Sigma}}(1 - G^{RE}) < 1$, finding $\beta_1 < 0$ implies that $G_1 > G_1^{RE}$.

To implement regression (14), we use the lagged consensus $\overline{E}_{t-1}(x_{t+h})$ – the average of the individual forecasts made in the previous quarter about the same future date $t+h$ – as a proxy for the public signal g_t.[13] The previous quarter's consensus is a newly available signal for quarter t, because the survey is distributed to participants at the end of the first

[13]For analytical convenience, in our model we assume there is a continuum of agents and we further assume that agents do not observe the consensus forecast, current or lagged. This is the standard assumption in the literature, because with a continuum of agents, averaging over equation (11) implies that the consensus perfectly reveals x_{t+h}, and hence observing the consensus makes the forecasting game and problem trivial. In real life, however, the actual panel dimension in the SPF is limited (around 30 forecasters per period). Therefore, the idiosyncratic signal noise will not completely average out, rendering the lagged consensus just a noisy signal about x_{t+h} as we treat it in the empirical analysis. We elaborate further on this subtle point in Appendix L, where we solve the model for a finite number of forecasters.

month of each quarter and results are publicized after these forecasts are submitted, at the end of the second month of the quarter. Hence, the first time the quarter $t-1$ survey results are available to the agents is when they are asked to submit forecasts for the quarter t survey. Second, this is a public signal because the history of the survey is available to all participants. And third, it has been well documented that consensus forecasts are indeed highly informative and salient signals that all survey participants are likely taking into account (see for example Kohlhas and Robertson (2022)).

In Table 1 we report both the panel regression estimates of (14) (with individual fixed effects) and the median estimates from running the regression forecaster-by-forecaster in the SPF. Both specifications display a consistent finding of $\beta_1 < 0$ and $\beta_2 > 0$ across variables, with our key regression coefficient β_2 being clearly positive and statistically significant for all forecast series. Figure 2 provides a graphical representation of our estimates.

This means that while idiosyncratic signals are indeed overweighted (as we might expect from the BGMS findings), there is in fact a relative *underreaction* to common signals. This result confirms the key implication of the model with strategic diversification incentives, and lead us to conclude that deviations from rationality are not necessarily needed to explain the puzzling features of forecast surveys. More generally, these empirical results provide a refined understanding of the over and under-reaction of forecasts – some information (i.e. public) is underweighted relative to other information – and this gives us a new informative empirical moment for models to take into account.

Figure 3, panel (a) shows that the estimates of regression (14), remain virtually the same when using the non-anonymous Blue Chip survey data.[14] Overall, forecasts are similarly estimated to underreact to public information and overreact to idiosyncratic information. Moreover, for variables that appear in both the SPF and Blue Chip (e.g. RGDP, CPI), the estimated β_1 and β_2 coefficients are numerically very similar across both data sets. In Figure 3 panel (b) we also consider an additional interesting variation on our regression,

[14]Moreover, Appendix F presents Tables and Figures that replicate all results in the paper using the BlueChip.

where we use the individual forecast errors and revision from the BC survey but as a measure of the public signal we use the lagged consensus from the SPF (so the consensus from a different survey). The results remain very much the same again.

Lastly, econometrically, it is also interesting to observe that the point estimates of β_1 and β_2 are very consistent across variables and are also tightly estimated, unlike the BGMS coefficients which are significantly noisier (see Figure 2). Moreover, our β_1 estimate is consistently more negative than the corresponding β_{BGMS} estimates. Thus, once we separately identify the overreaction to idiosyncratic information, we find that it is much stronger than the overall overreaction that BGMS estimate. This suggest that controlling separately for the public signal on the right hand side of the forecast error predictability regressions is useful even just for limiting estimation noise, as otherwise the differential under- and over-weighting of the different types of signals is averaged out in the BGMS regression.[15]

3.3. Further tests

While the differential under- and overreaction to common relative to idiosyncratic information is inconsistent with behavioral models of overreaction where agents overreact to all new information, it can be consistent with "overconfidence" models where agents overestimate the precision of their idiosyncratic signals relative to public signals (e.g. Daniel et al., 1998; Eyster et al., 2019; Broer and Kohlhas, 2022).

In order to distinguish between such behavioral models and strategic diversification theories, we bring in data on the Greenbook forecasts by the Federal Reserve Board of Governors. Differently from the surveys of professional forecasters, the Greenbook forecasts are not widely publicized but are prepared for internal FOMC purposes, hence they are less likely to be subject to the strategic incentives facing professional forecasters that compete with each other for clients. The Greenbook forecasts are also only published with a 5 year lag, which further diminishes any potential strategic forces.[16] Thus, under the null

[15]In Appendix C we also show that our empirical results are robust to much stricter treatment of outliers, than in our benchmark approach in which we follow Bordalo et al. (2020).

[16]Unfortunately, it is not straightforward to also compare against household or consumer surveys, since com-

hypothesis that the forecast biases in the surveys of professional forecasters are driven by strategic incentives, we would expect that such biases are not present in the Greenbook.

We treat the Greenbook as a time series of individual forecasts, and estimate both the BGMS regression and our main regression (14) on it. While one can argue that multiple economists contribute to the Greenbook forecast, the assumption we make is that they do so together, sharing their information sets, in which case the Greenbook projections are forecasts made under a single information set, and hence the rationality tests apply readily.

Data on the Greenbook (GB) projections is provided by the Federal Reserve Bank of Philadelphia. We use that data to construct a forecast dataset with the same structure and coverage as the SPF. Namely, while the GB projections are produced before each FOMC meeting, we group the forecasts at the quarterly level by keeping only the last forecasts of the quarter. The forecasts are produced for up to 9 quarters in the future, but we only keep up to 4 quarters ahead, in-line with the SPF forecasts. There are 15 forecasted variables, and we keep only the 11 appearing also in the SPF. We follow the same procedure as with the SPF and transform all series into forecasts of the implied annual growth rate from $t-1$ to $t+3$. Appendix A provides a detailed description of variable construction, and within that Table A.3 presents the summary statistics for each series in this dataset.

No overreaction overall. Panel (a) of Figure 4 plots the coefficients estimated by running the BGMS regression, equation (7), on the GB forecast data. The results show that with the only exception of forecasts of government consumption, the GB forecasts do not exhibit a BGMS overreaction bias to new information. Table E.9 reports full details on the BGMS regression estimates for the GB survey at both 3 and 2 quarters horizons.

No underreaction to public information. As in our benchmark case, we use the lagged consensus forecast from the SPF as a public signal since it is similarly available to the Fed's economists at the time of their projection. This also makes the regression directly

mon surveys like the Michigan Survey of Consumer Expectations have a very limited panel component (with households rotating out of the survey) or even only contain repeated cross-sections. Our tests need a substantial time series of forecasts of the same individual.

comparable to our results in the previous section.

Panel (b) of Figure 4 reports the estimates from regression (14) implemented on the GB forecast data. Again, the results show that in large part the GB forecasts exhibit neither overreaction to new idiosyncratic information ($\beta_1 \approx 0$, red bars), nor underreaction to public information ($\beta_2 \approx 0$, blue bars). The only exceptions are, again, government purchases forecasts. Table E.9 provides full estimation details for our benchmark horizon $h = 3$, and Table E.10 in the Appendix shows that results remain the same for $h = 2$.

Thus, we conclude that the Greenbook's forecasts do not display the biases documented in professional forecasters surveys – neither overall overreaction to new information, nor the differential under- and overreaction to idiosyncratic relative to common signals. This finding is consistent with the strategic diversification theory we propose, but not with behavioral explanations of the biases in professional forecast surveys, unless only the professionals, but not the Fed, suffer from behavioral biases in expectations formation.

Furthermore, we perform two additional tests in Appendix B. First, we show that professional forecasters underweight the lagged consensus substantially more than another commonly observed signal, suggesting that forecasters are especially sensitive to differentiating themselves from the consensus. And second, we show that the underweighting of common information is significantly more pronounced for the variables for which the precision of the available common information is higher, as predicted by the model.

3.4. A univariate regression approach

Alternatively to our benchmark specification (14), one can also run an univariate version of the regression with only the public signal surprise as a regressor,

$$fe^{(i)}_{t+h,t} = \alpha + \beta_{BK_{dm}} pi^{(i)}_{t+h,t} + err^{(i)}_t \tag{15}$$

Broer and Kohlhas (2022) estimate a similar specification on SPF data, and while both approaches agree under the null of RE without strategic incentives (i.e. $\beta_2 = 0$ in (14) and $\beta_{BK_{dm}} = 0$), estimating $\beta_2 \neq 0$ and $\beta_{BK_{dm}} \neq 0$ imply interestingly different notions of over and underreaction.

As Proposition 3 showed, a positive $\beta_2 > 0$ directly implies that public signals are underweighted relative to private signals, which is our key test of strategic incentives. Instead, a $\beta_{BK_{dm}} > 0$ roughly speaks to underweighting of the public signal relative to the *prior* forecast (i.e. $\widetilde{E}^{(i)}_{t-1}[x_{t+h}]$). We show this formally in Proposition 4 below.

Proposition 4. *When forecasts follow a linear structure as in (3) and x_t is stationary*

$$\beta_{BK_{dm}} > 0 \iff \frac{1-G}{1-G^{RE}} > \frac{G_2}{G_2^{RE}} \frac{\Sigma^{RE}}{\widetilde{\Sigma}}$$

where $\Sigma^{RE} = Var(x_{t+h} - \mathbb{E}^{(i)}_{t-1}[x_{t+h}])$ *and* $\widetilde{\Sigma} = va(x_{t+h} - \widetilde{E}^{(i)}_{t-1}[x_{t+h}])$

However, while estimating $\beta_{BK_{dm}} \neq 0$ clearly rejects RE and no strategic incentives, a non-zero estimate does not necessarily imply under or overweighting of public information relative to the prior, that is $\frac{1-G}{1-G^{RE}} \neq \frac{G_2}{G_2^{RE}}$. For example, a $\beta_{BK_{dm}} > 0$ can arise even when $\frac{1-G}{1-G^{RE}} = \frac{G_2}{G_2^{RE}}$, as long as both the public signal and the prior are *both underweighted*, i.e. $G_2 < G_2^{RE}$ and $1-G < 1-G^{RE}$. This is simply due the fact that non-rational forecasts imply a higher variance of errors, hence $\Sigma^{RE} < \widetilde{\Sigma}$ and by Proposition 4 the estimated $\beta_{BK_{dm}} > 0$ even though the weight on public info relative to the prior in the underlying forecasts is the same. On the other hand, in our benchmark regression (14) $\beta_2 > 0$ is directly informative about the relative weighting of public and private signals, and this is one of the key reasons we base our benchmark analysis on the multivariate regression.

Nevertheless, the univariate regression is an additional interesting moment which intuitively speaks to over/underweighting relative to the prior. And indeed when we estimate this univariate regression in our data we again consistently find evidence of underreaction $\beta_{BK_{dm}} > 0$ across all series – we report these results in Figure 5, panel (b). Thus, the univariate approach also consistently finds underreaction, although a different notion of underreaction – an underweighting of new public information relative to the prior.

And this result is actually also consistent with our strategic diversification framework. While the analytical model of Section 3 implies $\beta_{BK_{dm}} = 0$, this is only a consequence of the simplifying assumption we make in that Section that x_t is iid. In that case, agents share identical priors ($\mathbb{E}_{t-1}^{(i)} = \mu$), so both the prior and public signal are equally underweighted—i.e., $\frac{1-G}{1-G^{RE}} = \frac{G_2}{G_2^{RE}}$. Moreover, with x_t iid $\Sigma^{RE} = \widetilde{\Sigma}$, hence $\beta_{BK_{dm}} = 0$.

However, this knife-edge result changes once we allow x_t to be autocorrelated, as we do in the quantitative analysis below. The key intuition is that when x_t is autocorrelated, then the agents' information is long-lived (signals from previous periods are still informative about x_{t+h}), and hence there is heterogeneity in the agents' prior beliefs $\mathbb{E}_{t-1}^{(i)}[x_{t+h}]$. As a result, the new public signal g_t is optimally underweighted both relative to the heterogeneous priors and the new private signal, so we have $\frac{1-G}{1-G^{RE}} > \frac{G_2}{G_2^{RE}}$ and $\beta_{BK_{dm}} > 0$.[17]

Lastly, we note that Broer and Kohlhas (2022) run a slightly different univariate regression than (15) – while we use the *surprise* in the lagged consensus as the regressor, they use the level of the lagged consensus (i.e. just $\overline{E}_{t-1}[x_{t+h}]$). The deviations of the resulting regression coefficient β_{BK} from zero have a similar general interpretation (over/underweighting relative to the prior), but it is a distinct moment that for completeness we discuss in detail in Appendix C. Interestingly, even in the quantitative version of our model where we actually have underweighting of the public signal in equilibrium, i.e. $\frac{1-G}{1-G^{RE}} > \frac{G_2}{G_2^{RE}}$, the model implies $\beta_{BK} = 0$. So the framework's implications in terms of the univariate regressions is nuanced. The strategic diversification model is consistent with $\beta_{BK_{dm}} > 0$ as we robustly find in the data, but it does imply $\beta_{BK} = 0$, while the estimates of this coefficient in the data are dispersed around zero across the different variables in the SPF (some are positive, some negative, and some insignificant – panel (a) in Figure 5).

4. Quantitative Analysis

Our results indicate that strategic incentives make surveys of professional forecasts a biased measure of the true underlying beliefs of agents, and thus do not speak directly

[17] As a side note, our analytical model with iid fundamental in Section 3 would also imply $\beta_{BK_{dm}} > 0$ if we allow for heterogeneous unconditional (or long-run) priors μ_i. We show this in Appendix K, where we include heterogeneous priors for the long-run mean following Patton and Timmermann (2010). In that case, agents optimally put a lower weight on the public signal relative to their priors, i.e. $\frac{1-G}{1-G^{RE}} > \frac{G_2}{G_2^{RE}}$, and hence we can have both $\beta_{BK_{dm}} > 0$ and $\beta_2 > 0$ in the setting of the analytical model as well.

to the actual level of information frictions and noise in the forecasters' information sets. Nevertheless, the literature regularly turns to surveys to estimate such deep characteristics of interest as information frictions and the degree of uncertainty both in the time-series and the cross-section (see also the discussion in Coibion and Gorodnichenko (2015)).

In order to overcome this apparent problem, and uncover the true underlying beliefs, we estimate a quantitative version of the strategic model of Section 3.1, where we generalize the time series process for x_t to a AR(1) process and estimate all key parameters from the data. Agents are rational, and understand that x_t follows the AR(1) process

$$x_t = \rho x_{t-1} + u_t \tag{16}$$

with $u_t \sim N(0, \chi^{-1})$. They observe the two signals in equation (2), and form posterior beliefs about x_{t+h} according to the Kalman filter

$$\mathbb{E}_t^{(i)}[x_{t+h}] = \mathbb{E}_{t-1}^{(i)}[x_{t+h}] + G_1^{RE}(s_t^{(i)} - \mathbb{E}_{t-1}^{(i)}[x_{t+h}]) + G_2^{RE}(g_t - \mathbb{E}_{t-1}^{(i)}[x_{t+h}])$$

where the optimal Kalman gain weights are given by $G_1^{RE} = \frac{\tau}{1/\Sigma^{RE}+\nu+\tau}$ and $G_2^{RE} = \frac{\nu}{1/\Sigma^{RE}+\nu+\tau}$, and the forecast error variance Σ^{RE} is derived in closed form in Appendix I.

Strategic interactions The agents face the same objective function as in Section 3.1, equation (8). Solving the first order condition above involves finding a fixed point, and we do so by following the approach in Woodford (2001) and Coibion and Gorodnichenko (2012). We leave the details to Appendix I, and here only note that the reported forecast of agent i follows a Kalman-like recursion, with gain coefficients that deviate from optimality in a similar way as to the analytical model presented in Section 3.1. Specifically,

$$\widetilde{E}_t^{(i)}[x_{t+h}] = \widetilde{E}_{t-1}^{(i)}[x_{t+h}] + G_1(s_t^i - \widetilde{E}_{t-1}^{(i)}[x_{t+h}]) + G_2(g_t - \widetilde{E}_{t-1}^{(i)}[x_{t+h}]), \tag{17}$$

with

$$G_1 = \frac{G_1^{RE} - \lambda K_{(2,2)}}{1-\lambda} \text{ and } G_2 = \frac{G_2^{RE} - \lambda K_{(2,1)}}{1-\lambda}$$

where $K_{(2,2)}$ and $K_{(2,1)}$ are terms of a Kalman-gain matrix computed in the Appendix.

Omitting the algebraic details from the main text for the sake of brevity, we want to highlight two things. First, similar to the analytical framework we analyzed in Section 3.1, when $\lambda > 0$ the reported forecasts deviate from optimality by overweighting the private signal at the expense under-weighting the public signal in the sense of $\frac{G_1}{G_1^{RE}} > \frac{G_2}{G_2^{RE}}$. The reason is that, as we show numerically, $G_2^{RE} < K_{(2,1)}$ but $G_1^{RE} > K_{(2,2)}$.

Second, the structure of the reported forecasts $\widetilde{E}_t^{(i)}[x_{t+h}]$ falls within the general linear framework (3). Thus, given this now-familiar implication of relative under- and over-weighting of common and idiosyncratic signals, it follows that this more general model also implies $\beta_1 < 0$ and $\beta_2 > 0$ in regression (14), as we have found is true in the data.

4.1. Estimation

We estimate the model separately for each forecasted series in the SPF. For each series (e.g. GDP growth), we first estimate an AR(1) process on its realized values and recover the autoregressive coefficient ρ and the fundamental disturbance variance χ^{-1} that best describe each series. Given those parameters, we use simulated method of moments (SMM) to estimate the remaining parameters of the model: the public noise variance $\sigma_e^2 \equiv \nu^{-1}$, the private noise variance $\sigma_\eta^2 \equiv \tau^{-1}$ and the strategic incentive parameter λ.

The data moments we target for the SMM step are the average cross-sectional dispersion of forecast errors, the coefficient β_1 from our key regression in equation (14), and estimated information gain coefficient G from regression (6) We choose these three moments as they are differently affected by the three parameters to be estimated.

First, as foreshadowed in Section 3.1, a larger strategic incentive parameter λ increases the elasticity of posted forecasts to new information G, since the new information (i.e. g_t and $\eta_t^{(i)}$) is partly idiosyncratic and thus gets overweighted as a whole (see first result in Proposition 1) . Similarly, G is increasing in the precision of both the public and the private signals, ν and τ respectively, since G_1^{RE} and G_2^{RE} are increasing in those. On the other hand, as per Proposition 3, β_1 is directly informative about the weight put on the idiosyncratic signal $\eta_t^{(i)}$, G_1, which is increasing in the precision of the private signal τ but decreasing in the precision of the public signal ν, since those have opposite effects on the underlying optimal weight G_1^{RE}. Lastly, the dispersion of forecasts is increasing in λ (since that increases the weight put on the idiosyncratic signal, G_1), but decreasing in the precision of public signal ν (since that lowers the weight on the public signal, G_2).

Table 2 reports the estimated parameters, and in Table 3 we report how the model fits both the targeted moments, and a number of untargeted moments. As we can see, the model is able to essentially perfectly match all targeted moments (columns 1-6).

We evaluate the model against several untargeted moments, by considering the model's fit of the influential regression coefficients of CG and BGMS, and also the key β_2 coefficient in our key regression, all of which we had left untargeted in the estimation.

We find that the model does an excellent job of fitting the BGMS coefficients for virtually all series. This shows that the model is very much able to generate empirically relevant amounts of over-reaction in individual forecasts, without the need to assume any deviations from rationality. Second the model also generally fits the CG coefficients well, except for the case of the interest rate series. And third, the model also consistently generates positive β_2 coefficients, and if anything underestimates the degree to which public information is underweighted in the data. Thus, our estimation is in fact conservative in terms of its implications of the underweighting of public info.

Having confirmed that the estimated model fits the data, we use it to recover the underlying true beliefs of forecasters. A first important finding of this analysis is that the true individual beliefs of forecasters are significantly less precise than what might be inferred from the reported forecasts. To showcase this we report several moments of interest in Table 4. First, we compute the actual elasticity to new information of the true underlying beliefs, which we call G^{Honest} and is by definition equal to G^{RE} – the true Kalman gain used by the rational agents in the model. We differentiate this quantity from the elasticity of the reported forecasts, which we have labeled G all along.

In column 1, we report the elasticity G of the reported forecasts, as implied by the estimated model (as this was a targeted moment, the model's implications here simply match what we had already estimated from the data in reduced form). In column 2, instead, we report the true underlying elasticity to new information, $G^{Honest} \equiv G^{RE}$, that is implied by our estimated model. We call this the "honest" or true G. Comparing the two series of estimates, we find that the true beliefs are significantly "stickier", and put a much smaller weight on new information as compared to the raw reported forecasts. The average G^{RE} of the true beliefs is about 25% lower than that of the reported forecasts.

The reason is intuitive. The strategic diversification incentives give reason for agents to overweight their idiosyncratic signals and underweight the public signals in their reported forecasts. However, as discussed before, in their reported forecasts agents optimally overweight the idiosyncratic signals by a higher degree than the degree to which they underweight public signals, and as a result the overall weight put on new information in reported forecasts is higher than for the true underlying beliefs of the agents.

A second, related, but distinct implication is that the consensus forecast $\overline{E}_t[x_{t+h}]$ is in fact a highly accurate predictor of x_{t+h}, even though the individual forecasts underlying that consensus are suboptimal and display predictable errors. As explained in Section 3.1, the reason is that the overweighted idiosyncratic errors wash out when we average across forecasters. Wen report the difference between the mean squared error of the average of reported forecasts (i.e. the "consensus") and the corresponding MSE of the average of the agent's underlying true beliefs in columns 4 and 5 of Table 4 and also plot the ratio visually in panel (a) of Figure 6. For some variables (nominal GDP, CPI, Housing Start, Ten-year and AAA bond rate) the true mean squared error is more than *150% larger* than the MSE of the reported consensus forecast. Thus, strategic behavior has the unintended consequences of making the consensus forecast a surprisingly good predictor, as it harnesses the agents' dispersed information better than the average of the true underlying expectations.

And finally, the overweighting of idiosyncratic signals also artificially increases the cross-sectional dispersion of both forecast errors and reported forecasts, which two moments are often used as a model-free proxies for uncertainty and disagreement in the literature (e.g. Kozeniauskas et al. 2018). We report both the dispersion of the reported forecasts and the dispersion of the true underlying beliefs in columns 7 and 8, and also plot them visually in panel (b) Figure 6. We find the striking result that the actual, true dispersion of agents' beliefs is often *less than half* of the dispersion in reported forecasts.

All three of these results suggest that we should be cautious about the use of SPF moments for directly inferring the information noise and frictions that agents actually face. Surveys of forecasts can still be very useful for inferring deep economic parameters of interest, such as signal precision, but the survey data needs to be filtered to uncover the true underlying beliefs by taking into account the strategic incentives facing forecasters.

5. Conclusion

This paper evaluated whether strategic consideration might bias survey-reported forecasts away from the underlying true beliefs of agents, and thus complicate inference on the agents' true expectations. First, we show that a model with strategic diversification incentives is consistent with the seminal empirical findings of Coibion and Gorodnichenko (2015) and Bordalo et al. (2020). Second, we show that a characteristic and differentiating prediction of such a model is that common noisy signals should be *underweighted*, even though new information as a whole should be overweighted. We take this prediction to the data and find strong and robust evidence that common signals are indeed underweighted.

Our conclusion that survey of expectations is affected by strategic incentives has far reaching implications about the use of such expectational surveys in the broader economic literature. Most obviously, the surveys are not a direct and unbiased measure of expectations, as such the raw moments cannot be directly targeted to calibrate information frictions or uncertainty parameters in models. Still, the true beliefs and the likely true parameters of the underlying information sets of agents can be estimated once the strategic incentives are taken into account, and we do so and provide such estimates in the paper.

References

ANGELETOS, G.-M., Z. HUO, AND K. A. SASTRY (2021): “Imperfect macroeconomic expectations: Evidence and theory,” *NBER Macroeconomics Annual*, 35, 1–86.

BALEY, I. AND J. TURÉN (2023): “Lumpy Forecasts,” *Available at SSRN 4988612*.

BALEY, I. AND L. VELDKAMP (2023): “Bayesian learning,” in *Handbook of Economic Expectations*, Elsevier, 717–748.

BORDALO, P., N. GENNAIOLI, Y. MA, AND A. SHLEIFER (2020): “Overreaction in macroeconomic expectations,” *American Economic Review*, 110, 2748–82.

BORN, B., Z. ENDERS, M. MENKHOFF, G. J. MÜLLER, AND K. NIEMANN (2023): “Firm expectations and news: Micro v macro,” Tech. rep., Working Papers of the Priority Programme 1859” Experience and Expectation . . . .

BROER, T. AND A. N. KOHLHAS (2022): “Forecaster (mis-) behavior,” *Review of Economics and Statistics*, 1–45.

COIBION, O. AND Y. GORODNICHENKO (2012): “What can survey forecasts tell us about information rigidities?” *Journal of Political Economy*, 120, 116–159.

——— (2015): “Information rigidity and the expectations formation process: A simple framework and new facts,” *American Economic Review*, 105, 2644–78.

CROWE, C. (2010): “Consensus forecasts and inefficient information aggregation,” *IMF Working Papers*, 1–43.

DANIEL, K., D. HIRSHLEIFER, AND A. SUBRAHMANYAM (1998): “Investor psychology and security market under-and overreactions,” *the Journal of Finance*, 53, 1839–1885.

DRISCOLL, J. C. AND A. C. KRAAY (1998): “Consistent covariance matrix estimation with spatially dependent panel data,” *Review of economics and statistics*, 80, 549–560.

EUROPEAN CENTRAL BANK (2014): “Results of the Second Special Questionnaire for Participants in the ECB Survey of Professional Forecasters,” .

EYSTER, E., M. RABIN, AND D. VAYANOS (2019): “Financial markets where traders neglect the informational content of prices,” *The Journal of Finance*, 74, 371–399.

GOLDSTEIN, N. (2023): “Tracking inattention,” *Journal of the European Economic Association*, jvad022.

GOLDSTEIN, N. AND Y. GORODNICHENKO (2022): “Expectations Formation and Forward Information,” Tech. rep., National Bureau of Economic Research.

JUODIS, A. AND S. KUČINSKAS (2023): “Quantifying noise in survey expectations,” *Quantitative Economics*, 14, 609–650.

KOHLHAS, A. N. AND D. ROBERTSON (2022): “Cautious Expectations,” Tech. rep., Mimeo, Institute for International Economic Studies.

KOHLHAS, A. N. AND A. WALTHER (2021): "Asymmetric attention," *American Economic Review*, 111, 2879–2925.

KOZENIAUSKAS, N., A. ORLIK, AND L. VELDKAMP (2018): "What are uncertainty shocks?" *Journal of Monetary Economics*, 100, 1–15.

KUČINSKAS, S. AND F. S. PETERS (2022): "Measuring under-and overreaction in expectation formation," *Review of Economics and Statistics*, 1–45.

LASTER, D., P. BENNETT, AND I. S. GEOUM (1999): "Rational bias in macroeconomic forecasts," *The Quarterly Journal of Economics*, 114, 293–318.

LICHTENDAHL, K. C., Y. GRUSHKA-COCKAYNE, AND P. E. PFEIFER (2013): "The wisdom of competitive crowds," *Operations Research*, 61, 1383–1398.

MARINOVIC, I., M. OTTAVIANI, AND P. SORENSEN (2013): "Forecasters' objectives and strategies," in *Handbook of economic forecasting*, Elsevier, vol. 2, 690–720.

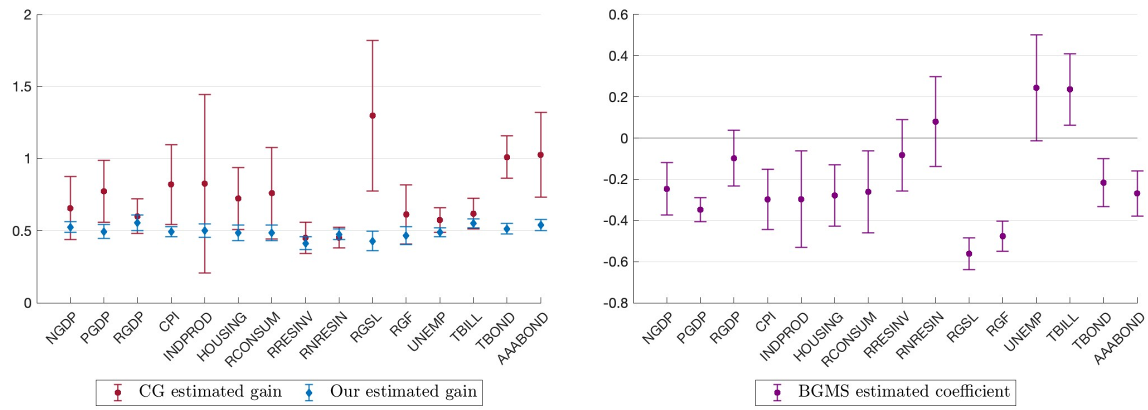
MORRIS, S. AND H. S. SHIN (2002): "Social value of public information," *american economic review*, 92, 1521–1534.

OTTAVIANI, M. AND P. N. SØRENSEN (2006): "The strategy of professional forecasting," *Journal of Financial Economics*, 81, 441–466.

PATTON, A. J. AND A. TIMMERMANN (2010): "Why do forecasters disagree? Lessons from the term structure of cross-sectional dispersion," *Journal of Monetary Economics*, 57, 803–820.

WOODFORD, M. (2001): "Imperfect common knowledge and the effects of monetary policy," Tech. rep., National Bureau of Economic Research.

Tables and figures



(a) Estimated gains G: our measure vs CG

(b) BGMS regression coefficient

Figure 1: Under- and over- reaction in SPF forecasts, $h = 3$ quarters

Notes: Panel (a): The red circles represent the implied gain from the CG regression, i.e. $\frac{1}{1+\beta_{CG}}$. Standard errors are Newey-West (1994), with optimal bandwidth selection. The blue diamonds represent the gain estimated from (6) with individual and time fixed effects. Standard errors are Driscoll and Kraay (1998). Panel (b): Panel estimate of β_{BGMS} with forecaster fixed effects. In both panels, bars reports the 90% confidence interval for the estimated coefficients and the forecast are at horizon h=3 quarters.

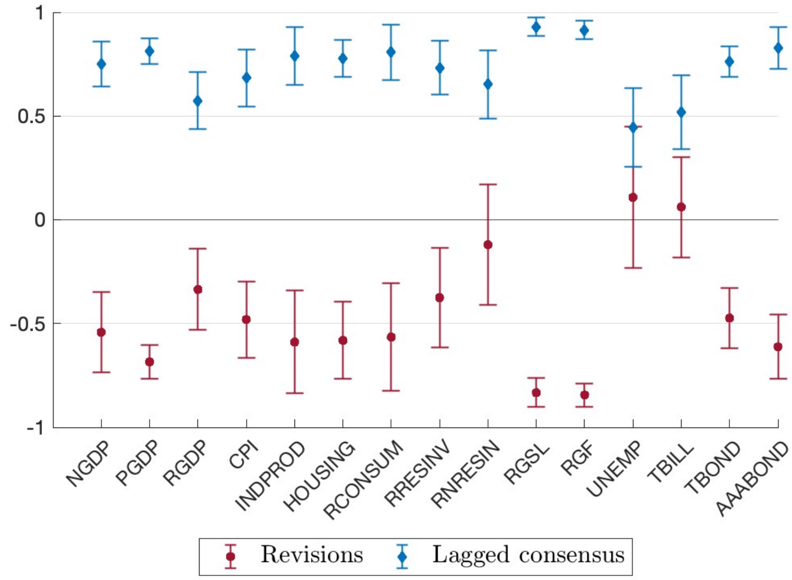
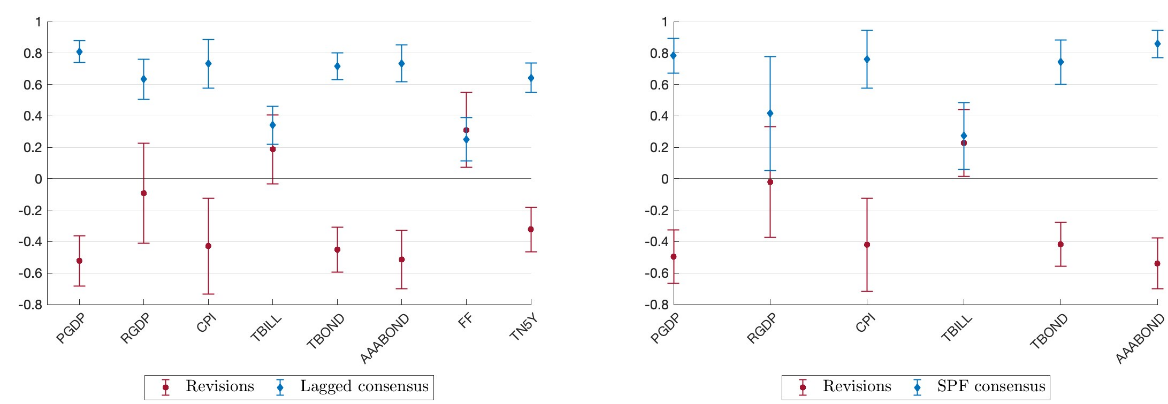


Figure 2: Forecast errors on forecast revisions and public information, SPF

Notes: this figure plots the coefficients from the panel regression (14) with individual fixed effect and horizon h=3. The red circles represent the coefficient β_1 while the blue diamonds represent the coefficient β_2. Bars reports the 90% confidence interval for the estimated coefficients. Standard errors are Driscoll and Kraay (1998).

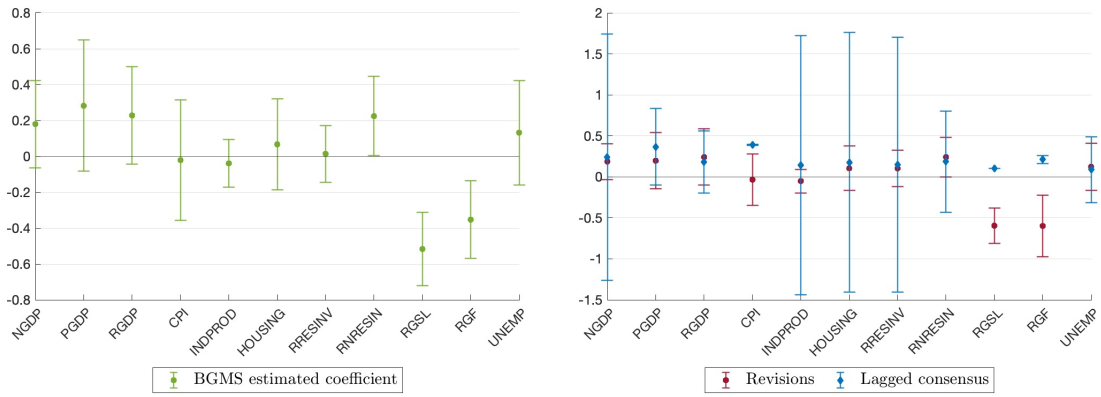


(a) Blue chip individual forecasts and consensus

(b) Blue chip individual forecasts and SPF consensus

Figure 3: Blue Chip survey data

Notes: this figure plots the panel estimates of regressions (14) with individual fixed effect and horizon h=3. Panel (a) uses only data from the Blue Chip survey. Panel (b) uses individual forecast data from the Blue Chip survey and the lagged consensus forecast data from the SPF. Bars reports the 90% confidence interval for the estimated coefficients. Standard errors are Driscoll and Kraay (1998).

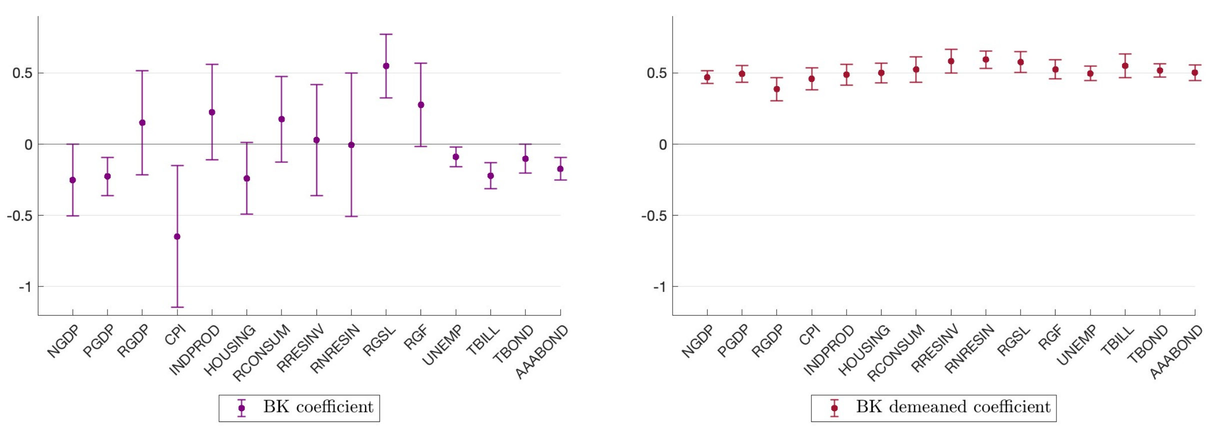


(a) BGMS coefficients

(b) Our regression (14)

Figure 4: Estimated coefficients on the Greenbook survey

Notes: Panel (a) reports the coefficient from the BGMS regression using the Greenbook projections. Panel (b) plots the estimates of our regression (14) using forecasts from the Greenbook. The blue diamonds represent the coefficient β_1 while the red circles represent the coefficient β_2. In both panels, bars report the 90% confidence interval for the estimated coefficients and the forecast are at horizon h=3. Standard errors are Newey and West (1994) with optimal bandwidth selection.

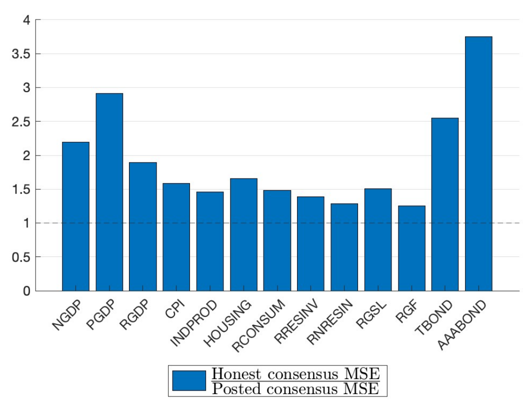


(a) Not demeaned (BK)

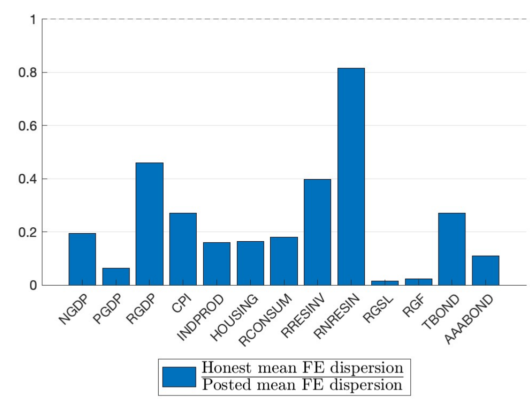
(b) Demeaned

Figure 5: Forecast errors on lagged consensus

Notes: this figure plots the panel estimates of regressions (24)and (15), in panel (a) and panel (b) respectively, with individual fixed effect and horizon h=3. Bars reports the 90% confidence interval for the estimated coefficients. Standard errors are Driscoll and Kraay (1998).



(a) Mean Squared Errors



(b) Forecast dispersion

Figure 6: Honest and posted forecasts

Table 1: Private and public information, SPF

Panel A: 3 quarters horizon

	Revision				Public signal			
	β_1	SE	p-value	Median	β_2	SE	p-value	Median
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Nominal GDP	-0.54	0.12	0.00	-0.44	0.75	0.07	0.00	0.76
GDP price index inflation	-0.68	0.05	0.00	-0.64	0.81	0.04	0.00	0.83
Real GDP	-0.34	0.12	0.01	-0.18	0.57	0.08	0.00	0.61
Consumer Price Index	-0.48	0.11	0.00	-0.46	0.68	0.08	0.00	0.69
Industrial production	-0.59	0.15	0.00	-0.60	0.79	0.08	0.00	0.78
Housing Start	-0.58	0.11	0.00	-0.53	0.78	0.06	0.00	0.71
Real Consumption	-0.57	0.16	0.00	-0.58	0.81	0.08	0.00	0.81
Real residential investment	-0.38	0.15	0.01	-0.39	0.73	0.08	0.00	0.66
Real nonresidential investment	-0.12	0.18	0.50	-0.10	0.65	0.10	0.00	0.51
Real state and local government consumption	-0.83	0.04	0.00	-0.81	0.93	0.03	0.00	0.89
Real federal government consumption	-0.84	0.03	0.00	-0.77	0.91	0.03	0.00	0.87
Unemployment rate	0.11	0.21	0.60	-0.02	0.44	0.11	0.00	0.42
Three-month Treasury rate	0.06	0.15	0.68	0.11	0.52	0.11	0.00	0.38
Ten-year Treasury rate	-0.47	0.09	0.00	-0.40	0.76	0.04	0.00	0.83
AAA Corporate Rate Bond	-0.61	0.09	0.00	-0.67	0.83	0.06	0.00	0.87

Panel B: 2 quarters horizon

	Revision				Public signal			
	β_1	SE	p-value	Median	β_2	SE	p-value	Median
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Nominal GDP	-0.35	0.09	0.00	-0.27	0.62	0.06	0.00	0.63
GDP price index inflation	-0.55	0.06	0.00	-0.50	0.70	0.04	0.00	0.66
Real GDP	-0.26	0.13	0.05	-0.14	0.54	0.08	0.00	0.54
Consumer Price Index	-0.38	0.09	0.00	-0.36	0.52	0.08	0.00	0.52
Industrial production	-0.16	0.12	0.19	-0.14	0.49	0.08	0.00	0.50
Housing Start	-0.15	0.08	0.07	-0.15	0.54	0.05	0.00	0.56
Real Consumption	-0.37	0.11	0.00	-0.29	0.64	0.07	0.00	0.69
Real residential investment	-0.13	0.11	0.22	-0.16	0.49	0.07	0.00	0.43
Real nonresidential investment	-0.02	0.10	0.81	-0.04	0.41	0.07	0.00	0.44
Real state and local government consumption	-0.63	0.08	0.00	-0.51	0.79	0.04	0.00	0.72
Real federal government consumption	-0.71	0.06	0.00	-0.64	0.80	0.04	0.00	0.74
Unemployment rate	0.09	0.15	0.56	0.03	0.39	0.10	0.00	0.37
Three-month Treasury rate	0.02	0.11	0.89	0.10	0.48	0.10	0.00	0.39
Ten-year Treasury rate	-0.46	0.11	0.00	-0.45	0.71	0.07	0.00	0.67
AAA Corporate Rate Bond	-0.49	0.08	0.00	-0.52	0.70	0.06	0.00	0.71

Notes: this table reports the coefficients of regression (14) (individual forecast errors on individual revisions and public information). Columns 1 to 3 show coefficient β_1 (forecast revision) from the panel regression with individual fixed effect, with standard errors and corresponding p-values. Standard errors are robust and clustered by time and forecaster. Column 4 shows the median coefficient of the same regression at the individual level. Columns 5 to 7 show coefficient β_2 (public information) from the panel regression with individual fixed effect, with standard errors and corresponding p-values. Standard errors are Driscoll and Kraay (1998). Column 8 shows the median coefficient of the same regression at the individual level. Panel A uses forecast at 3 quarters horizon and panel B uses forecast at 2 quarters horizon.

Table 2: Estimated parameters

Variable	ρ (1)	$\frac{\sigma_e}{\sigma_u}$ (2)	$\frac{\sigma_\eta}{\sigma_u}$ (3)	λ (4)
Nominal GDP	0.93	2.20	2.90	0.74
GDP price index inflation	0.93	2.57	4.52	0.88
Real GDP	0.88	2.47	1.84	0.55
Consumer Price Index	0.78	1.91	2.54	0.61
Industrial production	0.85	1.64	3.44	0.68
Housing Start	0.85	1.89	3.27	0.70
Real Consumption	0.87	1.77	3.40	0.67
Real residential investment	0.89	2.44	3.04	0.49
Real nonresidential investment	0.89	5.63	1.64	0.25
Real state and local government consumption	0.89	1.73	7.76	0.90
Real federal government consumption	0.80	1.65	8.42	0.87
Ten-year Treasury rate	0.83	3.28	2.44	0.72
AAA Corporate Rate Bond	0.85	3.08	3.33	0.87

Table 3: Moments in data and model

	Targeted moments						Untargeted moments					
	Mean Dispersion		C		β_1		β_{CG}		β_{BGMS}		β_2	
Variable	Data (1)	Model (2)	Data (3)	Model (4)	Data (5)	Model (6)	Data (7)	Model (8)	Data (9)	Model (10)	Data (11)	Model (12)
Nominal GDP	1.49	1.49	0.53	0.53	-0.54	-0.54	0.52	0.41	-0.25	-0.31	0.75	0.21
GDP price index inflation	0.33	0.33	0.49	0.49	-0.68	-0.68	0.29	0.50	-0.35	-0.44	0.81	0.31
Real GDP	0.93	0.93	0.56	0.56	-0.33	-0.33	0.67	0.51	-0.10	-0.22	0.57	0.09
Consumer Price Index	0.31	0.31	0.49	0.49	-0.48	-0.48	0.22	0.38	-0.30	-0.24	0.67	0.16
Industrial production	3.71	3.71	0.50	0.50	-0.59	-0.59	0.21	0.22	-0.30	-0.22	0.79	0.26
Housing Start	110.04	110.04	0.49	0.49	-0.58	-0.58	0.38	0.32	-0.28	-0.28	0.78	0.23
Real Consumption	0.51	0.51	0.49	0.49	-0.56	-0.56	0.31	0.25	-0.26	-0.23	0.80	0.23
Real residential investment	27.03	27.03	0.41	0.41	-0.37	-0.37	1.22	0.40	-0.08	-0.17	0.73	0.11
Real nonresidential investment	7.38	7.38	0.48	0.48	-0.12	-0.12	1.21	0.94	0.08	-0.10	0.65	0.01
Real state and local government consumption	1.41	1.41	0.47	0.47	-0.84	-0.84	0.63	0.17	-0.48	-0.41	0.91	0.45
Real federal government consumption	6.40	6.40	0.43	0.43	-0.83	-0.83	-0.23	0.12	-0.56	-0.35	0.93	0.37
Ten-year Treasury rate	0.17	0.17	0.51	0.51	-0.47	-0.47	-0.01	0.69	-0.22	-0.38	0.76	0.09
AAA Corporate Rate Bond	0.34	0.34	0.54	0.54	-0.61	-0.61	-0.03	0.62	-0.27	-0.48	0.83	0.18

Table 4: Posted and honest moments

	G			Consensus MSE			Dispersion		
	Posted	Honest	Ratio	Posted	Honest	Ratio	Posted	Honest	Ratio
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Nominal GDP	0.53	0.40	0.76	0.49	1.07	2.19	1.49	0.29	0.19
GDP price index inflation	0.49	0.32	0.66	0.05	0.14	2.92	0.33	0.02	0.06
Real GDP	0.56	0.45	0.80	0.37	0.71	1.89	0.93	0.43	0.46
Consumer Price Index	0.49	0.40	0.82	0.23	0.36	1.58	0.31	0.08	0.27
Industrial production	0.50	0.44	0.87	3.51	5.11	1.46	3.71	0.60	0.16
Housing Start	0.49	0.40	0.82	69.95	115.75	1.65	110.04	18.10	0.16
Real Consumption	0.49	0.42	0.86	0.46	0.68	1.49	0.51	0.09	0.18
Real residential investment	0.41	0.36	0.87	29.60	40.95	1.38	27.03	10.76	0.40
Real nonresidential investment	0.48	0.43	0.90	4.12	5.30	1.29	7.38	6.01	0.82
Real state and local government consumption	0.47	0.40	0.86	0.54	0.81	1.51	1.41	0.02	0.02
Real federal government consumption	0.43	0.39	0.90	5.96	7.49	1.26	6.40	0.14	0.02
Ten-year Treasury rate	0.51	0.33	0.64	0.04	0.11	2.55	0.17	0.05	0.27
AAA Corporate Rate Bond	0.54	0.29	0.54	0.04	0.14	3.75	0.34	0.04	0.11

Appendix to "Biased Surveys"

Appendix A Data and variable construction

The forecast data come from three different datasets: the Survey of Professional Forecasters, collected by the Federal Reserve Bank of Philadelphia, the Tealbook/Greenbook of the Federal Reserve Board of Governors, and the Blue Chip Financial Indicators survey.

Survey of Professional Forecasters All surveys are collected around the 3rd week of the middle month in the quarter. In this section, x_t indicate the actual value and $F_t x_{t+h}$ the forecast provided in t about horizon h. All actual values of macroeconomic series (1-12) use the first release level, which are available to forecasters in the following quarter. We transform the macroeconomic level in year-over-year growth, following the literature.

1. NGDP
   - Variable: nominal GDP.
   - Question: The level of nominal GDP in the current quarter and the next 4 quarters.
   - Forecast: Nominal GDP growth from end of quarter $t-1$ to end of quarter $t+3$: $\frac{F_t x_{t+3}}{x_{t-1}} - 1$
   - Revision: $\frac{F_t x_{t+3}}{x_{t-1}} - \frac{F_{t-1} x_{t+3}}{F_{t-1} x_{t-1}}$
   - Actual: $\frac{x_{t+3}}{x_{t-1}} - 1$
   - Sample: 1969q1-2017q4
2. RGDP
   - Variable: real GDP.
   - Question: The level of real GDP in the current quarter and the next 4 quarters.
   - Forecast: real GDP growth from end of quarter $t-1$ to end of quarter $t+3$: $\frac{F_t x_{t+3}}{x_{t-1}} - 1$

- Revision: $\frac{F_t x_{t+3}}{x_{t-1}} - \frac{F_{t-1} x_{t+3}}{F_{t-1} x_{t-1}}$
- Actual: $\frac{x_{t+3}}{x_{t-1}} - 1$
- Sample: 1969q1-2018q3

3. PGDP

- Variable: GDP deflator.
- Question: The level of GDP deflator in the current quarter and the next 4 quarters.
- Forecast: GDP price deflator inflation from end of quarter $t-1$ to end of quarter $t+3$: $\frac{F_t x_{t+3}}{x_{t-1}} - 1$
- Revision: $\frac{F_t x_{t+3}}{x_{t-1}} - \frac{F_{t-1} x_{t+3}}{F_{t-1} x_{t-1}}$
- Actual: $\frac{x_{t+3}}{x_{t-1}} - 1$
- Sample: 1984q1-2017q4

4. CPI

- Variable: Consumer Price Index.
- Question: CPI growth rate in the current quarter and the next 4 quarters.
- Forecast: CPI inflation from end of quarter $t-1$ to end of quarter $t+3$: $F_t(z_t/4+1) * F_t(z_{t+1}/4+1) * F_t(z_{t+2}/4+1) * F_t(z_{t+3}/4+1)$, where z is the annualized quarterly CPI inflation in quarter t.
- Revision: $F_t(z_t/4+1) * F_t(z_{t+1}/4+1) * F_t(z_{t+2}/4+1) * F_t(z_{t+3}/4+1) - F_{t-1}(z_t/4+1) * F_{t-1}(z_{t+1}/4+1) * F_{t-1}(z_{t+2}/4+1) * F_{t-1}(z_{t+3}/4+1)$
- Actual: $\frac{x_{t+3}}{x_{t-1}} - 1$. Real time data is not available before 1994Q3. For actual periods prior to this date, we use data published in 1994Q3 to measure the actual outcome.
- Sample: 1993q3-2018q2

5. RCONSUM

- Variable: Real consumption.
- Question: The level of real consumption in the current quarter and the next 4 quarters.
- Forecast: GDP price deflator inflation from end of quarter $t-1$ to end of quarter $t+3$: $\frac{F_t x_{t+3}}{x_{t-1}} - 1$
- Revision: $\frac{F_t x_{t+3}}{x_{t-1}} - \frac{F_{t-1} x_{t+3}}{F_{t-1} x_{t-1}}$
- Actual: $\frac{x_{t+3}}{x_{t-1}} - 1$
- Sample: 1981q4-2018q3

6. INDPROD

- Variable: Industrial production index.
- Question: The average level of the industrial production index in the current quarter and the next 4 quarters.
- Forecast: Growth of the industrial production index from quarter $t-1$ to quarter $t+3$: $\frac{F_t x_{t+3}}{x_{t-1}} - 1$
- Revision: $\frac{F_t x_{t+3}}{x_{t-1}} - \frac{F_{t-1} x_{t+3}}{F_{t-1} x_{t-1}}$
- Actual: $\frac{x_{t+3}}{x_{t-1}} - 1$
- Sample: 1970q1-2018q1

7. RNRESIN

- Variable: Real non-residential investment.
- Question: The level of real non-residential investment in the current quarter and the next 4 quarters.
- Forecast: Growth of real non-residential investment from quarter $t-1$ to quarter $t+3$: $\frac{F_t x_{t+3}}{x_{t-1}} - 1$
- Revision: $\frac{F_t x_{t+3}}{x_{t-1}} - \frac{F_{t-1} x_{t+3}}{F_{t-1} x_{t-1}}$
- Actual: $\frac{x_{t+3}}{x_{t-1}} - 1$

- Sample: 1981q4-2018q3

8. RRESIN
   - Variable: Real residential investment.
   - Question: The level of real residential investment in the current quarter and the next 4 quarters.
   - Forecast: Growth of real residential investment from quarter $t-1$ to quarter $t+3$: $\frac{F_t x_{t+3}}{x_{t-1}} - 1$
   - Revision: $\frac{F_t x_{t+3}}{x_{t-1}} - \frac{F_{t-1} x_{t+3}}{F_{t-1} x_{t-1}}$
   - Actual: $\frac{x_{t+3}}{x_{t-1}} - 1$
   - Sample: 1981q4-2018q3
9. RGF
   - Variable: Real federal government consumption.
   - Question: The level of real federal government consumption in the current quarter and the next 4 quarters.
   - Forecast: Growth of real federal government consumption from quarter $t-1$ to quarter $t+3$: $\frac{F_t x_{t+3}}{x_{t-1}} - 1$
   - Revision: $\frac{F_t x_{t+3}}{x_{t-1}} - \frac{F_{t-1} x_{t+3}}{F_{t-1} x_{t-1}}$
   - Actual: $\frac{x_{t+3}}{x_{t-1}} - 1$
   - Sample: 1981q4-2018q4
10. RGSL
   - Variable: Real state and local government consumption.
   - Question: The level of real state and local government consumption in the current quarter and the next 4 quarters.
   - Forecast: Growth of real state and local government consumption from quarter $t-1$ to quarter $t+3$: $\frac{F_t x_{t+3}}{x_{t-1}} - 1$

- Revision: $\frac{F_t x_{t+3}}{x_{t-1}} - \frac{F_{t-1} x_{t+3}}{F_{t-1} x_{t-1}}$
- Actual: $\frac{x_{t+3}}{x_{t-1}} - 1$
- Sample: 1981q4-2018q4

11. HOUSING
    - Variable: Housing starts.
    - Question: The level of housing starts in the current quarter and the next 4 quarters.
    - Forecast: Growth of housing starts from quarter $t-1$ to quarter $t+3$: $\frac{F_t x_{t+3}}{x_{t-1}} - 1$
    - Revision: $\frac{F_t x_{t+3}}{x_{t-1}} - \frac{F_{t-1} x_{t+3}}{F_{t-1} x_{t-1}}$
    - Actual: $\frac{x_{t+3}}{x_{t-1}} - 1$
12. UNEMP
    - Variable: Unemployment rate.
    - Question: The level of average unemployment rate in the current quarter and the next 4 quarters.
    - Forecast: Average quarterly unemployment rate in quarter $t+3$: $F_t x_{t+3}$
    - Revision: $F_t x_{t+3} - F_{t-1} x_{t+3}$
    - Actual: x_{t+3}
    - Sample: 1969q1-2018q4
13. TB3M
    - Variable: 3-month Treasury rate.
    - Question: The level of average 3-month Treasury rate in the current quarter and next 4 quarters.
    - Forecast: Average quarterly 3-month Treasury rate in quarter $t+3$: $F_t x_{t+3}$
    - Revision: $F_t x_{t+3} - F_{t-1} x_{t+3}$

- Actual: x_{t+3}
- Sample: 1981q4-2017q4

14. TN10Y
    - Variable: 10-year Treasury rate.
    - Question: The level of average 10-year Treasury rate in the current quarter and next 4 quarters.
    - Forecast: Average quarterly 10-year Treasury rate in quarter $t+3$: $F_t x_{t+3}$
    - Revision: $F_t x_{t+3} - F_{t-1} x_{t+3}$
    - Actual: x_{t+3}
    - Sample: 1992q2-2017q4
15. AAA
    - Variable: AAA corporate bond rate.
    - Question: The level of average AAA corporate bond rate in the current quarter and next 4 quarters.
    - Forecast: Average quarterly AAA corporate bond rate in quarter $t+3$: $F_t x_{t+3}$
    - Revision: $F_t x_{t+3} - F_{t-1} x_{t+3}$
    - Actual: x_{t+3}
    - Sample: 1981q4-2017q4

Tealbook/Greenbook projections The Tealbook/Greenbook is produced by the Research staff at the Federal Reserve Board of Governors before each meeting of the Federal Open Market Committee. We aggregate at quarterly frequency by considering only the last projection available in any quarter. The projections from the Tealbook/Greenbook are released to the public with a lag of five years. We consider the variables forecasted also in the SPF, meaning all the macroeconomic variables with the exception of real consumption.

For most variables (NGDP, PGDP, RGDP, CPI, INDPROD, RRESINV, RNRESIN, RGF, RGSL) the projections are provided in quarter-over-quarter annualized growth. We transform the annualized quarter-on-quarter growth forecast in not-annualized. That is, $F_t z_{t+h} = (F_t \tilde{z}_{t+h} + 1)^{(1/4)} - 1$. where $F_t \tilde{z}_{t+h}$ indicates the forecast provided in t about the annualized quarterly growth rate between quarter $t - h - 1$ and $t + h$. HOUSING is provided in level. We transform all of them in year-over-year growth in order to compare them with SPF forecasts. Similarly, we keep projections for UNEMP in level. The actuals are the same as for the SPF data.

1. NGDP
   - Variable: nominal GDP.
   - Question: nominal GDP growth annualized rate in the current quarter and the next quarters.
   - Forecast: nominal GDP growth from end of quarter $t-1$ to end of quarter $t+3$: $(F_t z_t + 1) * (F_t z_{t+1} + 1) * (F_t z_{t+2} + 1) * (F_t z_{t+3} + 1) - 1$.
   - Revision:$[(F_t z_t + 1) * (F_t z_{t+1} + 1) * (F_t z_{t+2} + 1) * (F_t z_{t+3} + 1) - 1] - [(F_{t-1} z_t + 1) * (F_{t-1} z_{t+1} + 1) * (F_{t-1} z_{t+2} + 1) * (F_{t-1} z_{t+3} + 1) - 1]$
   - Actual: $\frac{x_{t+3}}{x_{t-1}} - 1$
   - Sample: 1970q1-2016q4
2. RGDP
   - Variable: real GDP.
   - Question: Real GDP growth annualized rate in the current quarter and the next quarters.
   - Forecast: real GDP growth from end of quarter $t-1$ to end of quarter $t+3$: $(F_t z_t + 1) * (F_t z_{t+1} + 1) * (F_t z_{t+2} + 1) * (F_t z_{t+3} + 1) - 1$.
   - Revision:$[(F_t z_t + 1) * (F_t z_{t+1} + 1) * (F_t z_{t+2} + 1) * (F_t z_{t+3} + 1) - 1] - [(F_{t-1} z_t + 1) * (F_{t-1} z_{t+1} + 1) * (F_{t-1} z_{t+2} + 1) * (F_{t-1} z_{t+3} + 1) - 1]$

- Actual: $\frac{x_{t+3}}{x_{t-1}} - 1$
- Sample: 1970q1-2016q4

3. PGDP

- Variable: GDP deflator.
- Question: GDP deflator growth annualized rate in the current quarter and the next quarters.
- Forecast: GDP deflator growth from end of quarter $t-1$ to end of quarter $t+3$: $(F_t z_t + 1) * (F_t z_{t+1} + 1) * (F_t z_{t+2} + 1) * (F_t z_{t+3} + 1) - 1$.
- Revision:$[(F_t z_t + 1) * (F_t z_{t+1} + 1) * (F_t z_{t+2} + 1) * (F_t z_{t+3} + 1) - 1] - [(F_{t-1} z_t + 1) * (F_{t-1} z_{t+1} + 1) * (F_{t-1} z_{t+2} + 1) * (F_{t-1} z_{t+3} + 1) - 1]$
- Actual: $\frac{x_{t+3}}{x_{t-1}} - 1$
- Sample: 1984q1-2016q4

4. CPI

- Variable: Consumer Price Index.
- Question: CPI growth annualized rate in the current quarter and the next quarters.
- Forecast: CPI growth from end of quarter $t-1$ to end of quarter $t+3$: $(F_t z_t + 1) * (F_t z_{t+1} + 1) * (F_t z_{t+2} + 1) * (F_t z_{t+3} + 1) - 1$.
- Revision:$[(F_t z_t + 1) * (F_t z_{t+1} + 1) * (F_t z_{t+2} + 1) * (F_t z_{t+3} + 1) - 1] - [(F_{t-1} z_t + 1) * (F_{t-1} z_{t+1} + 1) * (F_{t-1} z_{t+2} + 1) * (F_{t-1} z_{t+3} + 1) - 1]$
- Actual: $\frac{x_{t+3}}{x_{t-1}} - 1$
- Sample: 1993q3-2016q4

5. INDPROD

- Variable: Industrial production index.

- Question: The average level of the industrial production index in the current quarter and the next 4 quarters.
- Forecast: INDPROD growth from end of quarter $t-1$ to end of quarter $t+3$: $(F_t z_t + 1) * (F_t z_{t+1} + 1) * (F_t z_{t+2} + 1) * (F_t z_{t+3} + 1) - 1$.
- Revision:$[(F_t z_t + 1) * (F_t z_{t+1} + 1) * (F_t z_{t+2} + 1) * (F_t z_{t+3} + 1) - 1] - [(F_{t-1} z_t + 1) * (F_{t-1} z_{t+1} + 1) * (F_{t-1} z_{t+2} + 1) * (F_{t-1} z_{t+3} + 1) - 1]$
- Actual: $\frac{x_{t+3}}{x_{t-1}} - 1$
- Sample: 1970q1-2016q4

6. RNRESIN

- Variable: Real non-residential investment.
- Question: The level of real non-residential investment in the current quarter and the next 4 quarters.
- Forecast: RNRESIN growth from end of quarter $t-1$ to end of quarter $t+3$: $(F_t z_t + 1) * (F_t z_{t+1} + 1) * (F_t z_{t+2} + 1) * (F_t z_{t+3} + 1) - 1$.
- Revision:$[(F_t z_t + 1) * (F_t z_{t+1} + 1) * (F_t z_{t+2} + 1) * (F_t z_{t+3} + 1) - 1] - [(F_{t-1} z_t + 1) * (F_{t-1} z_{t+1} + 1) * (F_{t-1} z_{t+2} + 1) * (F_{t-1} z_{t+3} + 1) - 1]$
- Actual: $\frac{x_{t+3}}{x_{t-1}} - 1$
- Sample: 1978q3-2016q4

7. RRESIN

- Variable: Real residential investment.
- Question: The level of real residential investment in the current quarter and the next 4 quarters.
- Forecast: RRESIN growth from end of quarter $t-1$ to end of quarter $t+3$: $(F_t z_t + 1) * (F_t z_{t+1} + 1) * (F_t z_{t+2} + 1) * (F_t z_{t+3} + 1) - 1$.
- Revision:$[(F_t z_t + 1) * (F_t z_{t+1} + 1) * (F_t z_{t+2} + 1) * (F_t z_{t+3} + 1) - 1] - [(F_{t-1} z_t + 1) * (F_{t-1} z_{t+1} + 1) * (F_{t-1} z_{t+2} + 1) * (F_{t-1} z_{t+3} + 1) - 1]$

- Actual: $\frac{x_{t+3}}{x_{t-1}} - 1$
- Sample: 1978q3-2016q4

8. RGF
    - Variable: Real federal government consumption.
    - Question: The level of real federal government consumption in the current quarter and the next 4 quarters.
    - Forecast: RGF growth from end of quarter $t-1$ to end of quarter $t+3$: $(F_t z_t + 1) * (F_t z_{t+1} + 1) * (F_t z_{t+2} + 1) * (F_t z_{t+3} + 1) - 1$.
    - Revision:$[(F_t z_t + 1) * (F_t z_{t+1} + 1) * (F_t z_{t+2} + 1) * (F_t z_{t+3} + 1) - 1] - [(F_{t-1} z_t + 1) * (F_{t-1} z_{t+1} + 1) * (F_{t-1} z_{t+2} + 1) * (F_{t-1} z_{t+3} + 1) - 1]$
    - Actual: $\frac{x_{t+3}}{x_{t-1}} - 1$
    - Sample: 1978q3-2016q4

9. RGSL
    - Variable: Real state and local government consumption.
    - Question: The level of real state and local government consumption in the current quarter and the next 4 quarters.
    - Forecast: RGSL growth from end of quarter $t-1$ to end of quarter $t+3$: $(F_t z_t + 1) * (F_t z_{t+1} + 1) * (F_t z_{t+2} + 1) * (F_t z_{t+3} + 1) - 1$.
    - Revision:$[(F_t z_t + 1) * (F_t z_{t+1} + 1) * (F_t z_{t+2} + 1) * (F_t z_{t+3} + 1) - 1] - [(F_{t-1} z_t + 1) * (F_{t-1} z_{t+1} + 1) * (F_{t-1} z_{t+2} + 1) * (F_{t-1} z_{t+3} + 1) - 1]$
    - Actual: $\frac{x_{t+3}}{x_{t-1}} - 1$
    - Sample: 1978q3-2016q4

10. HOUSING
    - Variable: Housing starts.

- Question: The level of housing starts in the current quarter and the next 4 quarters.
- Forecast: Growth of housing starts from quarter $t-1$ to quarter $t+3$: $\frac{F_t x_{t+3}}{x_{t-1}} - 1$
- Revision: $\frac{F_t x_{t+3}}{x_{t-1}} - \frac{F_{t-1} x_{t+3}}{F_{t-1} x_{t-1}}$
- Actual: $\frac{x_{t+3}}{x_{t-1}} - 1$
- Sample: 1970q1-2016q4

11. UNEMP

- Variable: Unemployment rate.
- Question: The level of average unemployment rate in the current quarter and the next 4 quarters.
- Forecast: Average quarterly unemployment rate in quarter $t+3$: $F_t x_{t+3}$
- Revision: $F_t x_{t+3} - F_{t-1} x_{t+3}$
- Actual: x_{t+3}
- Sample: 1970q1-2016q4

Blue Chip survey As we consider the survey published in the last month of the quarter, the answers were submitted at the end of the middle month in the quarter/beginning of the last month in the quarter. In this section, x_t indicate the actual value and for financial variables $F_t x_{t+h}$ indicate the forecast about the actual value in $t+h$. For macro variables, we transform the annualized quarter-on-quarter growth forecast in not-annualized. That is, $F_t z_{t+h} = (F_t \tilde{z}_{t+h} + 1)^{(1/4)} - 1$. where $F_t \tilde{z}_{t+h}$ indicates the forecast provided in t about the annualized quarterly growth rate between quarter $t-h-1$ and $t+h$.

All actual values of macroeconomic series use the first release level, which are available to forecasters in the following quarter. We transform the macroeconomic forecasts in year-over-year growth, following the literature.

1. RGDP

- Variable: real GDP.
- Question: Real GDP growth annualized rate in the current quarter and the next 4 to 5 quarters.
- Forecast: real GDP growth from end of quarter $t-1$ to end of quarter $t+3$: $(F_t z_t + 1) * (F_t z_{t+1} + 1) * (F_t z_{t+2} + 1) * (F_t z_{t+3} + 1) - 1$. To compute forecast for shorter horizons, we proxy unobserved back-casting $F_t z_{t-1}$, $F_t z_{t-2}$, etc., with the realized past quarterly real GDP growth rate, $z_{t-1} = \left(\frac{x_{t-1}}{x_{t-2}}\right) - 1$, $z_{t-2} = \left(\frac{x_{t-2}}{x_{t-3}}\right) - 1$, etc.
- Revision:$[(F_t z_t + 1) * (F_t z_{t+1} + 1) * (F_t z_{t+2} + 1) * (F_t z_{t+3} + 1) - 1] - [(F_{t-1} z_t + 1) * (F_{t-1} z_{t+1} + 1) * (F_{t-1} z_{t+2} + 1) * (F_{t-1} z_{t+3} + 1) - 1]$
- Actual: $\frac{x_{t+3}}{x_{t-1}} - 1$
- Sample: 1993q2-2018q3

2. PGDP

- Variable: GDP deflator.
- Question: GDP deflator growth annualized rate in the current quarter and the next 4 to 5 quarters.
- Forecast: GDP deflator growth from end of quarter $t-1$ to end of quarter $t+3$: $(F_t z_t + 1) * (F_t z_{t+1} + 1) * (F_t z_{t+2} + 1) * (F_t z_{t+3} + 1) - 1$. To compute forecast for shorter horizons, we proxy unobserved back-casting $F_t z_{t-1}$, $F_t z_{t-2}$, etc., with the realized past quarterly GDP deflator growth rate, $z_{t-1} = \left(\frac{x_{t-1}}{x_{t-2}}\right) - 1$, $z_{t-2} = \left(\frac{x_{t-2}}{x_{t-3}}\right) - 1$, etc.
- Revision:$[(F_t z_t + 1) * (F_t z_{t+1} + 1) * (F_t z_{t+2} + 1) * (F_t z_{t+3} + 1) - 1] - [(F_{t-1} z_t + 1) * (F_{t-1} z_{t+1} + 1) * (F_{t-1} z_{t+2} + 1) * (F_{t-1} z_{t+3} + 1) - 1]$
- Actual: $\frac{x_{t+3}}{x_{t-1}} - 1$
- Sample: 1993q2-2017q4

3. CPI

- Variable: Consumer Price Index.
- Question: CPI growth annualized rate in the current quarter and the next 4 to 5 quarters.
- Forecast: CPI growth from end of quarter $t-1$ to end of quarter $t+3$: $(F_t z_t+1)*(F_t z_{t+1}+1)*(F_t z_{t+2}+1)*(F_t z_{t+3}+1)-1$. To compute forecast for shorter horizons, we proxy unobserved back-casting $F_t z_{t-1}$, $F_t z_{t-2}$, etc., with the realized past quarterly CPI growth rate, $z_{t-1}=\left(\frac{x_{t-1}}{x_{t-2}}\right)-1$, $z_{t-2}=\left(\frac{x_{t-2}}{x_{t-3}}\right)-1$, etc.
- Revision:$[(F_t z_t+1)*(F_t z_{t+1}+1)*(F_t z_{t+2}+1)*(F_t z_{t+3}+1)-1]-[(F_{t-1} z_t+1)*(F_{t-1} z_{t+1}+1)*(F_{t-1} z_{t+2}+1)*(F_{t-1} z_{t+3}+1)-1]$
- Actual: $\frac{x_{t+3}}{x_{t-1}}-1$
- Sample: 2001q2-2018q2

4. TB3M

- Variable: 3-month Treasury rate.
- Question: The level of average 3-month Treasury rate in the current quarter and next 4 quarters.
- Forecast: Average quarterly 3-month Treasury rate in quarter $t+3$: $F_t x_{t+3}$
- Revision: $F_t x_{t+3}-F_{t-1} x_{t+3}$
- Actual: x_{t+3}
- Sample: 1993q2-2019q1

5. TN10Y

- Variable: 10-year Treasury rate.
- Question: The level of average 10-year Treasury rate in the current quarter and next 4 quarters.
- Forecast: Average quarterly 10-year Treasury rate in quarter $t+3$: $F_t x_{t+3}$
- Revision: $F_t x_{t+3}-F_{t-1} x_{t+3}$

- Actual: x_{t+3}
- Sample: 1993q2-2019q1

6. TN5Y
   - Variable: 5-year Treasury rate.
   - Question: The level of average 5-year Treasury rate in the current quarter and next 4 quarters.
   - Forecast: Average quarterly 5-year Treasury rate in quarter $t+3$: $F_t x_{t+3}$
   - Revision: $F_t x_{t+3} - F_{t-1} x_{t+3}$
   - Actual: x_{t+3}
   - Sample: 1993q2-2019q1
7. AAA
   - Variable: AAA corporate bond rate.
   - Question: The level of average AAA corporate bond rate in the current quarter and next 4 quarters.
   - Forecast: Average quarterly AAA corporate bond rate in quarter $t+3$: $F_t x_{t+3}$
   - Revision: $F_t x_{t+3} - F_{t-1} x_{t+3}$
   - Actual: x_{t+3}
   - Sample: 1993q2-2019q1
8. FF
   - Variable:Federal funds rate.
   - Question: The level of average FF corporate bond rate in the current quarter and next 4 quarters.
   - Forecast: Average quarterly AAA corporate bond rate in quarter $t+3$: $F_t x_{t+3}$
   - Revision: $F_t x_{t+3} - F_{t-1} x_{t+3}$

Table A.1: Summary Statistics: SPF

	Consensus					Individual		
	Errors			Revisions		Forecast dispersion	Nonrev share	Pr(< 80% revise same direction)
	Mean	SD	SE	Mean	SD			
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Nominal GDP	-0.26	1.69	0.19	-0.14	0.68	1.00	0.02	0.80
GDP price index inflation	-0.28	0.58	0.08	-0.08	0.25	0.49	0.07	0.85
Real GDP	-0.26	1.64	0.19	-0.16	0.58	0.78	0.02	0.74
Consumer Price Index	-0.08	1.04	0.15	-0.11	0.68	0.54	0.06	0.66
Industrial production	-0.83	3.94	0.46	-0.49	1.19	1.57	0.01	0.72
Housing Start	-3.36	17.79	2.20	-2.31	5.93	8.34	0.00	0.68
Real Consumption	0.32	1.10	0.15	-0.06	0.41	0.61	0.03	0.78
Real residential investment	-0.46	8.32	1.19	-0.61	2.33	4.37	0.04	0.87
Real nonresidential investment	0.20	5.60	0.79	-0.22	1.71	2.31	0.03	0.74
Real state and local government consumption	0.04	2.96	0.38	0.14	1.10	2.09	0.07	0.91
Real federal government consumption	0.02	1.10	0.15	-0.05	0.33	0.98	0.11	0.93
Unemployment rate	0.01	0.68	0.08	0.05	0.32	0.30	0.18	0.66
Three-month Treasury rate	-0.51	1.14	0.16	-0.19	0.51	0.43	0.15	0.59
Ten-year Treasury rate	-0.48	0.73	0.11	-0.12	0.36	0.37	0.11	0.55
AAA Corporate Rate Bond	-0.46	0.82	0.11	-0.11	0.38	0.49	0.09	0.66

Notes: Columns 1 to 5 show statistics for consensus forecast errors and revisions. Forecast errors are defined as actual realizations minus forecasts. Revisions are forecast provided in t minus forecasts provided in $t-1$ about the same horizon. Columns 6 to 8 show statistics for individual forecasts. Forecast dispersion is the average cross-sectional standard deviation of individual forecasts. The share of nonrevisions is the average quarterly share of instances in which forecast revision is less than 0.01 percentage points. The final column shows the fraction of quarters where less than 80 percent of the forecasters revise in the same direction.

- Actual: x_{t+3}
- Sample: 1993q2-2019q1

A.1 Summary Statistics.

SPF Table A.1 presents summary statistics for the key data series. Columns 1-5 report statistics for the consensus errors and revisions. Average forecast errors are statistically indistinguishable from zero for most series, except most notably for interest rates, for which the forecasts are systematically above realizations. As argued by BGMS, this is likely due to the downward secular trend in interest rates.

Columns 6-8 reports the summary statistics of the individual forecasts, including forecasts dispersion, share of forecasts with no meaningful revisions and the probability that less than 80 percent of forecasters revise in the same direction.[18] The large dispersion of forecasts (column (6)) and the fact that forecast revisions often go in different directions (column (8)) suggest a role for dispersed information among forecasters, which (as is standard) we allow for in our conceptual framework below. The share of not meaningful revisions is also often quite small, suggesting that forecasts are not "stale".

Blue Chip Similarly, Table A.2 presents the same summary statistics for the Blue Chip survey.

Table A.2: Summary Statistics: Blue Chip

	Consensus					Individual		
	Errors			Revisions		Forecast dispersion	Nonrev share	Pr(< 80% revise same direction)
	Mean	SD	SE	Mean	SD			
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
GDP price index	-0.21	0.56	0.09	-0.05	0.17	1.00	0.13	0.78
Real GDP	0.00	1.24	0.20	-0.10	0.46	1.24	0.06	0.65
CPI	-0.04	1.16	0.20	-0.05	0.43	1.19	0.05	0.65
Three-month Treasury rate	-0.42	0.85	0.14	-0.15	0.46	0.82	0.18	0.45
Ten-year Treasury rate	-0.44	0.74	0.11	-0.13	0.38	0.98	0.11	0.39
AAA corporate bond rate	-0.40	0.65	0.10	-0.12	0.30	1.05	0.10	0.54
Fed funds rate	-0.30	0.82	0.14	-0.14	0.47	0.73	0.29	0.38
Five-year Treasury rate	-0.46	0.83	0.13	-0.15	0.43	1.00	0.10	0.39

Notes: Columns 1 to 5 show statistics for consensus forecast errors and revisions. Forecast errors are defined as actual realizations minus forecasts. Revisions are forecast provided in t minus forecasts provided in $t-1$ about the same horizon. Columns 6 to 8 show statistics for individual forecasts. Forecast dispersion is the average cross-sectional standard deviation of individual forecasts. The share of nonrevisions is the average quarterly share of instances in which forecast revision is less than 0.01 percentage points. The final column shows the fraction of quarters where less than 80 percent of the forecasters revise in the same direction.

Green Book Table A.3 presents the summary statistics for each series in the Greenbook forecasts.

Appendix B Additional tests

B.1 Comparison between different public signals

The key intuition of the strategic diversification hypothesis is that the forecasters want differentiate themselves from the average forecast in the survey. A direct implication of

[18] We follow BGMS and denote a forecast as not a meaningful revision if its quarterly change is less than 0.01%.

Table A.3: Summary Statistics: Greenbook forecasts

	Errors			Revisions	
	Mean	SD	SE	Mean	SD
Variable	(1)	(2)	(3)	(4)	(5)
Nominal GDP	-0.06	1.46	0.16	-0.10	0.87
GDP price index inflation	-0.07	0.58	0.09	0.06	0.47
Real GDP	-0.13	1.55	0.18	-0.15	0.82
Consumer Price Index	0.15	1.08	0.16	0.00	0.69
Industrial production	-0.94	3.58	0.38	-0.46	1.69
Housing Start	-2.14	15.64	1.83	-2.97	9.51
Real residential investment	0.60	7.13	0.87	-0.92	4.85
Real nonresidential investment	0.68	4.93	0.62	-0.36	2.84
Real state and local government consumption	0.87	3.06	0.34	0.22	1.70
Real federal government consumption	-0.19	1.30	0.17	-0.08	0.89
Unemployment rate	-0.09	0.59	0.06	0.01	0.41

Notes: Columns 1 to 5 show statistics for consensus forecast errors and revisions using forecasts from the Federal Reserve Green Book dataset. Errors are defined as actual realizations of forecasted variables minus forecasts. Revisions are the forecast provided in t minus the forecast provided in $t-1$ about the same future time period $t+h$.

this, as we saw above, is that agents would underweight common signals in their reported forecasts, as those would be overrepresented in the average forecast.

More generally, however, when x_t is not i.i.d. the need to predict the average response of all other forecasters (i.e. $\mathbb{E}_t^{(i)}(\overline{E}_t[x_{t+h}])$) generates an infinite regress of higher order beliefs (we will also see this directly in our quantitative model in Section 4 below). In that case, the lagged consensus $\overline{E}_{t-1}[x_{t+h}]$ is in fact a very *special* common signal, as it is especially highly correlated with the slow moving higher order beliefs that underlie the current period consensus forecast. As such, the lagged consensus is in fact likely to be more highly under-weighted relative to other commonly observed signals.

Thus, in this section we consider if there is such differential underweighting relative to other public signals. A natural alternative public signal is most recent, newly released data on the actual realizations of the forecasted variable x_t.[19] In the case of financial variables, this is the realization in the second month of the quarter (when the survey is collected), in

[19]Observing the lagged realization would imply that in each period forecasters agree on the realization of the forecasted variable at some lag $t-k$. This is a slightly stronger implication than what our benchmark framework in equations (1) and (2) allow for, but it does not affect our inference method in any way. Nevertheless, if one wants to preserve an information structure with persistent disagreement across forecasters, we can generalize the simple autoregressive process in (1) to a latent factor structure, where the observed variable

case of macroeconomic variable this is the first release estimate of the previous quarter's activity. We construct the "surprise" in this alternative common signal in conceptually the same way as before, but this time using the fact that SPG provides "nowcasts' to construct the following alternative RHS variable for our main regression[20]

$$pi^{(i)}_{2,t+h,t} \equiv x_{t-1} - \widetilde{E}^{i}_{t-1}[x_{t-1}] \tag{19}$$

We test the hypothesis that this second type of common signal experiences less under-reaction in two different specification. First, we run our baseline regression (14) with this alternative public signal RHS variable $pi^{(i)}_{2,t+h,t}$ instead

$$fe^{(i)}_{t+h,t} = \alpha + \beta_1 fr^{(i)}_{t+h,t} + \beta_2 pi^{(i)}_{2,t+h,t} + err^{(i)}_t \tag{20}$$

Second, we then include both public signals in the following, augmented regression

$$fe^{i}_{t+h,t} = \alpha + \beta_1 fr^{i}_{t+h,t} + \beta_2 pi_{1,t+h,t} + \beta_3 pi_{2,t+h,t} + err^{i}_t \tag{21}$$

where $pi^{(i)}_{1,t+h,t} = \overline{E}_{t-1}[x_{t+h}] - \widetilde{E}^{(i)}_{t-1}[x_{t+h}]$ is the same variable as before.

x_t is a function of some autoregressive factor z_t with some noise, i.e.

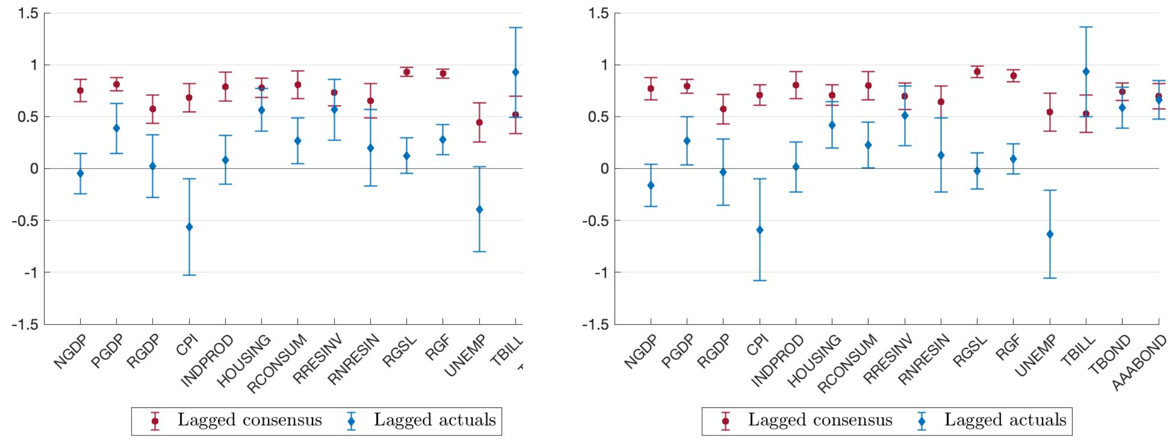
$$x_t = z_t + \varepsilon^x_t, \qquad z_t = \mu + \sum_{k=0}^{\infty} \psi_k \varepsilon^z_{t-k} \tag{18}$$

where ε^x_t and ε^z_t are i.i.d. standard normal variables and $\sum_{k=0}^{\infty} \psi_k^2 < \infty$. In this setting, the factor variable z_t is never observed and the persistent disagreement is preserved.

[20] If forecasters are able to observe the lagged realization x_{t-1}, then the vector $[x_{t+h}, s^{(i)}_t, x_{t-1}]$ conditional on the information set at time $t-1$ is multivariate Gaussian with a variance-covariance matrix that we label as $\hat{\Sigma}_{t-1,t+h}$. Therefore, we can apply the usual Bayesian updating formula to obtain $\mathbb{E}_t[x_{t+h}] = \mathbb{E}_{t-1}[x_{t+h}] + G^{RE}_1(s^{(i)}_t - \mathbb{E}_{t-1}[x_{t+h}]) + G^{RE}_2(x_{t-1} - \mathbb{E}_{t-1}[x_{t-1}])$, with $[G_1, G_2]' = \hat{\Sigma}^{-1}_{t-1,t+h}(\hat{\Sigma}^{(1,2:3)}_{t-1,t+h})'$ where $\hat{\Sigma}^{(1,2:3)}_{t-1,t+h}$ are the 2nd and 3rd elements of the first row of the variance-covariance matrix $\hat{\Sigma}_{t-1,t+h}$. We again assume that the reported forecasts have the same linear recursive structure as the optimal forecast, but with potentially different weights G_1 and G_2: $\widetilde{E}^{(i)}_t[x_{t+h}] = \widetilde{E}^{(i)}_{t-1}[x_{t+h}] + G_1(s^{(i)}_t - \widetilde{E}^{(i)}_{t-1}[x_{t+h}]) + G_2(x_{t-1} - \widetilde{E}^{(i)}_{t-1}[x_{t-1}])$. This structure implies that the forecast error can be expressed as

$$x_{t+h} - \widetilde{E}_t[x_{t+h}] = \frac{1-G_1}{G_1}\left(\widetilde{E}_t[x_{t+h}] - \widetilde{E}_{t-1}[x_{t+h}]\right) + \frac{G_2}{G_1}\left(x_{t-1} - \widetilde{E}_{t-1}[x_{t-1}]\right) - \eta^{(i)}_t$$

And this justifies the modified regression in equation (20).



(a) Different regressions

(b) Same regression

Figure B.1: Comparison between public signals

Notes: Panel (a): estimates of β_2 from our original regression (14) (in red), and from (20) (in blue). Panel(b): β_2 (in red) and β_3 (in blue) estimates from regression (21) using the SPF and h=3. Bars reports the 90% confidence interval for the estimated coefficients. Standard errors are Driscoll and Kraay (1998).

Under the strategic diversification framework, we expect that forecasters underweight the lagged consensus more than the latest realization of the forecasted variable, hence we have the two null hypotheses that (i): the β_2 estimate is larger in our original regression (14) than in (20), and (ii): $\beta_2 > \beta_3$ in regression (21).

The empirical estimates of both regressions are reported in Figure B.1 and indeed confirm both implications: underreaction to lagged consensus is larger than underreaction to lagged actuals, which is still weakly larger than zero with the exception of CPI inflation and unemployment. We leave the tables with individual estimates at both $h = 3$ and $h = 2$ horizons, and also using the Blue Chip data, to Appendix G.

B.2 Underreaction and the precision of common signals

Next, we investigate another intuitive implication of the strategic diversification model – when the available common information is more precise, and thus more important in all agents' expectations, then there will be a greater scope for underweighting it in reported forecasts and "standing out". Formally, we can show that in the setting of our analytical model the β_2 coefficient from our main regression (which measures underreaction to common information) has the following closed form:

$$\beta_2 = \lambda G_2^{RE} = \lambda \frac{\nu}{\chi + \tau + \nu}$$

Thus we see that indeed, as a comparative static, in the model β_2 is increasing in the precision of public information ν.

The key challenge to taking this intuitive comparative static to the data is that we do not directly observe the precision of information available to agents – that is, we do not directly observe ν. We propose a novel way of getting around this challenge by leveraging the fact that the Coibion and Gorodnichenko (2012) measure of stickiness, defined as

$$G_{CG} = \frac{1}{1 + \beta_{CG}}$$

is biased upwards when there are common errors (i.e. public signals) in the information set of the forecasters (as discussed in Section 2.2). And furthermore, the results in Figure 1, panel a) show not only that the CG measure is always biased upward compared to the true gain, i.e. $G_{CG} - G > 0$, but also that there is significant variation in the magnitude of this estimated bias across series, even though the overall information gain G is very similar at $G \approx 0.5$ for all series. This intuitively implies that the total precision of information (across both idiosyncratic and common signals) is relatively similar, but there is significant variation in the relative precision of public signals across different series.

To formalize this notion, in our analytical model we can in fact derive a closed form solution for the implied bias in the measured stickiness. Specifically, we have that

$$G_{CG} - G = G \left(\frac{G_2^2 \nu^{-1}}{G^2 \tilde{\tilde{\Sigma}}} \right) \tag{22}$$

furthermore, using $G_2 = \frac{G_2^{RE}}{(1-\lambda)+\lambda G_1^{RE}} = \frac{(1-\lambda)\nu}{(\nu+\tau+\chi)(1-\lambda)+\lambda\tau}$ and $G = \frac{G_1^{RE}+(1-\lambda)G_2^{RE}}{(1-\lambda)+\lambda G_1^{RE}} = \frac{\tau+(1-\lambda)\nu}{(\nu+\tau+\chi)(1-\lambda)+\lambda\tau}$

from the solution of the model, we can simplify the expression down to

$$G_{CG} - G = \frac{(1-G)^2}{G}\frac{\nu}{\chi} \tag{23}$$

Thus, the following specific transformation

$$(G_{CG} - G)\,\frac{G}{(1-G)^2} = \frac{\nu}{\chi}$$

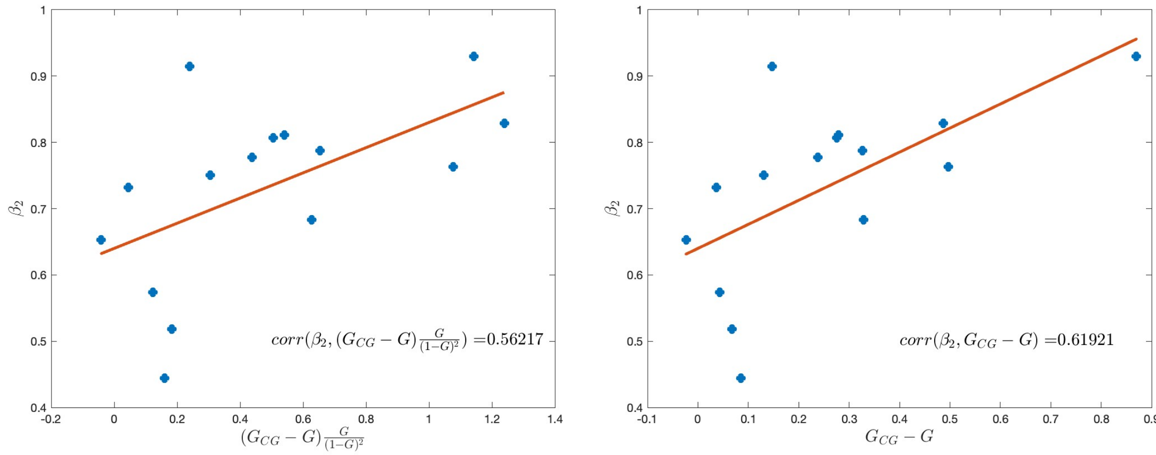
gives us a direct empirical measure of the precision of public information ν (relative to the agent's priors), since we can estimate both G and G_{CG} in the data. And in fact, more generally, since the estimated G is relatively constant across series in the data, in our case the raw bias itself, $G_{CG} - G$, should also be a good measure of ν.

So to test whether there is greater underreaction in cases of more precise public information (as the model implies), we regress the estimated β_2 for the different forecasted series on either the raw bias $G_{CG} - G$ or the rescaled bias measure $(G_{CG} - G)\,\frac{G}{(1-G)^2}$. We show the results in Figure B.2, and indeed find that in both cases β_2 is strongly increasing in the precision of public information, confirming the model's implications.

In both panels, we have the estimated β_2 on the y-axis, and in panel (a) we regress that on the strictest measure of public information precision, $(G_{CG} - G)\,\frac{G}{(1-G)^2}$ on the x-axis, and in panel (b) we regress it on the raw estimated bias $G_{CG} - G$ on the x-axis. In both cases we have substantial positive correlation (0.56 in panel a) and 0.62 in panel b)), which confirms this key comparative static of the model – the greater the precision of public information, the greater is the evidence of underweighting of common signals.

Appendix C Discussion

Outliers and Measurement Error. Juodis and Kučinskas (2023) make the powerful argument that the BGMS results could potentially be driven by measurement error in reported forecasts $\widetilde{E}_t^{(i)}[x_{t+h}]$, which show up on both the left and right hand sides of equation (7) and can thus mechanically generate a negative correlation. This is a concern that is



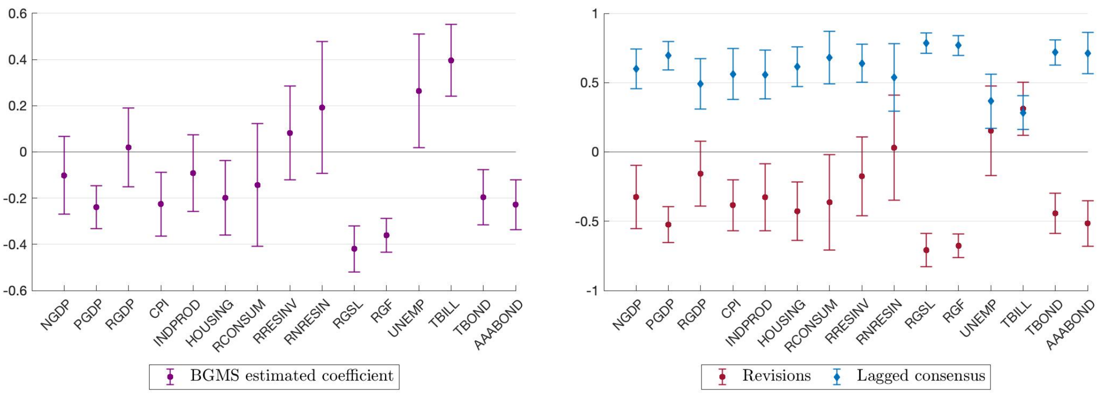
(a) β_2 on $(G_{CG}-G)\,\frac{G}{(1-G)^2}$

(b) β_2 on $(G_{CG}-G)$

Figure B.2

more broadly related to the observation of Angeletos et al. (2021) and Broer and Kohlhas (2022) that there are a substantial number of influential outliers in the SPF data, which can by themselves also substantially weaken the BGMS findings.

To the contrary, our results are robust to a variety of aggressive treatments of outliers. In our benchmark analysis up to this point we followed the same procedure to clean outliers as in BGMS in order to be comparable. However, our results also remain virtually unchanged under three alternative strategies suggested in the literature. First, as in Kohlhas and Walther (2021) we trim all observations outside of the first and the 99th percentile; second, we follow Angeletos et al. (2021) and drop forecasts that are more than 4 interquartile ranges away from the median of each horizon in each quarter; third, we use the median to measure consensus forecasts instead of the mean. To showcase the robustness of our results, in panel (b) of Figure C.1 we show that our benchmark estimates and the key conclusion of $\beta_2 > 0$ do not change even when we apply all three alternative data cleaning strategies *at the same time*, while in panel (a) we show the resulting BGMS estimates, which are indeed noisier and much closer to zero. In Appendix H, we show Tables and Figures that replicate all results in the paper under this stricter data cleaning procedure.



(a) BGMS regression

(b) Forecast errors on revisions and public information

Figure C.1: Regressions result with stricter data cleaning

Notes: this figure replicates Figure 1 (panel a) and Figure 2 (panel b) with a stricter data cleaning. Bars reports the 90% confidence interval for the estimated coefficients. Standard errors are Driscoll and Kraay (1998).

Alternative univariate regressions In a recent paper, Broer and Kohlhas (2022), run a univariate regression similar to regression (15) we discuss in Section 3.4, and find that survey of forecasts show both under- and overreaction to public signals, not just underreaction as we find. They run the following univariate regression

$$fe^{(i)}_{t+h,t} = \alpha + \beta_{BK}\overline{E}_{t-1}[x_{t+h}] + err^{(i)}_t \tag{24}$$

where they also consider the lagged consensus as the relevant public signal. Estimating β_{BK} across the different variables in our data, we indeed find estimates that are widely dispersed on both sides of zero, sometimes being positive, while other times negative often not significantly (see panel (a) of Figure 5). This contrasts to our findings in Section 3.4, where we estimate a slightly different univariate regression and find consistently positive $\beta_{BK_{dm}}$ for *all* forecasted variables.

The key difference across our two empirical methodologies is that in the original specification of Broer and Kohlhas (2022), the regressor is the lagged consensus alone(see equation 24), while in our version of the univariate regression (equation (15)) we "demean" it and construct the associated "surprise" in that signal at the individual forecaster level by defining: $pi^{(i)}_{t+h,t} = \overline{E}_{t-1}[x_{t+h}] - \widetilde{E}^{(i)}_{t-1}[x_{t+h}]$.

Under the null hypothesis of RE and honest reporting, both regressions imply $\beta_{BK} = \beta_{BK_{dm}} = 0$, however finding coefficients different from zero have different meanings in these two slightly different formulations of the regression. The interpretation of $\beta_{BK_{dm}}$ was discussed in Section 3.4 and summarized by Proposition 4. In the case of regression (24) instead, we can derive the following if and only if condition for β_{BK} to be positive.[21]

Proposition 5. *When forecasts follow a linear structure as in (3) and x_t is stationary*

$$\beta_{BK} > 0 \iff \frac{1-G}{1-G^{RE}} > \frac{G_2}{G_2^{RE}} \frac{\Sigma^{RE}}{\widetilde{\Sigma} + Cov(x_{t+h} - \widetilde{E}^{(i)}_{t-1}[x_{t+h}], \widetilde{E}^{(i)}_{t-1}[x_{t+h}])}$$

In certain special cases – like for example If x_t is i.i.d. over time – then $Cov(x_{t+h} - \widetilde{E}^{(i)}_{t-1}[x_{t+h}], \widetilde{E}^{(i)}_{t-1}[x_{t+h}]) = 0$, and hence any deviations from zero in β_{BK} have the same interpretation as in the case of $\beta_{BK_{dm}}$. But more generally the term $Cov(x_{t+h} - \widetilde{E}^{(i)}_{t-1}[x_{t+h}], \widetilde{E}^{(i)}_{t-1}[x_{t+h}])$ could be both positive and negative in the data, and hence we can find estimates of β_{BK} that are negative even when $\beta_{BK_{dm}}$ is positive. And this is indeed borne out in our empirical estimates (contrast panels (a) and (b) of Figure 5).[22,23]

Thus, the two univariate regressions lead to somewhat different results. For what it is worth, in our quantitative model (where public information is indeed underweighted relative to the prior, in the sense of $\frac{1-G}{1-G^{RE}} > \frac{G_2}{G_2^{RE}}$) we find that theoretically $\beta_{BK} = 0$ (as is also discussed in Appendix C.3 of Broer and Kohlhas (2022)) while $\beta_{BK_{dm}} > 0$. On the one hand, this confirms that the distinction between these two regression is nuanced and they speak to different moments. On the other hand, we think that the empirical evidence on these regressions is not necessarily inconsistent with the model – given that that the empirical β_{BK} estimates are dispersed around zero for the different forecast variables (with

[21] All proofs in the paper are collected together in Appendix D.

[22] Appendix E.4 reports all estimation details at both 3 and 2 quarters horizons, together with the set of estimates on Blue Chip data.

[23] As we discuss in Section 3.4, the interpretation of the univariate regression is different from our benchmark regression (14), as the former refers to underweighting public signals compared to prior information, while the latter refers to underweighting public signals compared to private information information.

relatively wide standard errors), while the $\beta_{BK_{dm}}$ estimates are consistently positive. Distinguishing further between these two empirical moments is a subtle and nuanced issue beyond the scope of this paper.

Anonymity. It is sometimes argued that strategic considerations do not apply to anonymous surveys like the SPF. However, to the contrary, the forecasting literature has noted that when the same forecasters participate in both anonymous and non-anonymous surveys, they simply submit the same forecast to both types of surveys, thus introducing strategic considerations in the anonymous survey as well. For example, Marinovic et al. (2013) notes that "According to industry experts, forecasters often seem to submit to the anonymous surveys the same forecasts they have already prepared for public (i.e. non-anonymous) release. There are two reasons for this. First, it might not be convenient for the forecasters to change their report, unless they have a strict incentive to do so. Second, the forecasters might be concerned that their strategic behavior could be uncovered by the editor of the anonymous survey."

We find direct evidence to this effect from a supplement to the European Central Bank's own Survey of Professional Forecasters. In that supplement, respondents were explicitly asked whether they "prepare a new forecast specifically for the SPF" (which is anonymous) or just submit the "latest available forecast" (p.8, European Central Bank (2014)). There are two waves of this questionnaire, and in both the respondents overwhelmingly indicated they do not produce a new forecast, but rather use one that is already available. In 2013, more than 80% of the panelists responded with "the last available", while in 2008 more than 90% gave the same answer (European Central Bank, 2014).

Appendix D Proofs

Proof of Proposition 1. From equation (11), using $G = G_1 + G_2$,

$$\widetilde{E}_t^{(i)}[x_{t+h}] = \mu + G(x_{t+h} - \mu) + G_1 \eta_t^{(i)} + G_2 e_t \tag{25}$$

and then we can express the forecast error as

$$\underbrace{(x_{t+h} - \widetilde{E}_t^{(i)}[x_{t+h}]}_{fe_t^{(i)}} = \frac{1-G}{G}(\underbrace{\widetilde{E}_t^{(i)}[x_{t+h}] - \mu}_{fr_t^{(i)}}) - \frac{G_1}{G}\eta_t^{(i)} - \frac{G_2}{G}e_t \tag{26}$$

From here, we can compute the large sample limit of β_{BGMS}:

$$\begin{aligned}\beta_{BGMS} &= \frac{1-G}{G} + \frac{Cov(\widetilde{E}_t^{(i)}[x_{t+h}] - \mu, -\frac{G_1}{G}\eta_t^{(i)} - \frac{G_2}{G}e_t)}{Var(\widetilde{E}_t^{(i)}[x_{t+h}] - \mu)} \\ &= \frac{1-G}{G} + \frac{-\frac{G_1^2}{G}\tau^{-1} - \frac{G_2^2}{G}\nu^{-1}}{G^2\chi^{-1} + G_1^2\tau^{-1} + G_2^2\nu^{-1}}\end{aligned} \tag{27}$$

substitute in the optimal choices $G_1 = \frac{G_1^{RE}}{(1-\lambda)+\lambda G_1^{RE}}$ and $G_2 = \frac{(1-\lambda)G_2^{RE}}{(1-\lambda)+\lambda G_1^{RE}}$, and the rational updating weights $G_1^{RE} = \frac{\tau}{\tau+\nu+\chi}$ and $G_2^{RE} = \frac{\nu}{\nu+\tau+\chi}$ to obtain

$$\hat{\beta}_{BGMS} = \frac{-\lambda\tau\chi}{([(1-\lambda)\nu+\tau]^2 + [(1-\lambda)^2\nu+\tau]\chi)} < 0 \iff \lambda > 0 \tag{28}$$

For the β_{CG}, integrate equation (26) across i to obtain

$$\underbrace{(x_{t+h} - \overline{E}_t[x_{t+h}]}_{\overline{fe}_t} = \frac{1-G}{G}(\underbrace{\overline{E}_t[x_{t+h}] - \mu}_{\overline{fr}_t}) - \frac{G_2}{G}e_t$$

Then we can compute the large sample limit as

$$\begin{aligned}\beta_{CG} &= \frac{1-G}{G} + \frac{Cov(\overline{E}_t[x_{t+h}] - \mu, -\frac{G_2}{G}e_t)}{Var(\overline{E}_t[x_{t+h}] - \mu)} \\ &= \frac{1-G}{G} + \frac{-\frac{G_2^2}{G}\nu^{-1}}{G^2\chi^{-1} + G_1^2\nu^{-1}}\end{aligned} \tag{29}$$

Substituting through with G_1, G_2, G_1^{RE} and G_2^{RE} and simplifying we get

$$\beta_{CG} = \frac{(1-\lambda)\tau\chi}{([(1-\lambda)\nu+\tau]^2 + [(1-\lambda)^2\nu+\tau]\chi)} \iff \lambda < 1$$

■

Proof of Proposition 2.

$$Var(x_{t+h} - \overline{\mathbb{E}}_t[x_{t+h}]) = Var((1 - (\underbrace{G_1^{RE} + G_2^{RE}}_{\equiv G^{RE}}))x_{t+h} + G_2^{RE} e_t)$$
$$= (1 - G^{RE})^2 \frac{1}{\chi} + \frac{(G_2^{RE})^2}{\nu} = \frac{\chi + \nu}{(\nu + \tau + \chi)^2}$$

Meanwhile, the MSE of the consensus of the reported forecasts is

$$Var(x_{t+h} - \overline{E}_t[x_{t+h}]) = Var((1 - G)x_{t+h} + G_2 e_t) = (1 - G)^2 \frac{1}{\chi} + \frac{(G_2)^2}{\nu}$$
$$= \frac{(1 - \lambda)^2}{(1 - \lambda + \lambda G_2^{RE})^2} \frac{\chi + \nu}{(\nu + \tau + \chi)^2} = \frac{(1 - \lambda)^2}{(1 - \lambda + \lambda G_2^{RE})^2} Var(x_{t+h} - \overline{\mathbb{E}}_t(x_{t+h}))$$

And thus

$$Var(x_{t+h} - \overline{E}_t[x_{t+h}]) < Var(x_{t+h} - \overline{\mathbb{E}}_t(x_{t+h}))$$

since $\frac{(1-\lambda)^2}{(1-\lambda+\lambda G_2^{RE})^2} < 1$ for $\lambda \in (0, 1)$.

■

Proof of Proposition 3. Rewriting (3) we can express forecast errors as

$$fe_{t+h,t}^{(i)} = \frac{1 - G_1}{G_1}(\widetilde{E}_t^{(i)}[x_{t+h}] - \widetilde{E}_{t-1}^{(i)}[x_{t+h}]) - \frac{G_2}{G_1}(g_t - \widetilde{E}_{t-1}^{(i)}[x_{t+h}]) - \eta_t^{(i)}$$

Let $pi_{t+h,t}^{(i)} = g_t - \widetilde{E}_{t-1}^{(i)}[x_{t+h}]$, then we can write regression (14) as

$$fe_{t,t}^{i} = X\beta + u_t^{(i)} \tag{30}$$

with $X = [fr_{t+h,t}^{i} \quad pi_{t+h,t}^{i}]$, $\beta = [\beta_1 \ \beta_2]'$ and $u_t^{(i)} = \eta_t^{(i)}$. The resulting plimit of β is

$$\beta = \begin{bmatrix} \frac{1-G_1}{G_1} \\ -\frac{G_2}{G_1} \end{bmatrix} + \Sigma_{XX}^{-1} \Sigma_{Xu} \tag{31}$$

where

$$\Sigma_{XX} = \begin{bmatrix} var(fr^i_{t+h,t}) & cov(fr^i_{t,t}, pi^i_{t+h,t}) \\ cov(fr^i_{t+h,t}, pi^i_{t,t}) & var(pi^i_{t+h,t}) \end{bmatrix}, \qquad \Sigma_{Xu} = \begin{bmatrix} cov(fr^i_{t+h,t}, u^{(i)}_t) \\ cov(pi^i_{t+h,t} u^{(i)}_t) \end{bmatrix}$$

Since $var(fr^i_{t+h,t}) = [G^2\widetilde{\Sigma} + G_1^2\tau^{-1} + G_2^2\nu^{-1}]$, $var(pi^i_{t+h,t}) = \widetilde{\Sigma} + \nu^{-1}$, $cov(fr^i_{t+h,t}, pi^i_{t,t}) = [G\widetilde{\Sigma} + G_2\nu^{-1}]$, $cov(fr^i_{t+h,t}, u^i_t) = -G_1\tau^{-1}$, $cov(pi^i_{t+h,t}, u^i_t) = 0$ and $\widetilde{\Sigma} = var(x_{t+h} - \widetilde{E}^{(i)}_{t-1}[x_{t+h}])$,

$$\beta_1 = \frac{1-G_1}{G_1} - \frac{(\widetilde{\Sigma} + \nu^{-1})G_1\frac{1}{\tau}}{(\widetilde{\Sigma} + \nu^{-1})(G^2\hat{\Sigma} + G_1^2\tau^{-1} + G_2^2\nu^{-1}) - (G\widetilde{\Sigma} + G_2\nu^{-1})^2}$$

$$= \frac{1}{G_1}\left(G_1^{RE}\frac{\nu + \tau + 1/\Sigma^{RE}}{\nu + \tau + 1/\widetilde{\Sigma}} - G_1\right) < 0 \iff G_1 > \frac{G_1^{RE}}{G^{RE} + \frac{\Sigma^{RE}}{\widetilde{\Sigma}}(1 - G^{RE})}$$

$$\beta_2 = -\frac{G_2}{G_1} + \frac{(G\widetilde{\Sigma} + G_2\nu^{-1})G_1\frac{1}{\tau}}{(\widetilde{\Sigma} + \nu^{-1})(G^2\hat{\Sigma} + G_1^2\tau^{-1} + G_2^2\nu^{-1}) - (G\widetilde{\Sigma} + G_2\nu^{-1})^2}$$

$$= \frac{1}{G_1}\frac{\nu + \tau + 1/\Sigma^{RE}}{\nu + \tau + 1/\widetilde{\Sigma}}\left(G_1 G_2^{RE} - G_2 G_1^{RE}\right) > 0 \iff \frac{G_1}{G_1^{RE}} > \frac{G_2}{G_2^{RE}}$$

Where we have used the definitions $G_1^{RE} = \frac{\tau}{\tau + \nu + 1/\Sigma^{RE}}$ and $G_2^{RE} = \frac{\nu}{\tau + \nu + 1/\Sigma^{RE}}$, and as already defined $\Sigma^{RE} = var(x_{t+h} - \mathbb{E}^{(i)}_t[x_{t+h}]$ is the rational posterior variance. ∎

Proof of Proposition 4. Using

$$fe^{(i)}_{t+h,t} = \frac{1-G_1}{G_1}(\widetilde{E}^{(i)}_t[x_{t+h}] - \widetilde{E}^{(i)}_{t-1}[x_{t+h}]) - \frac{G_2}{G_1}(g_t - \widetilde{E}^{(i)}_{t-1}[x_{t+h}]) - \eta^{(i)}_t$$

it follows that the probability limit of the $\beta_{BK_{dm}}$ coefficient is

$$
\begin{aligned}
\beta_{BK_{dm}} &= \frac{Cov(fe^{(i)}_{t+h,t}, g_t - \widetilde{E}^{(i)}_{t-1}[x_{t+h}])}{Var(g_t - \widetilde{E}^{(i)}_{t-1}[x_{t+h}])} \\
&= \frac{-\frac{G_2}{G_1}(\widetilde{\Sigma} + \nu^{-1}) + \frac{1-G_1}{G_1} Cov(G(x_{t+h} - \widetilde{E}^{(i)}_{t-1}[x_{t+h}]) + G_1\eta^{(i)}_t + G_2 e_t, g_t - \widetilde{E}^{(i)}_{t-1}[x_{t+h}])}{\widetilde{\Sigma} + \nu^{-1}} \\
&= \frac{-\frac{G_2}{G_1}(\widetilde{\Sigma} + \nu^{-1}) + \frac{1-G_1}{G_1}(G\widetilde{\Sigma} + G_2\nu^{-1})}{\widetilde{\Sigma} + \nu^{-1}} = \frac{(1-G)\widetilde{\Sigma} - G_2\nu^{-1}}{\widetilde{\Sigma} + \nu^{-1}} \\
&> 0 \iff (1-G)\widetilde{\Sigma} > G_2\nu^{-1} \iff \frac{1-G}{1-G^{RE}} > \frac{G_2}{G_2^{RE}} \frac{\Sigma^{RE}}{\widetilde{\Sigma}}
\end{aligned}
$$

■

Proof of Proposition 5. Using

$$
fe^{(i)}_{t+h,t} = \frac{1-G_1}{G_1}(\widetilde{E}^{(i)}_t[x_{t+h}] - \widetilde{E}^{(i)}_{t-1}[x_{t+h}]) - \frac{G_2}{G_1}(g_t - \widetilde{E}^{(i)}_{t-1}[x_{t+h}]) - \eta^{(i)}_t
$$

it follows that the probability limit of the β_{BK} coefficient is

$$
\begin{aligned}
\beta_{BK} &= \frac{Cov(fe^{(i)}_{t+h,t}, g_t)}{Var(g_t)} \\
&= \frac{-\frac{G_2}{G_1} Cov(g_t - \widetilde{E}^{(i)}_{t-1}[x_{t+h}], g_t) + \frac{1-G_1}{G_1} Cov(G(x_{t+h} - \widetilde{E}^{(i)}_{t-1}[x_{t+h}]) + G_1\eta^{(i)}_t + G_2 e_t, g_t)}{Var(x_t) + \nu^{-1}} \\
&= \frac{G_2}{G_1} \frac{\left(Cov(x_{t+h} - \widetilde{E}^{(i)}_{t-1}[x_{t+h}], x_{t+h})[(1-G_1)\frac{G}{G_2} - 1] - G_1\nu^{-1}\right)}{Var(x_t) + \nu^{-1}} \\
&= \frac{G_2}{G_1} \frac{\left((Cov(x_{t+h} - \widetilde{E}^{(i)}_{t-1}[x_{t+h}], \widetilde{E}^{(i)}_{t-1}[x_{t+h}]) + \widetilde{\Sigma})[(1-G_1)\frac{G}{G_2} - 1] - G_1\nu^{-1}\right)}{Var(x_t) + \nu^{-1}} \\
&> 0 \iff (Cov(x_{t+h} - \widetilde{E}^{(i)}_{t-1}[x_{t+h}], \widetilde{E}^{(i)}_{t-1}[x_{t+h}]) + \widetilde{\Sigma})[(1-G_1)\frac{G}{G_2} - 1] > G_1\nu^{-1} \\
&\iff \frac{1-G}{1-G^{RE}} > \frac{G_2}{G_2^{RE}} \frac{\Sigma^{RE}}{\widetilde{\Sigma} + Cov(x_{t+h} - \widetilde{E}^{(i)}_{t-1}[x_{t+h}], \widetilde{E}^{(i)}_{t-1}[x_{t+h}])}
\end{aligned}
$$

■

Appendix E Additional tables

E.1 *Full details on Coibion and Gorodnichenko (2015) regressions*

SPF results

Table E.1: Gain G regression estimates (SPF)

	3 quarters horizon				2 quarters horizon			
	G	SE	p-value	Median	G	SE	p-value	Median
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Nominal GDP	0.53	0.02	0.00	0.49	0.61	0.01	0.00	0.62
GDP price index inflation	0.49	0.03	0.00	0.52	0.63	0.02	0.00	0.68
Real GDP	0.56	0.03	0.00	0.52	0.63	0.02	0.00	0.62
Consumer Price Index	0.49	0.02	0.00	0.53	0.70	0.02	0.00	0.71
Industrial production	0.50	0.03	0.00	0.52	0.59	0.02	0.00	0.63
Housing Start	0.49	0.03	0.00	0.55	0.53	0.02	0.00	0.56
Real Consumption	0.49	0.03	0.00	0.48	0.63	0.03	0.00	0.62
Real residential investment	0.41	0.03	0.00	0.44	0.56	0.02	0.00	0.64
Real nonresidential investment	0.48	0.02	0.00	0.49	0.61	0.03	0.00	0.61
Real state and local government consumption	0.43	0.04	0.00	0.40	0.60	0.05	0.00	0.56
Real federal government consumption	0.47	0.04	0.00	0.48	0.62	0.03	0.00	0.62
Unemployment rate	0.49	0.02	0.00	0.54	0.56	0.02	0.00	0.62
Three-month Treasury rate	0.55	0.02	0.00	0.59	0.63	0.03	0.00	0.67
Ten-year Treasury rate	0.51	0.02	0.00	0.54	0.60	0.02	0.00	0.63
AAA Corporate Rate Bond	0.54	0.02	0.00	0.56	0.61	0.02	0.00	0.62

Notes: The table shows the result from regression (4) Columns (1)-(3) report coefficients, standard errors and p-values from the panel data regression with individual fixed effects. Column (4) reports the median coefficient from individual regressions using demeaned variables. Columns (5)-(8) reports the same statistics for the 2 quarters horizon. Standard errors are Driscoll and Kraay (1998).

Table E.2: Difference between estimated gains G and G_{CG}, horizon 3 quarters (SPF)

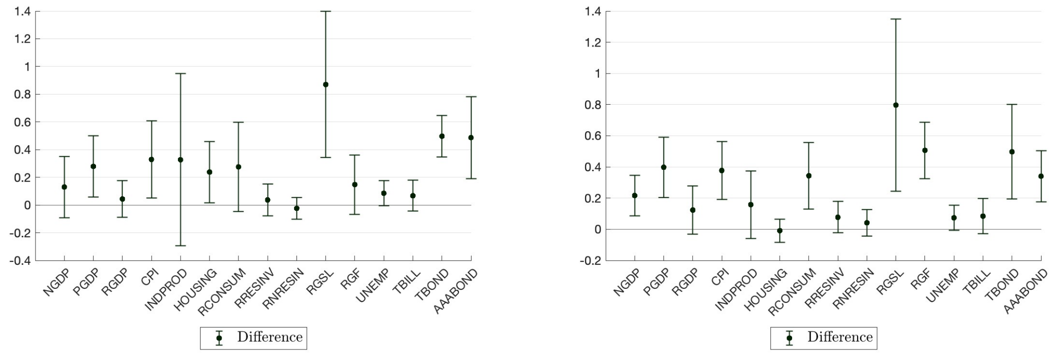
Variable	G_{CG} (1)	SE (2)	G (3)	SE (4)	Difference (5)	SE (6)	p-value (7)
Nominal GDP	0.66	0.13	0.53	0.02	0.13	0.13	0.17
GDP price index inflation	0.77	0.13	0.49	0.03	0.28	0.13	0.02
Real GDP	0.60	0.07	0.56	0.03	0.04	0.08	0.29
Consumer Price Index	0.82	0.17	0.49	0.02	0.33	0.17	0.03
Industrial production	0.83	0.38	0.50	0.03	0.33	0.38	0.19
Housing Start	0.72	0.13	0.49	0.03	0.24	0.13	0.04
Real Consumption	0.76	0.19	0.49	0.03	0.28	0.20	0.08
Real residential investment	0.45	0.07	0.41	0.03	0.04	0.07	0.30
Real nonresidential investment	0.45	0.04	0.48	0.02	-0.02	0.05	0.69
Real state and local government consumption	1.30	0.32	0.43	0.04	0.87	0.32	0.00
Real federal government consumption	0.61	0.12	0.47	0.04	0.15	0.13	0.13
Unemployment rate	0.57	0.05	0.49	0.02	0.08	0.05	0.06
Three-month Treasury rate	0.62	0.07	0.55	0.02	0.07	0.07	0.16
Ten-year Treasury rate	1.01	0.09	0.51	0.02	0.50	0.09	0.00
AAA Corporate Rate Bond	1.03	0.18	0.54	0.02	0.49	0.18	0.00

Notes: Columns (1)-(2) reports the implied gain from CG regressions of table 4. Columns (3)-(4) replicate the gain estimate from regression (6) Columns (5)-(8) reports the difference between column (1) and (3), its standard error and the probability of rejecting the null of column (5) lower or equal to zero.

Table E.3: Difference between estimated gains G and G_{CG}, horizon 2 quarters (SPF)

Variable	G_{CG} (1)	SE (2)	G (3)	SE (4)	Difference (5)	SE (6)	p-value (7)
Nominal GDP	0.83	0.08	0.61	0.01	0.22	0.08	0.00
GDP price index inflation	1.03	0.12	0.63	0.02	0.40	0.12	0.00
Real GDP	0.75	0.09	0.63	0.02	0.12	0.09	0.10
Consumer Price Index	1.08	0.11	0.70	0.02	0.38	0.11	0.00
Industrial production	0.75	0.13	0.59	0.02	0.16	0.13	0.12
Housing Start	0.52	0.04	0.53	0.02	-0.01	0.05	0.58
Real Consumption	0.97	0.12	0.63	0.03	0.34	0.13	0.00
Real residential investment	0.64	0.06	0.56	0.02	0.08	0.06	0.10
Real nonresidential investment	0.66	0.04	0.61	0.03	0.04	0.05	0.21
Real state and local government consumption	1.40	0.33	0.60	0.05	0.80	0.34	0.01
Real federal government consumption	1.12	0.11	0.62	0.03	0.51	0.11	0.00
Unemployment rate	0.63	0.05	0.56	0.02	0.07	0.05	0.06
Three-month Treasury rate	0.71	0.06	0.63	0.03	0.08	0.07	0.11
Ten-year Treasury rate	1.10	0.18	0.60	0.02	0.50	0.18	0.00
AAA Corporate Rate Bond	0.95	0.10	0.61	0.02	0.34	0.10	0.00

Notes: Columns (1)-(2) reports the implied gain from CG regressions of table 4. Columns (3)-(4) replicate the gain estimate from regression (6). Columns (5)-(8) reports the difference between column (1) and (3), its standard error and the probability of rejecting the null of column (5) lower or equal to zero.



(a) Horizon 3 quarters

(b) Horizon 2 quarters

Figure E.1: Estimated gains: our measure vs CG (SPF)

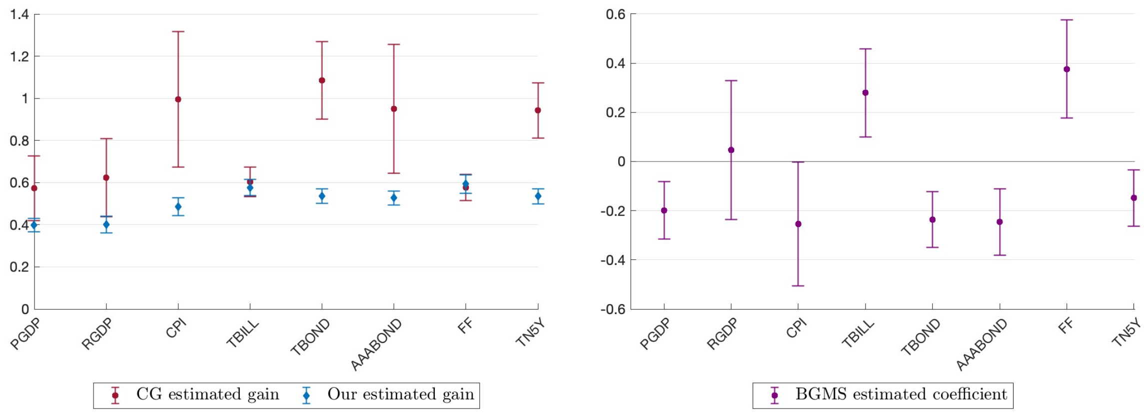
Notes: the Figure plot the difference between the gain G implied by the original CG regression (4) and our estimated coefficients in regression (6). Standard errors are robust to heteroskedasticity and Newey-West with the automatic bandwidth selection procedure of Newey and West (1994). Bars reports the 90% confidence interval for the estimated coefficients. Panel (a) reports the result for the 3 quarters horizon, panel (b) for the 2 quarters.

1 *Blue Chip results*

Table E.4: Standard CG regression estimates (Blue Chip)

	3 quarters horizon				2 quarters horizon			
	β_{CG}	SE	p-value	Median	β_{CG}	SE	p-value	Median
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
GDP price index	0.40	0.02	0.00	0.45	0.58	0.02	0.00	0.63
Real GDP	0.40	0.02	0.00	0.44	0.54	0.02	0.00	0.58
CPI	0.49	0.03	0.00	0.54	0.69	0.03	0.00	0.73
Three-month Treasury rate	0.58	0.02	0.00	0.65	0.69	0.02	0.00	0.72
Ten-year Treasury rate	0.54	0.02	0.00	0.56	0.61	0.02	0.00	0.64
AAA corporate bond rate	0.53	0.02	0.00	0.57	0.59	0.02	0.00	0.63
Fed funds rate	0.59	0.03	0.00	0.63	0.72	0.02	0.00	0.76
Five-year Treasury rate	0.53	0.02	0.00	0.57	0.62	0.02	0.00	0.65

Notes: The table shows the result from regression (4) Columns (1)-(3) report coefficients, standard errors and p-values from the panel data regression with individual fixed effects. Column (4) reports the median coefficient from individual regressions using demeaned variables. Columns (5)-(8) reports the same statistics for the 2 quarters horizon. Standard errors are Driscoll and Kraay (1998).



(a) Estimated gains G: our measure vs CG

(b) BGMS regression coefficient

Figure E.2: Under- and over- reaction in forecasts, $h = 3$ quarter (Blue Chip)

Notes: Panel (a): The red circles represent the implied gain from the CG regression, i.e. $\frac{1}{1+\beta_{CG}}$. Standard errors are Newey-West (1994), with optimal bandwidth selection. The blue diamonds represent the gain estimated from (6) with individual and time fixed effects. Standard errors are Driscoll and Kraay (1998). Panel (b): Panel estimate of β_{BGMS} with forecaster fixed effects. In both panels, bars reports the 90% confidence interval for the estimated coefficients and the forecast are at horizon h=3 quarters.

Table E.5: Difference between estimated gains G and G_{CG}, horizon 3 quarters (Blue Chip)

Variable	G_{CG} (1)	SE (2)	G (3)	SE (4)	Difference (5)	SE (6)	p-value (7)
GDP price index	0.57	0.09	0.40	0.02	0.18	0.10	0.03
Real GDP	0.62	0.11	0.40	0.02	0.22	0.12	0.03
CPI	0.99	0.20	0.49	0.03	0.51	0.20	0.00
Three-month Treasury rate	0.60	0.04	0.58	0.02	0.03	0.05	0.29
Ten-year Treasury rate	1.08	0.11	0.54	0.02	0.55	0.11	0.00
AAA corporate bond rate	0.95	0.19	0.53	0.02	0.42	0.19	0.01
Fed funds rate	0.58	0.04	0.59	0.03	-0.02	0.05	0.65
Five-year Treasury rate	0.94	0.08	0.53	0.02	0.41	0.08	0.00

Notes: Columns (1)-(2) reports the implied gain from CG regressions of table 4. Columns (3)-(4) replicate the gain estimate from regression (6) Columns (5)-(8) reports the difference between column (1) and (3), its standard error and the probability of rejecting the null of column (5) lower or equal to zero.

Table E.6: Difference between estimated gains G and G_{CG}, horizon 2 quarters (Blue Chip)

Variable	G_{CG} (1)	SE (2)	G (3)	SE (4)	Difference (5)	SE (6)	p-value (7)
GDP price index	0.88	0.12	0.58	0.02	0.30	0.12	0.01
Real GDP	0.77	0.11	0.54	0.02	0.23	0.11	0.02
CPI	1.06	0.20	0.69	0.03	0.37	0.20	0.03
Three-month Treasury rate	0.69	0.04	0.69	0.02	0.00	0.04	0.48
Ten-year Treasury rate	1.08	0.17	0.61	0.02	0.48	0.17	0.00
AAA corporate bond rate	0.89	0.12	0.59	0.02	0.31	0.12	0.01
Fed funds rate	0.69	0.04	0.72	0.02	-0.03	0.04	0.76
Five-year Treasury rate	1.01	0.14	0.62	0.02	0.40	0.14	0.00

Notes: Columns (1)-(2) reports the implied gain from CG regressions of table 4. Columns (3)-(4) replicate the gain estimate from regression (6). Columns (5)-(8) reports the difference between column (1) and (3), its standard error and the probability of rejecting the null of column (5) lower or equal to zero.

E.2 Full details on Bordalo et al. (2020) regressions

Table E.7: BGMS: Individual errors on revisions (SPF)

	3 quarters horizon				2 quarters horizon			
	β_{BGMS}	SE	p-value	Median	β_{BGMS}	SE	p-value	Median
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Nominal GDP	-0.25	0.08	0.00	-0.19	-0.11	0.06	0.09	-0.08
GDP price index inflation	-0.35	0.04	0.00	-0.35	-0.25	0.04	0.00	-0.26
Real GDP	-0.10	0.08	0.23	0.07	-0.07	0.10	0.45	0.02
Consumer Price Index	-0.30	0.09	0.00	-0.29	-0.24	0.07	0.00	-0.24
Industrial production	-0.30	0.14	0.04	-0.31	-0.01	0.10	0.94	0.03
Housing Start	-0.28	0.09	0.00	-0.28	0.12	0.05	0.03	0.07
Real Consumption	-0.26	0.12	0.03	-0.24	-0.16	0.08	0.06	-0.16
Real residential investment	-0.08	0.11	0.43	-0.07	0.07	0.08	0.40	0.02
Real nonresidential investment	0.08	0.13	0.55	0.15	0.10	0.07	0.17	0.10
Real state and local government consumption	-0.56	0.05	0.00	-0.52	-0.30	0.05	0.00	-0.26
Real federal government consumption	-0.48	0.04	0.00	-0.40	-0.28	0.04	0.00	-0.27
Unemployment rate	0.24	0.16	0.12	0.19	0.20	0.11	0.08	0.20
Three-month Treasury rate	0.24	0.11	0.03	0.29	0.14	0.08	0.09	0.21
Ten-year Treasury rate	-0.22	0.07	0.00	-0.24	-0.24	0.09	0.01	-0.27
AAA Corporate Rate Bond	-0.27	0.07	0.00	-0.32	-0.22	0.06	0.00	-0.29

Notes: The table reports the coefficients from the BGMS regression (individual forecast errors on individual revisions). Columns (1) to (3) shows the panel data with fixed effect coefficient with standard errors and corresponding p-values. Standard errors are Driscoll and Kraay (1998). Columns (7) shows the median coefficient of the BGMS regression at the individual level.

Table E.8: BGMS: Individual errors on revisions (Blue Chip)

	3 quarters horizon				2 quarters horizon			
Variable	β_{BGMS} (1)	SE (2)	p-value (3)	Median (4)	β_{BGMS} (5)	SE (6)	p-value (7)	Median (8)
GDP price index	-0.20	0.07	0.01	-0.26	-0.11	0.07	0.08	-0.16
Real GDP	0.05	0.17	0.78	0.10	0.05	0.13	0.73	-0.02
CPI	-0.25	0.15	0.10	-0.30	-0.18	0.11	0.10	-0.19
Three-month Treasury rate	0.28	0.11	0.01	0.22	0.22	0.08	0.01	0.18
Ten-year Treasury rate	-0.24	0.07	0.00	-0.27	-0.20	0.09	0.03	-0.20
AAA corporate bond rate	-0.25	0.08	0.00	-0.29	-0.17	0.08	0.04	-0.18
Fed funds rate	0.38	0.12	0.00	0.34	0.25	0.09	0.00	0.21
Five-year Treasury rate	-0.15	0.07	0.04	-0.17	-0.16	0.09	0.08	-0.16

Notes: The table reports the coefficients from the BGMS regression (individual forecast errors on individual revisions). Columns (1) to (3) shows the panel data with fixed effect coefficient with standard errors and corresponding p-values. Standard errors are Driscoll and Kraay (1998). Columns (4) shows the median coefficient of the BGMS regression at the individual level. Columns (5)-(8) do the same for the 2 quarters ahead horizon.

E.3 Regressions on Greenbook survey

Table E.9: Estimated coefficients on the Greenbook survey

	BGMS			Our regression (14)					
	β_{BGMS}	SE	p-value	β_1	SE	p-value	β_2	SE	p-value
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Nominal GDP	0.18	0.15	0.23	0.18	0.13	0.17	-0.03	0.24	0.91
GDP price index inflation	0.28	0.22	0.21	0.20	0.21	0.35	0.39	0.37	0.28
Real GDP	0.23	0.16	0.17	0.24	0.21	0.25	0.22	0.18	0.23
Consumer Price Index	-0.02	0.20	0.92	-0.03	0.19	0.86	1.11	0.39	0.01
Industrial production	-0.04	0.08	0.63	-0.05	0.09	0.56	0.01	0.14	0.96
Housing Start	0.07	0.15	0.66	0.11	0.16	0.52	0.01	0.18	0.96
Real residential investment	0.01	0.10	0.88	0.10	0.13	0.45	0.01	0.15	0.94
Real nonresidential investment	0.22	0.13	0.10	0.24	0.15	0.10	0.17	0.19	0.38
Real state and local government consumption	-0.52	0.12	0.00	-0.60	0.13	0.00	0.38	0.10	0.00
Real federal government consumption	-0.35	0.13	0.01	-0.60	0.23	0.01	0.46	0.21	0.03
Unemployment rate	0.13	0.18	0.46	0.12	0.18	0.48	-0.10	0.09	0.24

Notes: The table reports the coefficients from the BGMS regression (columns 1-3) and our specification (14) (columns 4-9) using Greenbook forecasts at horizon 3 quarters. Standard errors are Newey-West(1994) with optimal bandwidth selection.

Table E.10: Estimated coefficients on Greenbook survey: horizon 2 quarters

	BGMS			Our regression (14)					
	β_{BGMS}	SE	p-value	β_1	SE	p-value	β_2	SE	p-value
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Nominal GDP	0.09	0.10	0.36	0.10	0.10	0.32	0.07	0.11	0.50
GDP price index inflation	0.31	0.18	0.09	0.25	0.16	0.11	0.24	0.20	0.22
Real GDP	0.17	0.16	0.28	0.18	0.18	0.30	0.23	0.09	0.01
Consumer Price Index	0.16	0.08	0.03	0.18	0.09	0.04	0.59	0.26	0.02
Industrial production	0.12	0.12	0.33	0.12	0.14	0.38	0.18	0.10	0.08
Housing Start	0.02	0.10	0.83	0.03	0.10	0.79	-0.12	0.11	0.26
Real residential investment	0.02	0.09	0.80	0.09	0.12	0.48	0.01	0.10	0.96
Real nonresidential investment	0.18	0.12	0.15	0.24	0.15	0.12	0.11	0.11	0.31
Real state and local government consumption	-0.19	0.10	0.06	-0.26	0.08	0.00	0.43	0.09	0.00
Real federal government consumption	-0.41	0.11	0.00	-0.65	0.21	0.00	0.49	0.20	0.01
Unemployment rate	-0.03	0.14	0.83	-0.03	0.14	0.82	-0.02	0.12	0.84

Notes: The table reports the coefficients from the BGMS regression (columns 1-3) and our regression (14) (columns 4-9) using forecasts from the Federal Reserve Green Book at horizon 2 quarters. Standard errors are robust to heteroskedasticity and Newey-West with the automatic bandwidth selection procedure of Newey and West (1994).

E.4 Broer and Kohlhas (2022) regressions, additional details

Table E.11: Broer and Kohlhas (2022) original (not demeaned)

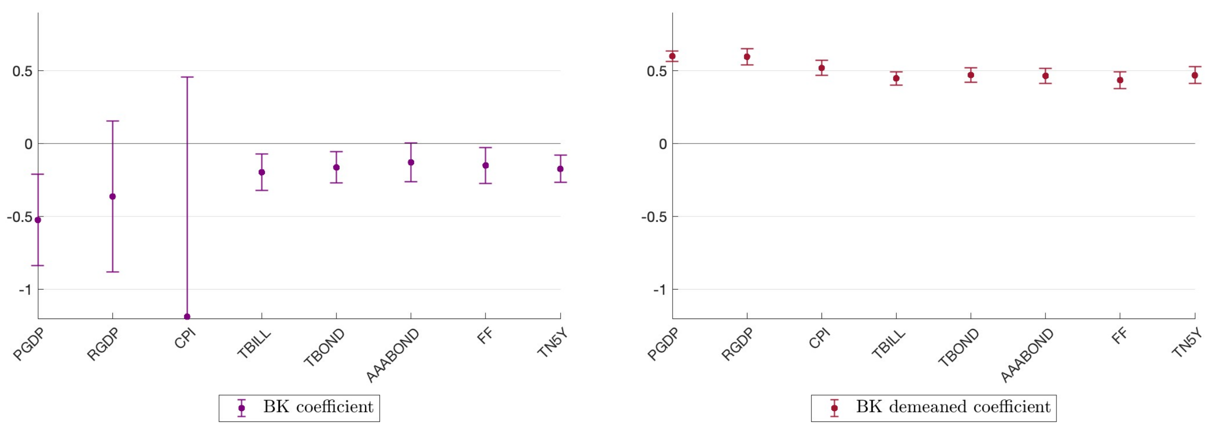
	3 quarters horizon			2 quarters horizon		
	β_{BK}	SE	p-value	β_{BK}	SE	p-value
Variable	(1)	(2)	(3)	(4)	(5)	(6)
Nominal GDP	-0.25	0.15	0.10	-0.10	0.08	0.21
GDP price index inflation	-0.23	0.08	0.01	-0.09	0.08	0.23
Real GDP	0.15	0.22	0.50	0.10	0.13	0.47
Consumer Price Index	-0.65	0.30	0.03	-0.48	0.22	0.03
Industrial production	0.22	0.20	0.27	0.16	0.14	0.24
Housing Start	-0.24	0.15	0.12	-0.09	0.13	0.51
Real Consumption	0.18	0.18	0.34	0.10	0.12	0.37
Real residential investment	0.03	0.24	0.90	0.12	0.17	0.46
Real nonresidential investment	-0.01	0.31	0.99	0.05	0.16	0.76
Real state and local government consumption	0.55	0.14	0.00	0.24	0.10	0.02
Real federal government consumption	0.28	0.18	0.12	0.20	0.13	0.12
Unemployment rate	-0.09	0.04	0.03	-0.05	0.03	0.10
Three-month Treasury rate	-0.22	0.06	0.00	-0.15	0.04	0.00
Ten-year Treasury rate	-0.10	0.06	0.10	-0.09	0.05	0.04
AAA Corporate Rate Bond	-0.17	0.05	0.00	-0.15	0.04	0.00

Notes: The table reports the coefficients from the BK regression (individual forecast errors on public signal not demeaned). Standard errors are Driscoll and Kraay (1998).

Table E.12: Broer and Kohlhas (2022) our modified version (demeaned)

	3 quarters horizon			2 quarters horizon		
	β_{BK}	SE	p-value	β_{BK}	SE	p-value
Variable	(1)	(2)	(3)	(4)	(5)	(6)
Nominal GDP	0.47	0.03	0.00	0.40	0.02	0.00
GDP price index inflation	0.49	0.04	0.00	0.37	0.03	0.00
Real GDP	0.39	0.05	0.00	0.37	0.03	0.00
Consumer Price Index	0.46	0.05	0.00	0.27	0.05	0.00
Industrial production	0.49	0.04	0.00	0.40	0.03	0.00
Housing Start	0.50	0.04	0.00	0.46	0.03	0.00
Real Consumption	0.53	0.05	0.00	0.41	0.05	0.00
Real residential investment	0.58	0.05	0.00	0.42	0.04	0.00
Real nonresidential investment	0.59	0.04	0.00	0.40	0.04	0.00
Real state and local government consumption	0.58	0.05	0.00	0.40	0.05	0.00
Real federal government consumption	0.53	0.04	0.00	0.37	0.03	0.00
Unemployment rate	0.50	0.03	0.00	0.44	0.03	0.00
Three-month Treasury rate	0.55	0.05	0.00	0.49	0.05	0.00
Ten-year Treasury rate	0.52	0.03	0.00	0.44	0.03	0.00
AAA Corporate Rate Bond	0.50	0.03	0.00	0.40	0.04	0.00

Notes: The table reports the coefficients from our version of the BK regression (individual forecast errors on public signal demeaned). Standard errors are Driscoll and Kraay (1998).



(a) Not demeaned (BK)

(b) Demeaned

Figure E.3: Forecast errors on lagged consensus (Blue Chip)

Notes: this figure plots the panel estimates of regressions (24)and (15), in panel (a) and panel (b) respectively, with individual fixed effect and horizon h=3. Bars reports the 90% confidence interval for the estimated coefficients. Standard errors are Driscoll and Kraay (1998).

Appendix F Regression Results with Blue Chip survey

Table F.1: Private and public information (Blue Chip)

Panel A: 3 quarters horizon	Revision				Public signal			
	β_1	SE	p-value	Median	β_2	SE	p-value	Median
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
GDP price index	-0.52	0.10	0.00	-0.55	0.81	0.04	0.00	0.83
Real GDP	-0.09	0.19	0.64	-0.07	0.63	0.08	0.00	0.60
CPI	-0.43	0.19	0.02	-0.46	0.73	0.09	0.00	0.90
Three-month Treasury rate	0.19	0.13	0.16	0.12	0.34	0.07	0.00	0.34
Ten-year Treasury rate	-0.45	0.09	0.00	-0.47	0.72	0.05	0.00	0.71
AAA corporate bond rate	-0.51	0.11	0.00	-0.49	0.73	0.07	0.00	0.71
Fed funds rate	0.31	0.15	0.04	0.26	0.25	0.08	0.00	0.25
Five-year Treasury rate	-0.32	0.09	0.00	-0.32	0.64	0.06	0.00	0.69

Notes: this table reports the coefficients of regression (14) (individual forecast errors on individual revisions and public information) using Blue Chip survey data. Columns 1 to 3 show coefficient β_1 (forecast revision) from the panel regression with individual fixed effect, with standard errors and corresponding p-values. Standard errors are Driscoll and Kraay (1998). Column 4 shows the median coefficient of the same regression at the individual level. Columns 5 to 7 show coefficient β_2 (public information) from the panel regression with individual fixed effect, with standard errors and corresponding p-values. Standard errors are Driscoll and Kraay (1998). Column 8 shows the median coefficient of the same regression at the individual level.

Table F.2: Private and public information (Blue Chip with SPF consensus)

Panel A: 3 quarters horizon

	Revision				Public signal			
	β_1	SE	p-value	Median	β_2	SE	p-value	Median
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
GDP price index	-0.50	0.10	0.00	-0.52	0.78	0.07	0.00	0.77
Real GDP	-0.02	0.21	0.92	-0.02	0.42	0.22	0.06	0.54
CPI	-0.42	0.18	0.02	-0.46	0.76	0.11	0.00	0.83
Three-month Treasury rate	0.23	0.13	0.08	0.12	0.27	0.13	0.04	0.28
Ten-year Treasury rate	-0.42	0.09	0.00	-0.42	0.74	0.09	0.00	0.74
AAA corporate bond rate	-0.54	0.10	0.00	-0.51	0.86	0.05	0.00	0.79

Notes: this table reports the coefficients of regression (14) (individual forecast errors on individual revisions and public information). The regression uses individual forecast data from the BC survey and consensus forecast from the SPF. Columns 1 to 3 show coefficient β_1 (forecast revision) from the panel regression with individual fixed effect, with standard errors and corresponding p-values. Standard errors are Driscoll and Kraay (1998). Column 4 shows the median coefficient of the same regression at the individual level. Columns 5 to 7 show coefficient β_2 (public information) from the panel regression with individual fixed effect, with standard errors and corresponding p-values. Standard errors are Driscoll and Kraay (1998). Column 8 shows the median coefficient of the same regression at the individual level.

Table F.3: Private and public information (Blue Chip from 1984)

Panel A: 3 quarters horizon

	Revision				Public signal			
	β_1	SE	p-value	Median	β_2	SE	p-value	Median
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
GDP price index	-0.52	0.10	0.00	-0.55	0.81	0.04	0.00	0.83
Real GDP	-0.09	0.19	0.64	-0.07	0.63	0.08	0.00	0.60
CPI	-0.43	0.19	0.02	-0.46	0.73	0.09	0.00	0.90
Three-month Treasury rate	-0.02	0.13	0.89	0.01	0.50	0.08	0.00	0.43
Ten-year Treasury rate	-0.47	0.08	0.00	-0.47	0.73	0.04	0.00	0.71
AAA corporate bond rate	-0.48	0.08	0.00	-0.47	0.74	0.05	0.00	0.71
Fed funds rate	0.03	0.17	0.88	0.10	0.48	0.10	0.00	0.36
Five-year Treasury rate	-0.35	0.08	0.00	-0.36	0.66	0.05	0.00	0.69

Notes: this table reports the coefficients of regression (14) (individual forecast errors on individual revisions and public information) using the full Blue Chip survey data from 1984. Columns 1 to 3 show coefficient β_1 (forecast revision) from the panel regression with individual fixed effect, with standard errors and corresponding p-values. Standard errors are Driscoll and Kraay (1998). Column 4 shows the median coefficient of the same regression at the individual level. Columns 5 to 7 show coefficient β_2 (public information) from the panel regression with individual fixed effect, with standard errors and corresponding p-values. Standard errors are Driscoll and Kraay (1998). Column 8 shows the median coefficient of the same regression at the individual level.

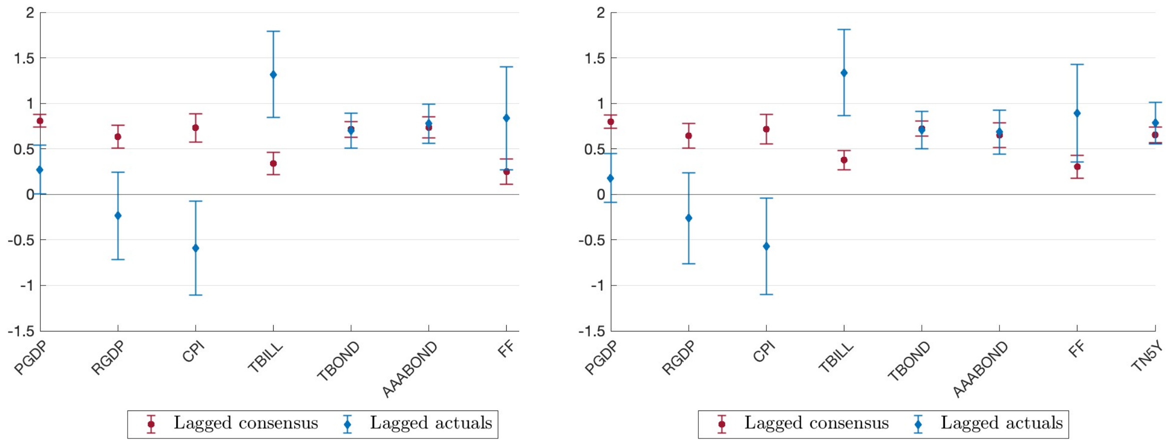
Table F.4: Private and public information (Blue Chip from 1984 with SPF consensus)

Panel A: 3 quarters horizon

	Revision				Public signal			
	β_1	SE	p-value	Median	β_2	SE	p-value	Median
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
GDP price index	-0.50	0.10	0.00	-0.52	0.78	0.07	0.00	0.77
Real GDP	-0.02	0.21	0.92	-0.02	0.42	0.22	0.06	0.54
CPI	-0.42	0.18	0.02	-0.46	0.76	0.11	0.00	0.83
Three-month Treasury rate	0.01	0.13	0.95	0.02	0.48	0.12	0.00	0.42
Ten-year Treasury rate	-0.43	0.09	0.00	-0.42	0.74	0.09	0.00	0.74
AAA corporate bond rate	-0.47	0.08	0.00	-0.47	0.75	0.06	0.00	0.76

Notes: this table reports the coefficients of regression (14) (individual forecast errors on individual revisions and public information). The regression uses individual forecast data from the full BC survey from 1984 and consensus forecast from the SPF. Columns 1 to 3 show coefficient β_1 (forecast revision) from the panel regression with individual fixed effect, with standard errors and corresponding p-values. Standard errors are Driscoll and Kraay (1998). Column 4 shows the median coefficient of the same regression at the individual level. Columns 5 to 7 show coefficient β_2 (public information) from the panel regression with individual fixed effect, with standard errors and corresponding p-values. Standard errors are Driscoll and Kraay (1998). Column 8 shows the median coefficient of the same regression at the individual level.

Appendix G Comparison between public signals



(a) Different regressions

(b) Same regression

Figure G.1: Comparison between public signals (Blue Chip)

Notes: Panel (a): estimates of β_2 from our original regression (14) (in red), and from (20) (in blue). Panel(b): β_2 (in red) and β_3 (in blue) estimates from regression (21) using the BC survey and h=3. Bars reports the 90% confidence interval for the estimated coefficients. Standard errors are Driscoll and Kraay (1998).

Table G.1: Idiosyncratic and public information: alternative measure of public information

Panel A: 3 quarters horizon

	Revision				Public signal			
	β_1	SE	p-value	Median	β_2	SE	p-value	Median
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Nominal GDP	-0.25	0.08	0.00	-0.18	-0.05	0.12	0.69	-0.13
GDP price index inflation	-0.40	0.04	0.00	-0.40	0.39	0.15	0.01	0.30
Real GDP	-0.10	0.08	0.22	0.06	0.02	0.18	0.89	-0.09
Consumer Price Index	-0.19	0.08	0.03	-0.14	-0.56	0.28	0.05	-0.52
Industrial production	-0.30	0.14	0.03	-0.35	0.08	0.14	0.56	0.11
Housing Start	-0.09	0.09	0.35	-0.13	0.57	0.13	0.00	0.37
Real Consumption	-0.30	0.12	0.01	-0.25	0.27	0.13	0.05	0.15
Real residential investment	-0.09	0.10	0.38	-0.07	0.57	0.18	0.00	0.48
Real nonresidential investment	0.06	0.14	0.65	0.18	0.20	0.22	0.37	0.14
Real state and local government consumption	-0.53	0.05	0.00	-0.53	0.12	0.10	0.23	0.17
Real federal government consumption	-0.47	0.04	0.00	-0.39	0.28	0.09	0.00	0.19
Unemployment rate	0.26	0.16	0.10	0.18	-0.39	0.25	0.12	-0.44
Three-month Treasury rate	-0.26	0.10	0.01	-0.31	0.93	0.26	0.00	1.30
Ten-year Treasury rate	-0.63	0.05	0.00	-0.64	0.61	0.11	0.00	0.62
AAA Corporate Rate Bond	-0.69	0.04	0.00	-0.78	0.80	0.10	0.00	0.75

Panel B: 2 quarters horizon

	Revision				Public signal			
	β_1	SE	p-value	Median	β_2	SE	p-value	Median
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Nominal GDP	-0.14	0.09	0.11	-0.10	0.12	0.12	0.33	0.04
GDP price index inflation	-0.41	0.04	0.00	-0.38	0.46	0.12	0.00	0.34
Real GDP	-0.13	0.10	0.20	0.07	0.22	0.17	0.21	-0.04
Consumer Price Index	-0.07	0.14	0.59	-0.12	-0.50	0.34	0.15	-0.54
Industrial production	-0.19	0.17	0.26	-0.15	0.35	0.22	0.11	0.32
Housing Start	0.03	0.06	0.67	-0.04	0.29	0.11	0.01	0.27
Real Consumption	-0.25	0.11	0.02	-0.21	0.21	0.13	0.11	0.14
Real residential investment	-0.09	0.09	0.31	-0.12	0.41	0.14	0.00	0.41
Real nonresidential investment	0.13	0.12	0.27	0.17	-0.02	0.20	0.94	-0.11
Real state and local government consumption	-0.40	0.04	0.00	-0.36	0.20	0.10	0.05	0.25
Real federal government consumption	-0.42	0.05	0.00	-0.33	0.29	0.10	0.00	0.08
Unemployment rate	0.22	0.12	0.06	0.20	-0.30	0.18	0.09	-0.28
Three-month Treasury rate	-0.33	0.14	0.02	-0.43	0.78	0.30	0.01	1.04
Ten-year Treasury rate	-0.80	0.06	0.00	-0.92	0.75	0.12	0.00	0.76
AAA Corporate Rate Bond	-0.77	0.05	0.00	-0.83	0.92	0.07	0.00	0.88

Notes: Estimates of regression (20) using the latest actual realization of the forecasted variable (i.e. x_t) as the public signal proxy. Columns 1 to 3 show coefficient β_1 (forecast revision) from the panel regression with individual fixed effect, with standard errors and corresponding p-values. Standard errors are Driscoll and Kraay (1998). Column 4 shows the median coefficient of the same regression at the individual level. Columns 5 to 7 show coefficient β_2 (public information) from the panel regression with individual fixed effect, with standard errors and corresponding p-values. Column 8 shows the median coefficient of the same regression at the individual level. Panel A uses forecast at 3 quarters horizon and panel B uses forecast at 2 quarters horizon.

Table G.2: Comparison between public signals: 3 quarters horizon

	Revision				Public signal: consensus				Public signal: realization			
	β_1	SE	p-value	Median	β_2	SE	p-value	Median	β_3	SE	p-value	Median
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Nominal GDP	-0.54	0.12	0.00	-0.46	0.77	0.06	0.00	0.79	-0.16	0.12	0.19	-0.15
GDP price index inflation	-0.71	0.04	0.00	-0.66	0.79	0.04	0.00	0.85	0.27	0.14	0.06	0.22
Real GDP	-0.33	0.12	0.01	-0.18	0.57	0.09	0.00	0.62	-0.03	0.19	0.86	-0.12
Consumer Price Index	-0.36	0.13	0.00	-0.30	0.71	0.06	0.00	0.77	-0.59	0.30	0.05	-0.56
Industrial production	-0.61	0.14	0.00	-0.70	0.80	0.08	0.00	0.82	0.02	0.15	0.92	0.14
Housing Start	-0.41	0.13	0.00	-0.41	0.71	0.06	0.00	0.68	0.42	0.14	0.00	0.31
Real Consumption	-0.59	0.15	0.00	-0.58	0.80	0.08	0.00	0.84	0.23	0.13	0.09	0.14
Real residential investment	-0.36	0.14	0.01	-0.32	0.70	0.08	0.00	0.68	0.51	0.17	0.00	0.34
Real nonresidential investment	-0.12	0.18	0.50	-0.12	0.64	0.09	0.00	0.50	0.13	0.22	0.55	0.06
Real state and local government consumption	-0.84	0.05	0.00	-0.79	0.93	0.03	0.00	0.85	-0.03	0.11	0.81	-0.02
Real federal government consumption	-0.83	0.03	0.00	-0.75	0.89	0.04	0.00	0.83	0.09	0.09	0.30	0.05
Unemployment rate	0.11	0.20	0.57	0.00	0.54	0.11	0.00	0.50	-0.63	0.26	0.01	-0.55
Three-month Treasury rate	-0.44	0.11	0.00	-0.55	0.53	0.11	0.00	0.49	0.93	0.26	0.00	1.26
Ten-year Treasury rate	-0.86	0.07	0.00	-0.83	0.74	0.05	0.00	0.80	0.59	0.12	0.00	0.60
AAA Corporate Rate Bond	-0.90	0.06	0.00	-0.96	0.70	0.07	0.00	0.77	0.66	0.11	0.00	0.58

Notes: this table reports the coefficients of regression (21) (individual forecast errors on individual revisions and both proxies for public information). Columns 1 to 3 show coefficient β_1 (forecast revision) from the panel regression with individual fixed effect, with standard errors and corresponding p-values. Standard errors are robust and clustered by forecaster and time. Column 4 shows the median coefficient of the same regression at the individual level. Columns 5 to 7 show coefficient β_2 (lagged consensus) from the panel regression with individual fixed effect, with standard errors and corresponding p-values. Standard errors are Driscoll and Kraay (1998). Column 8 shows the median coefficient of the same regression at the individual level. Panel A uses forecast at 3 quarters horizon and panel B uses forecast at 2 quarters horizon. Columns 9 to 11 show coefficient β_3 (latest actual realization of x_t) from the panel regression with individual fixed effect, with standard errors and corresponding p-values. Standard errors are Driscoll and Kraay (1998). Column 12 shows the median coefficient of the same regression at the individual level.

Table G.3: Comparison between public signals: 2 quarters horizon

	Revision				Public signal: consensus				Public signal: realization			
	β_1	SE	p-value	Median	β_2	SE	p-value	Median	β_3	SE	p-value	Median
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Nominal GDP	-0.35	0.11	0.00	-0.31	0.63	0.05	0.00	0.61	0.00	0.13	0.98	0.05
GDP price index inflation	-0.66	0.04	0.00	-0.62	0.68	0.05	0.00	0.65	0.35	0.12	0.00	0.26
Real GDP	-0.29	0.13	0.03	-0.05	0.53	0.09	0.00	0.47	0.15	0.18	0.43	-0.09
Consumer Price Index	-0.18	0.16	0.26	-0.23	0.59	0.08	0.00	0.64	-0.62	0.36	0.09	-0.57
Industrial production	-0.39	0.18	0.03	-0.43	0.55	0.08	0.00	0.50	0.41	0.22	0.07	0.34
Housing Start	-0.21	0.09	0.01	-0.20	0.52	0.06	0.00	0.53	0.25	0.12	0.03	0.20
Real Consumption	-0.49	0.13	0.00	-0.42	0.66	0.07	0.00	0.69	0.25	0.13	0.05	0.16
Real residential investment	-0.28	0.12	0.02	-0.26	0.49	0.07	0.00	0.46	0.40	0.14	0.00	0.37
Real nonresidential investment	0.02	0.14	0.91	0.06	0.45	0.06	0.00	0.44	-0.05	0.19	0.80	-0.11
Real state and local government consumption	-0.70	0.06	0.00	-0.60	0.77	0.05	0.00	0.69	0.15	0.09	0.10	0.16
Real federal government consumption	-0.79	0.04	0.00	-0.64	0.78	0.04	0.00	0.71	0.23	0.10	0.02	0.08
Unemployment rate	0.10	0.15	0.51	0.11	0.48	0.09	0.00	0.48	-0.52	0.18	0.00	-0.44
Three-month Treasury rate	-0.44	0.12	0.00	-0.53	0.46	0.13	0.00	0.35	0.76	0.29	0.01	1.05
Ten-year Treasury rate	-0.97	0.07	0.00	-1.04	0.67	0.07	0.00	0.76	0.70	0.13	0.00	0.69
AAA Corporate Rate Bond	-0.88	0.05	0.00	-0.95	0.49	0.07	0.00	0.57	0.80	0.09	0.00	0.73

Notes: this table reports the coefficients of regression (21) (individual forecast errors on individual revisions and both proxies for public information). Columns 1 to 3 show coefficient β_1 (forecast revision) from the panel regression with individual fixed effect, with standard errors and corresponding p-values. Standard errors are robust and clustered by forecaster and time. Column 4 shows the median coefficient of the same regression at the individual level. Columns 5 to 7 show coefficient β_2 (lagged consensus) from the panel regression with individual fixed effect, with standard errors and corresponding p-values. Standard errors are robust and clustered by forecaster and time. Column 8 shows the median coefficient of the same regression at the individual level. Panel A uses forecast at 3 quarters horizon and panel B uses forecast at 2 quarters horizon. Columns 9 to 11 show coefficient β_3 (latest actual realization of x_t) from the panel regression with individual fixed effect, with standard errors and corresponding p-values. Standard errors are robust and clustered by forecaster and time. Column 12 shows the median coefficient of the same regression at the individual level.

Appendix H Stricter treatment for outliers in the data

In the main text, we follow Bordalo et al. (2020) data trimming procedure, which consists in removing forecasts that are more than 5 interquartile ranges away from the median of each horizon in each quarters. In this section, we replicate our empirical results using a trimming procedure with stricter criteria than the one utilized in the text. First, we follow Kohlhas and Walther (2021) in dropping observations with forecast error is in the unconditional top or bottom 1 percentile. Second, we follow Angeletos et al. (2021) in removing forecasts that are more than 4 interquartile ranges away from the median of each horizon in each quarters. Third, we use median instead of mean forecasts to compute the consensus forecasts. We apply all of these three filters at the same time.

Table H.1: Summary Statistics

	Consensus					Individual		
	Errors			Revisions		Forecast dispersion	Nonrev share	Pr(< 80% revise same direction)
	Mean	SD	SE	Mean	SD			
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Nominal GDP	-0.29	1.61	0.19	-0.14	0.60	1.00	0.02	0.80
GDP price index inflation	-0.28	0.56	0.08	-0.08	0.22	0.47	0.07	0.83
Real GDP	-0.26	1.57	0.18	-0.15	0.53	0.78	0.02	0.74
Consumer Price Index	-0.08	0.99	0.15	-0.09	0.40	0.50	0.07	0.66
Industrial production	-0.94	3.72	0.44	-0.51	1.02	1.60	0.01	0.71
Housing Start	-3.04	16.81	2.08	-2.31	5.41	8.29	0.00	0.68
Real Consumption	0.32	1.06	0.15	-0.05	0.36	0.60	0.04	0.77
Real residential investment	-0.47	7.78	1.12	-0.64	1.97	4.50	0.04	0.85
Real nonresidential investment	0.20	5.35	0.75	-0.22	1.45	2.42	0.03	0.75
Real state and local government consumption	0.03	2.80	0.36	0.11	0.88	2.00	0.08	0.92
Real federal government consumption	0.00	1.03	0.14	-0.04	0.30	0.91	0.11	0.91
Unemployment rate	0.02	0.65	0.08	0.04	0.29	0.31	0.19	0.66
Three-month Treasury rate	-0.49	1.03	0.14	-0.18	0.44	0.45	0.15	0.58
Ten-year Treasury rate	-0.47	0.70	0.11	-0.12	0.33	0.39	0.11	0.54
AAA Corporate Rate Bond	-0.45	0.75	0.10	-0.11	0.33	0.51	0.09	0.68

Notes: Columns 1 to 5 show statistics for consensus forecast errors and revisions. Errors are defined as actuals minus forecasts, where actuals are the realized outcome corresponding to the variable forecasted. Revisions are forecast provided in t minus forecasts provided in $t-1$ about the same horizon. Columns 6 to 8 show statistics for individual forecasts, with Newey West (1994) standard errors. Forecast dispersion is the average standard deviation of individual forecasts at each quarter. The share of nonrevisions is the average quarterly share of instances in which forecast revision is less than 0.01 percentage points. The final column shows the fraction of quarters where less than 80 percent of the forecasters revise in the same direction.

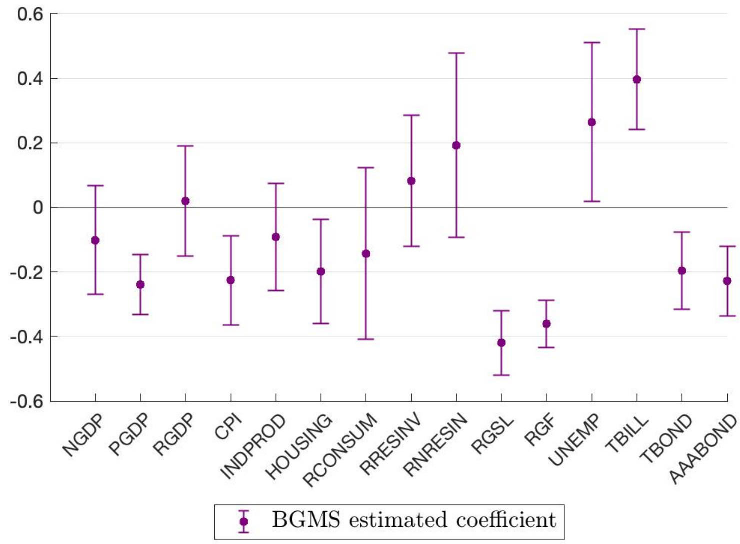


Figure H.1: Forecast errors on forecast revisions and public information

Notes: this figure plots the coefficient from panel data regression 7 with individual fixed effect and horizon h=3. Bars reports the 90% confidence interval for the estimated coefficients. Standard errors are Driscoll and Kraay (1998).

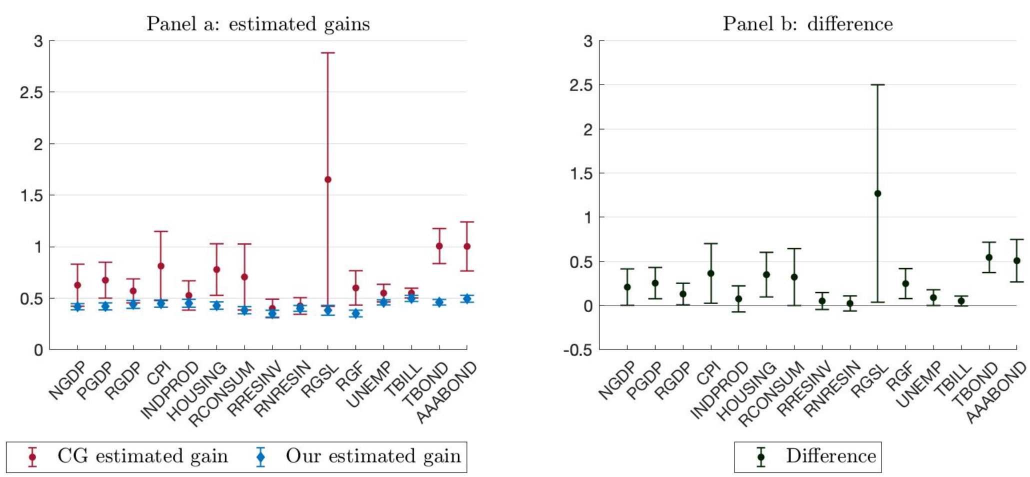


Figure H.2: Estimated gains: our measure vs CG

Notes: Panel (a): the red circles represent the implied gain from estimated coefficients in regression (4) at horizon h=3. Standard errors are robust to heteroskedasticity and Newey-West with the automatic bandwidth selection procedure of Newey and West (1994). The blue diamonds represent the gain estimated from (6) with individual fixed effect at horizon h=3. Standard errors are Driscoll and Kraay (1998). Bars reports the 90% confidence interval for the estimated coefficients. Panel (b): difference between the gains reported in panel (a).

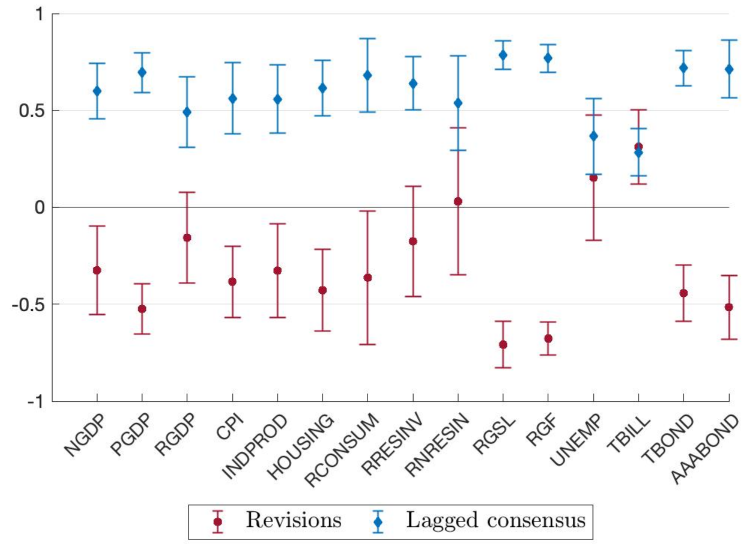


Figure H.3: Forecast errors on forecast revisions and public information

Notes: this figure plots the coefficient from panel data regression (14) with individual fixed effect and horizon h=3. The red circles represent the coefficient β_1 while the blue diamonds represent the coefficient β_2. Bars reports the 90% confidence interval for the estimated coefficients. Standard errors are Driscoll and Kraay (1998).

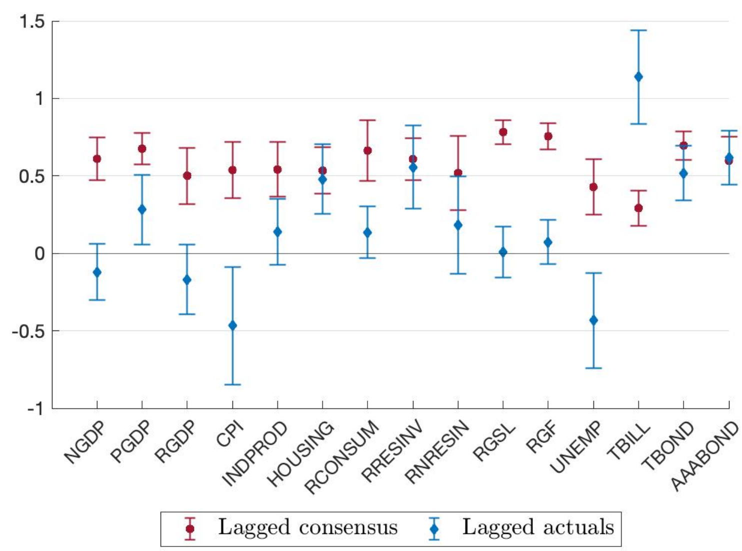


Figure H.4: Comparison between public signals

Notes: this figure plots the coefficient from a panel regression with fixed effects of (21) using forecasts from the Survey of Professional Forecasters and horizon h=3. The red circles represent the coefficient β_2 while the blue diamonds represent the coefficient β_3. Bars reports the 90% confidence interval for the estimated coefficients. Standard errors are Driscoll and Kraay (1998).

Table H.2: Individual errors on revisions

Variable	3 quarters horizon β_{BGMS} (1)	SE (2)	p-value (3)	Median (4)	2 quarters horizon β_{BGMS} (5)	SE (6)	p-value (7)	Median (8)
Nominal GDP	-0.10	0.10	0.33	-0.11	-0.02	0.08	0.86	0.00
GDP price index inflation	-0.24	0.06	0.00	-0.26	-0.18	0.05	0.00	-0.20
Real GDP	0.02	0.10	0.85	0.08	0.05	0.10	0.65	0.04
Consumer Price Index	-0.23	0.08	0.01	-0.18	-0.15	0.08	0.08	-0.16
Industrial production	-0.09	0.10	0.37	-0.11	0.13	0.11	0.24	0.09
Housing Start	-0.20	0.10	0.05	-0.29	0.23	0.08	0.00	0.16
Real Consumption	-0.14	0.16	0.39	-0.20	-0.07	0.10	0.45	-0.12
Real residential investment	0.08	0.12	0.51	0.02	0.16	0.09	0.10	0.03
Real nonresidential investment	0.19	0.17	0.28	0.20	0.22	0.10	0.04	0.21
Real state and local government consumption	-0.42	0.06	0.00	-0.45	-0.18	0.06	0.01	-0.18
Real federal government consumption	-0.36	0.04	0.00	-0.34	-0.22	0.05	0.00	-0.24
Unemployment rate	0.26	0.15	0.09	0.18	0.24	0.12	0.06	0.21
Three-month Treasury rate	0.40	0.09	0.00	0.39	0.30	0.07	0.00	0.31
Ten-year Treasury rate	-0.20	0.07	0.01	-0.23	-0.20	0.10	0.05	-0.22
AAA Corporate Rate Bond	-0.23	0.07	0.00	-0.30	-0.18	0.07	0.01	-0.24

Notes: The table reports the coefficients from the BGMS regression (individual forecast errors on individual revisions). Columns (1) to (3) shows the panel data with fixed effect coefficient with standard errors and corresponding p-values. Standard errors are Driscoll and Kraay (1998). Columns (7) shows the median coefficient of the BGMS regression at the individual level.

Table H.3: Stickiness estimation

Variable	3 quarters horizon β (1)	SE (2)	p-value (3)	Median (4)	2 quarters horizon β (5)	SE (6)	p-value (7)	Median (8)
Nominal GDP	0.42	0.02	0.00	0.47	0.53	0.01	0.00	0.58
GDP price index inflation	0.42	0.02	0.00	0.42	0.59	0.02	0.00	0.62
Real GDP	0.44	0.02	0.00	0.49	0.51	0.02	0.00	0.59
Consumer Price Index	0.45	0.02	0.00	0.52	0.65	0.02	0.00	0.69
Industrial production	0.45	0.02	0.00	0.50	0.54	0.02	0.00	0.60
Housing Start	0.43	0.02	0.00	0.49	0.45	0.02	0.00	0.49
Real Consumption	0.38	0.02	0.00	0.42	0.56	0.03	0.00	0.58
Real residential investment	0.35	0.02	0.00	0.39	0.52	0.02	0.00	0.58
Real nonresidential investment	0.40	0.02	0.00	0.46	0.53	0.02	0.00	0.57
Real state and local government consumption	0.38	0.03	0.00	0.35	0.53	0.02	0.00	0.52
Real federal government consumption	0.35	0.02	0.00	0.40	0.53	0.01	0.00	0.58
Unemployment rate	0.46	0.01	0.00	0.51	0.52	0.02	0.00	0.58
Three-month Treasury rate	0.50	0.02	0.00	0.55	0.58	0.02	0.00	0.63
Ten-year Treasury rate	0.46	0.02	0.00	0.50	0.55	0.02	0.00	0.58
AAA Corporate Rate Bond	0.49	0.02	0.00	0.55	0.56	0.02	0.00	0.61

Notes: The table shows the result from regression (6). Columns (1)-(3) report coefficients, standard errors and p-values from the panel data regression with time and individual fixed effect. Column (4) reports the median coefficient from individual regressions. Columns (5)-(8) reports the same statistics for the 2 quarters horizon. Standard errors are Driscoll and Kraay (1998).

Table H.4: Difference between estimated gains

Variable	G_{CG} (1)	SE (2)	G (3)	SE (4)	Difference (5)	SE (6)	p-value (7)
Nominal GDP	0.63	0.12	0.42	0.02	0.21	0.13	0.05
GDP price index inflation	0.67	0.11	0.42	0.02	0.25	0.11	0.01
Real GDP	0.57	0.07	0.44	0.02	0.13	0.07	0.04
Consumer Price Index	0.81	0.20	0.45	0.02	0.36	0.21	0.04
Industrial production	0.53	0.09	0.45	0.02	0.08	0.09	0.20
Housing Start	0.78	0.15	0.43	0.02	0.35	0.15	0.01
Real Consumption	0.71	0.19	0.38	0.02	0.32	0.20	0.05
Real residential investment	0.40	0.05	0.35	0.02	0.05	0.06	0.19
Real nonresidential investment	0.42	0.05	0.40	0.02	0.02	0.05	0.32
Real state and local government consumption	1.65	0.75	0.38	0.03	1.27	0.75	0.04
Real federal government consumption	0.60	0.10	0.35	0.02	0.25	0.10	0.01
Unemployment rate	0.55	0.05	0.46	0.01	0.09	0.05	0.05
Three-month Treasury rate	0.55	0.03	0.50	0.02	0.05	0.03	0.06
Ten-year Treasury rate	1.01	0.10	0.46	0.02	0.55	0.10	0.00
AAA Corporate Rate Bond	1.00	0.14	0.49	0.02	0.51	0.15	0.00

Notes: Columns (1)-(2) reports the implied gain from CG regressions. Columns (3)-(4) replicate the gain estimated from regression (6). Columns (5)-(8) reports the difference between column (1) and (3), its standard error and the probability of rejecting the null of column (5) lower or equal to zero.

Table H.5: Private and public information

Panel A: 3 quarters horizon

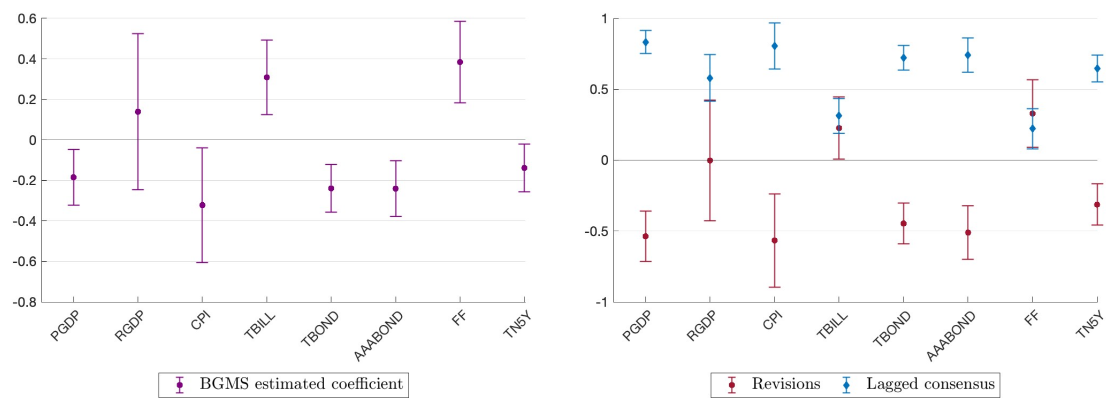
	Revision				Public signal			
	β_1	SE	p-value	Median	β_2	SE	p-value	Median
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Nominal GDP	-0.32	0.14	0.02	-0.35	0.60	0.09	0.00	0.63
GDP price index inflation	-0.52	0.08	0.00	-0.53	0.69	0.06	0.00	0.75
Real GDP	-0.16	0.14	0.28	-0.14	0.49	0.11	0.00	0.57
Consumer Price Index	-0.38	0.11	0.00	-0.40	0.56	0.11	0.00	0.59
Industrial production	-0.33	0.15	0.03	-0.37	0.56	0.11	0.00	0.63
Housing Start	-0.43	0.13	0.00	-0.49	0.61	0.09	0.00	0.69
Real Consumption	-0.36	0.21	0.09	-0.47	0.68	0.12	0.00	0.74
Real residential investment	-0.18	0.17	0.32	-0.31	0.64	0.08	0.00	0.66
Real nonresidential investment	0.03	0.23	0.90	-0.06	0.54	0.15	0.00	0.49
Real state and local government consumption	-0.71	0.07	0.00	-0.74	0.79	0.04	0.00	0.77
Real federal government consumption	-0.68	0.05	0.00	-0.63	0.77	0.04	0.00	0.76
Unemployment rate	0.15	0.20	0.44	0.01	0.37	0.12	0.00	0.40
Three-month Treasury rate	0.31	0.12	0.01	0.29	0.28	0.07	0.00	0.34
Ten-year Treasury rate	-0.44	0.09	0.00	-0.43	0.72	0.05	0.00	0.83
AAA Corporate Rate Bond	-0.52	0.10	0.00	-0.64	0.71	0.09	0.00	0.81

Panel B: 2 quarters horizon

	Revision				Public signal			
	β_1	SE	p-value	Median	β_2	SE	p-value	Median
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Nominal GDP	-0.18	0.11	0.11	-0.20	0.47	0.08	0.00	0.50
GDP price index inflation	-0.42	0.08	0.00	-0.45	0.59	0.06	0.00	0.55
Real GDP	-0.08	0.13	0.55	-0.11	0.43	0.09	0.00	0.43
Consumer Price Index	-0.24	0.11	0.05	-0.30	0.38	0.12	0.00	0.34
Industrial production	0.03	0.15	0.85	0.02	0.30	0.11	0.01	0.39
Housing Start	0.08	0.11	0.47	0.01	0.37	0.08	0.00	0.42
Real Consumption	-0.24	0.13	0.07	-0.28	0.52	0.09	0.00	0.57
Real residential investment	-0.01	0.12	0.95	-0.10	0.43	0.07	0.00	0.44
Real nonresidential investment	0.14	0.13	0.31	0.15	0.32	0.09	0.00	0.34
Real state and local government consumption	-0.44	0.08	0.00	-0.48	0.67	0.05	0.00	0.63
Real federal government consumption	-0.56	0.07	0.00	-0.56	0.68	0.05	0.00	0.68
Unemployment rate	0.15	0.16	0.36	0.05	0.32	0.10	0.00	0.33
Three-month Treasury rate	0.23	0.08	0.00	0.21	0.27	0.05	0.00	0.29
Ten-year Treasury rate	-0.38	0.11	0.00	-0.41	0.60	0.07	0.00	0.57
AAA Corporate Rate Bond	-0.38	0.10	0.00	-0.46	0.56	0.11	0.00	0.66

Notes: this table reports the coefficients of regression (14) (individual forecast errors on individual revisions and public information). Columns 1 to 3 show coefficient β_1 (forecast revision) from the panel regression with individual fixed effect, with standard errors and corresponding p-values. Standard errors are robust and clustered by forecaster and time. Column 4 shows the median coefficient of the same regression at the individual level. Columns 5 to 7 show coefficient β_2 (public information) from the panel regression with individual fixed effect, with standard errors and corresponding p-values. Standard errors are Driscoll and Kraay (1998). Column 8 shows the median coefficient of the same regression at the individual level. Panel A uses forecast at 3 quarters horizon and panel B uses forecast at 2 quarters horizon.

H.1 Blue Chip Data



(a) BGMS regression

(b) Forecast errors on revisions and public information

Figure H.5: Regressions result with stricter data cleaning (Blue Chip)

Notes: this figure replicates Figure 1 (panel a) and Figure 2 (panel b) with a stricter data cleaning. Bars reports the 90% confidence interval for the estimated coefficients. Standard errors are Driscoll and Kraay (1998).

Table H.6: BGMS: Individual errors on revisions (Blue Chip, stricter data cleaning)

	3 quarters horizon				2 quarters horizon			
	β_{BGMS}	SE	p-value	Median	β_{BGMS}	SE	p-value	Median
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
GDP price index	-0.18	0.08	0.03	-0.28	-0.07	0.07	0.32	-0.15
Real GDP	0.14	0.23	0.55	0.17	0.03	0.14	0.81	-0.05
CPI	-0.32	0.17	0.07	-0.39	-0.22	0.12	0.08	-0.22
Three-month Treasury rate	0.31	0.11	0.01	0.24	0.23	0.09	0.01	0.19
Ten-year Treasury rate	-0.24	0.07	0.00	-0.28	-0.20	0.10	0.05	-0.21
AAA corporate bond rate	-0.24	0.08	0.00	-0.29	-0.15	0.09	0.08	-0.17
Fed funds rate	0.38	0.12	0.00	0.34	0.27	0.09	0.00	0.22
Five-year Treasury rate	-0.14	0.07	0.06	-0.17	-0.15	0.10	0.11	-0.16

Notes: The table reports the coefficients from the BGMS regression (individual forecast errors on individual revisions). Columns (1) to (3) shows the panel data with fixed effect coefficient with standard errors and corresponding p-values. Standard errors are Driscoll and Kraay (1998). Columns (4) shows the median coefficient of the BGMS regression at the individual level. Columns (5)-(8) do the same for the 2 quarters ahead horizon.

Table H.7: Private and public information (Blue Chip, stricter data cleaning)

Panel A: 3 quarters horizon

	Revision				Public signal			
	β_1	SE	p-value	Median	β_2	SE	p-value	Median
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
GDP price index	-0.54	0.11	0.00	-0.62	0.83	0.05	0.00	0.83
Real GDP	0.00	0.26	0.99	-0.05	0.58	0.10	0.00	0.60
CPI	-0.57	0.20	0.01	-0.61	0.81	0.10	0.00	0.94
Three-month Treasury rate	0.23	0.13	0.09	0.15	0.31	0.07	0.00	0.35
Ten-year Treasury rate	-0.45	0.09	0.00	-0.47	0.72	0.05	0.00	0.71
AAA corporate bond rate	-0.51	0.11	0.00	-0.53	0.74	0.07	0.00	0.78
Fed funds rate	0.33	0.15	0.03	0.27	0.22	0.09	0.01	0.24
Five-year Treasury rate	-0.31	0.09	0.00	-0.33	0.65	0.06	0.00	0.70

Notes: this table reports the coefficients of regression (14) (individual forecast errors on individual revisions and public information) using Blue Chip survey data. Columns 1 to 3 show coefficient β_1 (forecast revision) from the panel regression with individual fixed effect, with standard errors and corresponding p-values. Standard errors are Driscoll and Kraay (1998). Column 4 shows the median coefficient of the same regression at the individual level. Columns 5 to 7 show coefficient β_2 (public information) from the panel regression with individual fixed effect, with standard errors and corresponding p-values. Standard errors are Driscoll and Kraay (1998). Column 8 shows the median coefficient of the same regression at the individual level.

Appendix I Details on solving the dynamic model

We average the expression for $\widetilde{E}_t^{(i)}[x_{t+h}]$ in equation (9) across agents and use repeated substitution in that same equation to obtain the following relation:

$$F_t \equiv -\frac{1}{1-\lambda}\sum_{k=0}^{\infty}\left(\frac{\lambda}{1-\lambda}\right)^k \bar{E}_t^{(k)}[x_{t+h}] = \frac{1}{1-\lambda}\bar{E}_t[x_{t+h}] - \frac{\lambda}{1-\lambda}\bar{\mathbb{E}}_t[\bar{E}_t[x_{t+h}]] \tag{32}$$

We then guess and verify that the law of motion for F_t and the other unobserved state variables is a VAR(1) process:[24]

$$Z_t \equiv \begin{bmatrix} x_{t+h} \\ F_t \\ w_t \end{bmatrix} = MZ_{t-1} + m \begin{bmatrix} u_{t+h} \\ e_t \end{bmatrix} \tag{33}$$

Where

$$M = \begin{bmatrix} \rho & 0 & 0 \\ M_{2,1} & M_{2,2} & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad m = \begin{bmatrix} 1 & 0 \\ m_{2,1} & m_{2,2} \\ 0 & 1 \end{bmatrix}. \tag{34}$$

We collect the two signals about x_{t+h} in the vector

$$V_t^i \equiv \begin{bmatrix} g_t \\ s_t^i \end{bmatrix} = HZ_t + \begin{bmatrix} 0 \\ \eta_t^i \end{bmatrix} \tag{35}$$

where

$$H = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \tag{36}$$

Agents use the conjectured law of motion (33) and the observables (35) to build the

[24] It is analytically convenient to introduce a third element of the state vector Z_t, $w_t \equiv e_t$, which is methodologically useful to handle the correlation of common shocks e_t in the signal g_t and F_t

optimal posterior estimate of the state vector

$$\begin{aligned}\mathbb{E}_t^{(i)}[Z_t] &= M\mathbb{E}_{t-1}^{(i)}[Z_{t-1}] + K(V_t^i - \mathbb{E}_{t-1}^{(i)}[V_t^{(i)}]) \\ &= (I-KH)M\mathbb{E}_{t-1}^{(i)}[Z_{t-1}] + KHMZ_{t-1} + KHm\begin{bmatrix}u_t\\e_t\end{bmatrix} + K\begin{bmatrix}0\\ \eta_t^i\end{bmatrix}\end{aligned} \tag{37}$$

where K is the rational 3x2 Kalman gain matrix. The first line of the vector equation above is the optimal forecast of x_{t+h}, thus naturally $G_1^{RE} = K_{(1,2)}$ and $G_2^{RE} = K_{(1,1)}$, where $K_{(i,j)}$ is the (i,j) entry of the optimal Kalman gain matrix K.

Lastly, average (37) to find the average posterior belief over the state vector.

$$\overline{\mathbb{E}}_t[Z_t] = (I-KH)M\overline{\mathbb{E}}_{t-1}[Z_{t-1}] + KHMZ_{t-1} + KHm\begin{bmatrix}u_t\\e_t\end{bmatrix} \tag{38}$$

At this point, the definition of F_t in (32) and equations (33) and (38) define a fixed point, which we can use to solve for the unknown coefficients in our conjecture. From the definition on F_t in (32) it follows that

$$\begin{aligned}F_t &= \begin{bmatrix}\frac{1}{1-\lambda} & -\frac{\lambda}{1-\lambda} & 0\end{bmatrix}\bar{\mathbb{E}}_t[Z_t] \equiv \xi\bar{\mathbb{E}}_t[Z_t] \\ &= \xi(I-KH)M\bar{\mathbb{E}}_{t-1}[Z_{t-1}] + \xi KHMZ_{t-1} + \xi KHm\begin{bmatrix}u_{t+h}\\e_t\end{bmatrix}\end{aligned} \tag{39}$$

$$\begin{aligned}F_t =&(-\frac{\lambda}{1-\lambda}\rho + \frac{1}{1-\lambda}M_{2,1})\bar{\mathbb{E}}_{t-1}[x_{t-1}] + \frac{1}{1-\lambda}M_{2,2}\bar{\mathbb{E}}_{t-1}[F_{t-1}] - G\rho\bar{\mathbb{E}}_{t-1}[x_{t-1}] \\ &+ G\rho x_{t-1} + G_2 e_t + Gu_{t+h} \\ =&[-\rho\frac{\lambda}{1-\lambda} + \frac{1}{1-\lambda}M_{2,1} + \frac{\lambda}{1-\lambda}M_{2,2} - G\rho]\bar{\mathbb{E}}_{t-1}[x_{t-1}] + \\ &+ M_{2,2}F_{t-1} + G\rho x_{t-1} + Gu_{t+h} + G_2 e_t\end{aligned} \tag{40}$$

where we used (32) to substitute

$$\frac{1}{1-\lambda}\bar{\mathbb{E}}_t[F_{t-1}] = F_{t-1} + \frac{\lambda}{1-\lambda}\bar{\mathbb{E}}_{t-1}[x_{t-1}]$$

and we defined

$$G_1 \equiv \frac{G_1^{RE} - \lambda K_{(2,2)}}{1-\lambda} \text{ and } G_2 \equiv \frac{G_2^{RE} - \lambda K_{(2,1)}}{1-\lambda}$$

Equation (40) must equal the second line of the perceived law of motion (33). The solution to the fixed point is given by $M_{2,1} = G\rho$, $m_{2,1} = G$, $m_{2,2} = G_2$ and $M_{2,2} = \rho - M_{2,1}$.

Given the law of motion of unobserved state 33 and the observable 35, the posterior variance of the forecast solves the following Ricatti equation

$$\begin{aligned} &\Sigma \equiv \mathbb{E}[(Z_t - Z^i_{t,t-1})(Z_t - Z^i_{t,t-1})'] \\ &\Sigma = M(\Sigma - \Sigma H' \left(H\Sigma H' + \begin{bmatrix} 0 & 0 \\ 0 & \tau^{-1} \end{bmatrix} \right)^{-1} H\Sigma)M' + m \begin{bmatrix} \chi^{-1} & 0 \\ 0 & \nu^{-1} \end{bmatrix} m' \end{aligned} \tag{41}$$

and the Kalman filter is

$$K = \Sigma H' \left(H\Sigma H' + \begin{bmatrix} 0 & 0 \\ 0 & \tau^{-1} \end{bmatrix} \right)^{-1} \tag{42}$$

Finally, this implies that the individual posted forecast is

$$\widetilde{E}^{(i)}_t[x_{t+h}] = \widetilde{E}^{(i)}_{t-1}[x_{t+h}] + G_1(s^i_t - \widetilde{E}^{(i)}_{t-1}[x_{t+h}]) + G_2(g_t - \widetilde{E}^{(i)}_{t-1}[x_{t+h}]), \tag{43}$$

which is equation 17 in the main text.

Appendix J Structural estimation at: 2 quarters horizon

Table J.1: Estimated parameters

Variable	ρ (1)	$\frac{\sigma_e}{\sigma_u}$ (2)	$\frac{\sigma_\eta}{\sigma_u}$ (3)	λ (4)
Nominal GDP	0.93	2.31	1.71	0.61
GDP price index inflation	0.94	2.41	2.47	0.84
Real GDP	0.88	2.23	1.39	0.48
Consumer Price Index	0.78	1.32	1.32	0.57
Industrial production	0.85	2.00	1.35	0.29
Housing Start	0.85	9.77	1.61	-0.11
Real Consumption	0.87	1.44	1.70	0.52
Real residential investment	0.89	2.11	1.45	0.23
Real nonresidential investment	0.89	9.99	1.08	0.14
Real state and local government consumption	0.89	1.41	3.12	0.83
Real federal government consumption	0.80	1.27	2.72	0.73
Unemployment rate	0.97	9.93	1.05	-0.28
Three-month Treasury rate	0.94	9.99	1.06	0.06
Ten-year Treasury rate	0.83	2.18	1.94	0.69
AAA Corporate Rate Bond	0.85	4.52	2.35	0.83

Table J.2: Moments in data and model

	Targeted moments						Untargeted moments					
	Mean Dispersion		C		β_1		β_{CG}		β_{BGMS}		β_2	
Variable	Data (1)	Model (2)	Data (3)	Model (4)	Data (5)	Model (6)	Data (7)	Model (8)	Data (9)	Model (10)	Data (11)	Model (12)
Nominal GDP	0.94	0.94	0.61	0.61	-0.35	-0.35	0.20	0.43	-0.11	-0.23	0.62	0.11
GDP price index inflation	0.20	0.20	0.63	0.63	-0.55	-0.55	-0.03	0.43	-0.25	-0.38	0.70	0.21
Real GDP	0.69	0.69	0.63	0.63	-0.26	-0.26	0.33	0.42	-0.07	-0.17	0.54	0.07
Consumer Price Index	0.21	0.21	0.70	0.70	-0.38	-0.38	-0.07	0.25	-0.24	-0.18	0.52	0.17
Industrial production	2.49	2.49	0.59	0.59	-0.16	-0.16	0.33	0.41	-0.01	-0.09	0.49	0.05
Housing Start	75.76	63.74	0.53	0.39	-0.15	0.05	0.91	1.42	0.12	0.05	0.54	0.00
Real Consumption	0.34	0.34	0.63	0.63	-0.37	-0.37	0.03	0.26	-0.16	-0.15	0.64	0.17
Real residential investment	16.76	16.76	0.56	0.56	-0.13	-0.13	0.56	0.42	0.07	-0.07	0.49	0.04
Real nonresidential investment	5.03	5.02	0.61	0.60	-0.02	-0.05	0.52	0.66	0.10	-0.05	0.41	0.00
Real state and local government consumption	0.93	0.93	0.62	0.62	-0.71	-0.71	-0.11	0.23	-0.28	-0.34	0.80	0.40
Real federal government consumption	4.42	4.42	0.60	0.60	-0.63	-0.63	-0.28	0.19	-0.30	-0.25	0.78	0.33
Unemployment rate	0.09	0.06	0.56	0.55	0.09	0.08	0.59	0.78	0.20	0.08	0.39	0.00
Three-month Treasury rate	0.21	0.12	0.63	0.60	0.02	-0.02	0.40	0.66	0.14	-0.02	0.48	0.00
Ten-year Treasury rate	0.12	0.12	0.60	0.60	-0.46	-0.46	-0.09	0.45	-0.24	-0.31	0.71	0.14
AAA Corporate Rate Bond	0.25	0.25	0.61	0.61	-0.49	-0.49	0.05	0.58	-0.22	-0.44	0.70	0.07

Table J.3: Posted and honest moments

	Gain			Consensus MSE			Dispersion		
	Posted	Honest	Ratio	Posted	Honest	Ratio	Posted	Honest	Ratio
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Nominal GDP	0.61	0.49	0.79	0.26	0.60	2.25	0.94	0.41	0.44
GDP price index inflation	0.63	0.41	0.65	0.02	0.08	4.17	0.20	0.03	0.17
Real GDP	0.63	0.53	0.84	0.25	0.46	1.85	0.69	0.40	0.57
Consumer Price Index	0.70	0.60	0.86	0.08	0.17	2.00	0.21	0.08	0.39
Industrial production	0.59	0.54	0.92	1.71	2.29	1.34	2.49	1.80	0.72
Housing Start	0.39	0.41	1.05	65.45	60.27	0.92	63.74	68.66	1.08
Real Consumption	0.63	0.55	0.89	0.23	0.37	1.62	0.34	0.14	0.40
Real residential investment	0.56	0.53	0.94	13.62	16.90	1.24	16.76	13.02	0.78
Real nonresidential investment	0.60	0.57	0.95	2.06	2.41	1.17	5.02	4.65	0.93
Real state and local government consumption	0.62	0.50	0.81	0.25	0.54	2.17	0.93	0.06	0.07
Real federal government consumption	0.60	0.52	0.87	2.85	4.63	1.63	4.42	0.57	0.13
Unemployment rate	0.55	0.60	1.08	0.04	0.03	0.78	0.06	0.07	1.12
Three-month Treasury rate	0.60	0.59	0.98	0.05	0.05	1.07	0.12	0.11	0.97
Ten-year Treasury rate	0.60	0.44	0.73	0.03	0.08	2.48	0.12	0.04	0.29
AAA Corporate Rate Bond	0.61	0.31	0.51	0.02	0.10	4.58	0.25	0.06	0.24

Appendix K Strategic diversification with heterogeneous priors

In this section we present an extension to the baseline model of Section 3.1 allowing for heterogeneous long-run mean priors μ_i.

Suppose forecasters have the same objective function and receive the same signals as in section 3.1, but with heterogeneous priors. For tractability we will assume i.i.d x_t process, and for simplicity of exposition we will drop all time subscripts t, since when x_t is iid the model is effectively static. Thus, the forecaster's optimal response is again

$$\hat{x}^i = \frac{1}{1-\lambda}\mathbb{E}^i[x] - \frac{\lambda}{1-\lambda}\mathbb{E}^i[\bar{\hat{x}}] \tag{44}$$

where we label the reported forecast that agent i submits to the survey as $\hat{x}^i$ and $\bar{\hat{x}} = \int \hat{x}^i di$ is the consensus forecast. Similarly to the main text, agents received a private and a public signal, both unbiased and centered around the true x with some noise.

$$\begin{aligned} s^i &= x + \eta^i \\ g &= x + e \end{aligned} \tag{45}$$

with $\eta^i \sim N(0, \tau^{-1})$ and $e \sim N(0, \nu^{-1})$.

Differently from the main text, forecasters have a dispersed priors about the mean of x. We model this by assuming that agents know the unconditional distribution of the fundamental $x \sim N(\mu_x, \chi^{-1})$, but they do not know the mean value μ_x. Their prior is that $\mu_x \sim N(0, \omega^{-1})$ and each forecasters has a private signal about it, $\mu^i = \mu_x + v^i$, with $v^i \sim N(0, \xi^{-1})$.[25]

[25] One can think of a setting where agents use an iid sample of t realizations of x to learn about its mean (Baley and Veldkamp, 2023). The accuracy of their private belief ξ will be proportional to the size of their observed sample, which we assume to be the same for every agent.

Posterior on fundamental Their true posterior beliefs about the fundamental are

$$\begin{aligned} \mathbb{E}[\mu_x|\mu^i] &= \frac{\xi}{\xi+\omega}\mu^i \\ Var[\mu_x|\mu^i] &= (\xi+\omega)^{-1} \end{aligned} \tag{46}$$

which implies

$$\begin{aligned} \mathbb{E}[x|\mu^i] &= \frac{\xi}{\xi+\omega}\mu^i \\ Var[x|\mu^i] &= (\xi+\omega)^{-1}+\chi^{-1} \end{aligned} \tag{47}$$

as a result

$$\mathbb{E}[x|s^i, g^i, \mathbb{E}[x|\mu^i]] = (1-\gamma_1-\gamma_2)\mathbb{E}[x|\mu^i] + \gamma_1 s^i + \gamma_2 g \tag{48}$$

where $\gamma_1 = \frac{\tau}{\tau+\nu+Var[x|\mu^i]^{-1}}$ and $\gamma_2 = \frac{\nu}{\tau+\nu+Var[x|\mu^i]^{-1}}$. If priors are not dispersed, that is $\xi \to \infty$ and we are back to the same posterior belief as in our baseline case we analyze in the main text.

Posterior on mean Similarly, their posterior beliefs about the mean are

$$\begin{aligned} \mathbb{E}[\mu_x|s^i, g] &= \frac{\tau}{\tau+\nu}s^i + \frac{\nu}{\tau+\nu}g \\ Var[\mu_x|s^i, g] &= (\tau+\nu)^{-1}+\chi^{-1} \end{aligned} \tag{49}$$

which implies

$$\mathbb{E}[\mu_x|s^i, g^i, \mathbb{E}[x|\mu^i]] = (1-f)\mathbb{E}[x|\mu^i] + f(hg+(1-h)s^i) \tag{50}$$

where $f = \frac{Var[\mu_x|s^i,g]^{-1}}{\xi+\omega+Var[\mu_x|s^i,g]^{-1}}$ and $h = \frac{\nu}{\nu+\tau}$. If priors are not dispersed, $\xi \to \infty$, therefore $\mathbb{E}[\mu_x|s^i, g^i, \mathbb{E}[x|\mu^i]] = \mu_x$, meaning we are in the common prior baseline case.

Posted forecast We guess the following linear solution for the posted forecast

$$\hat{x}^i = \delta_1 s^i + \delta_2 g + \delta_3 \mu^i \tag{51}$$

The solution of the fixed point between 51 and 44 is given by the following system

$$\begin{aligned}
\delta_3 &= \frac{(1-\lambda)(1-\gamma_1-\gamma_2)}{(1-\lambda)+\lambda\gamma_2}\left[\frac{\xi+\omega}{\xi}+\frac{\lambda(1-f)}{1-\lambda}-\frac{\lambda^2 f(1-h)(1-\lambda)(1-\gamma_1-\gamma_2)}{(1-\lambda)+\lambda\gamma_2}\right]^{-1} \\
\delta_2 &= (1-\lambda\delta_2)\gamma_1 - \lambda\delta_3 f h \\
\delta_1 &= \frac{\gamma_2-\lambda\delta_3 f(1-h)}{(1-\lambda)+\lambda\gamma_2}
\end{aligned} \tag{52}$$

First, one can see that in the case of no strategic incentives $\lambda = 0$, the posted forecast would coincide with the honest belief 48. Second, this is a generalization of the common prior case described in the main text. To see this, consider the case where prior is common, $\xi \to \infty$ which implies $f \to 0$. Then the solution for 51 coincides with the one in main text 11.

Simulated coefficient Since the model with heterogeneous prior is not as tractable as the one in the main text, we use numerical simulation. Figure K.1 reports the coefficients estimated from running regressions 4, 7, 14 and 15 in simulated data for different value of individual prior accuracy ξ^{-1}. If $\xi^{-1} = 0$, the results coincide with the proposition in the main text and $\hat{\beta}$ in regression 15 equal zero. However, when priors are dispersed, $\xi^{-1} > 0$, we find $\hat{\beta} > 0$ while the sign in all the other coefficients is unaffected. This is consistent with our finding in Figure 5.

Appendix L Analytical model with a finite number of forecasters

In Section 8 we consider a game featuring a continuum of atomistic forecasters with mass 1. Here we generalize this result by considering instead a finite number of forecasters N. The main implication is that observing the lagged consensus forecast $\overline{E}_{t-1}[x_{t+h}]$ does not reveal the fundamental x_{t+h}, while the main results of Section 8 are unchanged.

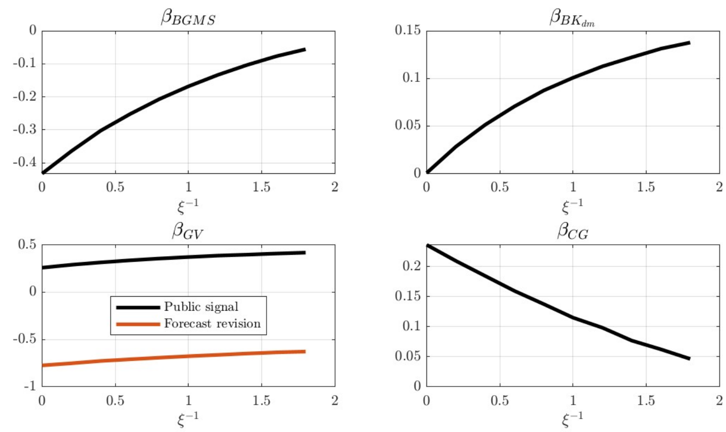


Figure K.1: Simulated coefficients

Notes: This figure reports the coefficients of regressions 4, 7, 14 and 15 in simulated data for different value of individual prior accuracy ξ. The rest of the parameters are set as follow: $\tau = 0.05$, $\nu = 0.5$, $\chi = 1$, $\omega = 0.25$ and $\lambda = 0.8$.

First, consider the lagged consensus forecast $\overline{E}_{t-1}[x_{t+h}]$. With a finite number of forecasters $N < \infty$, it equals

$$\overline{E}_{t-1}[x_{t+h}] \equiv \frac{1}{N}\sum_{i=1}^{N}\widetilde{E}^{(i)}_{t-1}[x_{t+h}] = \mu + G_1(\frac{1}{N}\sum_{i=1}^{N}s^{(i)}_{t-1} - \mu) + G_2(g_{t-1} - \mu) \tag{53}$$

where $\frac{1}{N}\sum_{i=1}^{N}s^{(i)}_{t-1} = \frac{s^{(i)}_{t-1}+\sum_{j\neq i}s^{(j)}_{t-1}}{N} = \frac{1}{N}s^{(i)}_{t-1} + \frac{N-1}{N}x_{t+h} + \frac{1}{N}\sum_{j\neq i}\eta^{(j)}_{t-1}$. As a result, since forecasters i observes μ, g_{t-1} and $s^{(i)}_{t-1}$, the lag consensus $\overline{E}_{t-1}[x_{t+h}]$ is a signal about $\frac{N-1}{N}x_{t+h}$ with noise $\frac{1}{N}\sum_{j\neq i}\eta^{(j)}_{t-1}$. If $N \to \infty$, then $\overline{E}_{t-1}[x_{t+h}]$ is a fully revealing signal. But more generally, it contains the noise $\frac{\tau^{-1}}{N}$.

Second, relaxing the assumption of infinite number of forecasters does not significantly affect the solution of the strategic game in equation (9). Notice that expectations of forecasters i about average forecast equals

$$\mathbb{E}^{(i)}_t(\overline{E}_t[x_{t+h}]) = \mu + G_1(\frac{1}{N}s^{(i)}_t + \frac{N-1}{N}\mathbb{E}^{(i)}_t[x_{t+h}] + \frac{1}{N}\sum_{j\neq i}\mathbb{E}^{(i)}_t[\eta^{(j)}_t] - \mu) + G_2(g_t - \mu) \tag{54}$$

where $\mathbb{E}_t^{(i)}[\eta_t^{(j)}] = 0$. Guessing the same linear solution as in (11) yields

$$G_1 = \frac{G_1^{RE}}{(1-\lambda) + \lambda \frac{1+(N-1)G_1^{RE}}{N}}$$

$$G_2 = \frac{(1-\lambda) + \lambda \frac{1}{N}}{(1-\lambda) + \lambda \frac{1+(N-1)G_1^{RE}}{N}} G_2^{RE}$$

If $N \to \infty$, the solution equals the one in Section 8. Importantly, even in the general case of $N < \infty$ we still have

$$\frac{G_1}{G_1^{RE}} > \frac{G_2}{G_2^{RE}} \iff \lambda > 0 \tag{55}$$

which is the main implication we want to test in the data. So the basic implication of the model is the same regardless of N, but in order to be comparable with the previous literature we present the model for the case of $N \to \infty$. When we go to the data, however, we use the lagged consensus as the observed signal since in reality, with $N < \infty$, the consensus is indeed a noisy signal.

Appendix M Forecast Smoothing

An alternative, reputation-based explanation for predictable forecast errors in surveys is that forecasters might have an incentive to avoid reporting drastic period-to-period changes in their forecasts to their customers. This would lead forecasters to effectively "smooth" their reported forecasts over time. For example, Coibion and Gorodnichenko (2015) consider this type of reputational incentive to show that it indeed leads to predictable errors in reported forecasts.

In this appendix we present an extension of our model in which we consider the possibility that in addition to the strategic incentives that are the main focus of our paper, forecasters also have an incentive to "smooth" their own forecasts over time. It turns out that these two different types of incentives, while both are motivated from a reputational standpoint, have distinct empirical implications. The strategic incentives we focus on lead forecasters to change the relative under- and overweighting of idiosyncratic information relative to public information. For example, when faced with strategic diversification incentives in particular we have shown that forecasters will optimally overweight idiosyncratic signals and underweight common, publicly observed signals.

In contrast, in this Appendix we show that an incentive to smooth forecasts over time leads forecasters to *underweight all new information* equally. Furthermore, we show that even when agents optimally smooth their forecasts, our key implication that idiosyncratic information is relatively overweighted relative to common information remains true if and only if the forecasters face strategic diversification incentives. Thus, our main empirical exercise (Section 3.2), which is specifically designed to test the relative under- and overweighting of idiosyncratic and public information can also be used to distinguish between strategic and forecast smoothing incentives.

To derive these results, we consider an augmented version of the analytical model presented in Section 3.1. The key differences with Section 3.1 are two-fold. First, the objective function of the forecasters is augmented with an additional term that captures the incentive to smooth the reported forecast over time. Second, to make this smoothing incentive non-trivial, we now assume that each period forecasters make forecasts for two consecutive future horizons. To keep things simple we assume that these are one and two step ahead forecasts. Thus, at time t the forecasters are providing a prediction for both x_{t+1} and x_{t+2}.

We model the incentive to smooth forecasts following Coibion and Gorodnichenko (2015) and assume that when agents report their time t forecast of tomorrow, $\widetilde{E}_t(x_{t+1})$ they want to make sure that this does not deviate too much from the prediction about x_{t+1} they made yesterday – i.e. $\widetilde{E}_{t-1}(x_{t+1})$. In addition, the forecasters also face the same strategic incentives we have already discussed. This results in the objective function:

$$\min_{\widetilde{E}_t^{(i)}[x_{t+1}],\widetilde{E}_t^{(i)}[x_{t+2}]} \sum_{h=1}^{2} \mathbb{E}_t \left[(\widetilde{E}_t^{(i)}[x_{t+h}] - x_{t+h})^2 - \lambda(\widetilde{E}_t^{(i)}[x_{t+h}] - \overline{E}_t[x_{t+h}])^2 \right] \\ + \alpha(\widetilde{E}_t^{(i)}[x_{t+1}] - \widetilde{E}_{t-1}^{(i)}[x_{t+1}])^2 + \alpha\mathbb{E}_t \left[\widetilde{E}_t^{(i)}[x_{t+2}] - \widetilde{E}_{t+1}^{(i)}[x_{t+2}] \right]$$

The first line of this objective function models strategic incentives in the same way as already discussed in the main text. The only difference is that now we consider a richer problem, where each period agents make forecasts at two different horizons – one and two-periods in the future. The agents then trade-off an accuracy incentive (captured by the expected mean squared error term) and the strategic incentive to either herd or differentiate, as controlled by $\lambda \in (-1, 1)$.

The forecast smoothing incentive is modeled with the two terms on the second line. Specifically, when $\alpha > 0$, the forecasters want to make sure their current one-period ahead forecast is not too different from the two period forecast they made yesterday, since both of these forecasts are about the same x_{t+1}. Similarly, agents are also forward looking, and hence when they make their two-period ahead forecast today they want to make sure it is not too different from what they expect to report tomorrow.

This gives us two first order conditions, one for the one-period ahead forecast and one for the two-periods ahead forecast the agents make at time t:

$$\widetilde{E}_t^{(i)}[x_{t+1}] = \frac{1}{1-\lambda+\alpha}\mathbb{E}_t(x_{t+1}) - \frac{\lambda}{1-\lambda+\alpha}\mathbb{E}_t\left(\overline{E}_t(x_{t+1})\right) + \frac{\alpha}{1-\lambda+\alpha}\widetilde{E}_{t-1}^{(i)}[x_{t+1}]$$

$$\widetilde{E}_t^{(i)}[x_{t+2}] = \frac{1}{1-\lambda+\alpha}\mathbb{E}_t(x_{t+2}) - \frac{\lambda}{1-\lambda+\alpha}\mathbb{E}_t\left(\overline{E}_t(x_{t+2})\right) + \frac{\alpha}{1-\lambda+\alpha}\mathbb{E}_t\left(\widetilde{E}_{t+1}^{(i)}[x_{t+2}]\right)$$

From these conditions, we can directly see that now the reported forecast $\widetilde{E}_t^{(i)}[x_{t+h}]$ is going to be biased away from the rational expectation of the agent not only by an incentive to stray farther away from the consensus (for $\lambda > 0$), but also with an incentive to stay closer to this agent's own *last period* forecast $\widetilde{E}_{t-1}^{(i)}[x_{t+h}]$.

To make the two-period head forecasting problem non-trivial, we augment the information structure of Section 3.1 with the assumption that each period the agents observe two sets of signals – one about x_{t+1} and one about x_{t+2}:

$$s_{1t}^{(i)} = x_{t+1} + \eta_{1t}^{(i)}$$
$$g_{1t} = x_{t+1} + e_{1t}$$

$$s_{2t}^{(i)} = x_{t+2} + \eta_{2t}^{(i)}$$
$$g_{2t} = x_{t+2} + e_{2t}$$

The Rational Expectations of x_{t+1} follow a familiar recursive structure, with

$$\mathbb{E}_t(x_{t+1}) = \mathbb{E}_{t-1}(x_{t+1}) + G_1^{RE}(s_{1t}^{(i)} - \mathbb{E}_{t-1}(x_{t+1})) + G_2^{RE}(g_{1t} - \mathbb{E}_{t-1}(x_{t+1}))$$

$$\mathbb{E}_{t-1}(x_{t+1}) = \mu + \bar{G}_1^{RE}(s_{2,t-1}^{(i)} - \mu) + \bar{G}_2^{RE}(g_{2,t-1} - \mu)$$

Consistent with our previous notation, G_1^{RE} and G_2^{RE} are the rational Kalman gain coefficients of the time t expectation, and we introduce the bar notation $\bar{G}_1^{RE}$ and $\bar{G}_2^{RE}$ to denote the weights put on the time $t-1$ signals in the $t-1$ rational expectation.

Finally, to solve the FOCs of the agents we guess that the reported forecasts, $\widetilde{E}_t^{(i)}[x_{t+h}]$ similarly follow a recursive structure, given by

$$\widetilde{E}_t^{(i)}[x_{t+1}] = \widetilde{E}_{t-1}^{(i)}[x_{t+1}] + G_1(s_{1t}^{(i)} - \widetilde{E}_{t-1}^{(i)}[x_{t+1}]) + G_2(g_{1t} - \widetilde{E}_{t-1}^{(i)}[x_{t+1}])$$
$$\widetilde{E}_{t-1}^{(i)}[x_{t+1}] = \mu + \bar{G}_1(s_{2,t-1}^{(i)} - \mu) + \bar{G}_2(g_{2,t-1} - \mu)$$

Plugging this guess in the FOCs, we can then solve the fixed point for the coefficients $G_1, G_2, \bar{G}_1, \bar{G}_2$. It is instructive to consider the following four equations that implicitly define these coefficients as proportional to their rational counter-parts, but with weights different than one:

$$G_1 = \frac{1 - \lambda G_1 - \lambda(1-G)\bar{G}_1}{1-\lambda+\alpha} G_1^{RE}$$

$$G_2 = \frac{1 - \lambda G_1 - \lambda(1-G)\bar{G}_1}{1+\alpha} G_2^{RE}$$

$$\bar{G}_1 = \frac{(1-\lambda+\alpha)(1-\lambda\bar{G}_1) - \alpha\lambda(G+(1-G)\bar{G}_1) + \alpha}{(1-\lambda)(1-\lambda+2\alpha)} \bar{G}_1^{RE}$$

$$\bar{G}_2 = \frac{(1-\lambda+\alpha)(1-\lambda\bar{G}_1) - \alpha\lambda(G+(1-G)\bar{G}_1) + \alpha}{(1-\lambda)(1-\lambda+2\alpha) + \lambda(1-\lambda+\alpha+\alpha(1-G))} \bar{G}_2^{RE}$$

With these relationships at hand we can derive the following key results. First, notice that if there are no strategic incentives ($\lambda = 0$), then solution simplifies down to reported forecasts *underweighting equally* all new information. Specifically, given $\lambda = 0$ we have:

$$G_1 = \frac{1}{1+\alpha} G_1^{RE}$$

$$G_2 = \frac{1}{1+\alpha} G_2^{RE}$$

So the smoothing incentive, by itself, leads agents to report forecasts that put less weight on all new information. These "dampened" forecasts underweight equally both the public and the idiosyncratic sources of information, as they simply put more weight on the beginning of period priors.

When both strategic and smoothing incentives are active, we can again derive a sharp result that strategic incentives change the *relative* over- and under-weighting of idiosyncratic and common information. Specifically, we can show again, in parallel to the same result in our main text, that idiosyncratic information is overweighted relative to public information if and only if $\lambda > 0$ (i.e. only when agents face strategic diversification incentives). We can see this from the fact that the relative weights on idiosyncratic and public information can be expressed as

$$\frac{G_1}{G_2} = \frac{1+\alpha}{1-\lambda+\alpha}\frac{G_1^{RE}}{G_2^{RE}}$$

$$\frac{\bar{G}_1}{\bar{G}_2} = \frac{(1-\lambda)(1-\lambda+2\alpha)+\lambda(1-\lambda+\alpha+\alpha(1-G))}{(1-\lambda)(1-\lambda+2\alpha)}\frac{\bar{G}_1^{RE}}{\bar{G}_2^{RE}}$$

And from here it follows directly that

$$\frac{G_1}{G_2} > \frac{G_1^{RE}}{G_2^{RE}} \iff \lambda > 0$$
$$\frac{\bar{G}_1}{\bar{G}_2} > \frac{\bar{G}_1^{RE}}{\bar{G}_2^{RE}} \iff \lambda > 0$$

Thus, with and without the smoothing incentive, the prediction that idiosyncratic information is overweighted relative to public information is unique to the strategic diversification incentives case ($\lambda > 0$). Intuitively, smoothing incentives $\alpha > 0$ on their own lower the weight put on all new information. Strategic diversification incentives, on the other hand, lead agents to underweight common information relative to idiosyncratic information, even in the presence of smoothing incentives.

Thus, our benchmark empirical methodology (Section 3.2) remains valid and can detect strategic incentives even in this more general case, where agents also smooth their forecasts over time.

Appendix N Winner-take-all Forecasting Game

In this Appendix we analyze a formal winner-take-all game being played among forecasters. The basic motivation is that there are a number of well publicized and salient "Best Forecaster Awards" (e.g. the Lawrence Klein Best Forecasting Award) that economic forecasting and consulting firms care about. Such awards generate a lot of valuable publicity for the winners, but by the nature of a competition, only the best forecaster in any given period wins this reputational gain. As such, this creates strategic incentives among forecaster to "stand out". The key takeaway of the microfounded model presented in this

Appendix is that the implied equilibrium behavior is very similar to that of our benchmark model analyzed in Sections 3.1 and 4. Intuitively, the Morris and Shin type of quadratic strategic framework we analyze in the main text could be thought of as a more tractable, reduced form representation of the microfounded model presented here.

The setting is the same as in the analytical model of Section 3, and can be viewed as an extension to the winner-take-all model in Ottaviani and Sørensen (2006), with one of the key extensions being the introduction of a common, public signal g_t observed by all forecasters. There is a measure one of forecasters i that are trying to forecast an iid Gaussian variable x_{t+h} with mean μ and variance χ^{-1}. The forecasters have the same information structure as in the main text, where each sees a private signal with idiosyncratic noise $s_t^{(i)}$, with precision τ, and a commonly observed public signal g_t, with precision ν. The forecasters submit a forecast $\widetilde{E}_{it}[x_{t+h}]$ and the forecaster that is closest to the actual realization of x_{t+h} wins a prize. If more than one forecasters guesses the same winning value the prize is divided equally among all winners.

Normalizing the value of the prize to one, since there is a continuum of forecasters which make a continuum of forecasts, the winning forecasters are the ones that guess the exact realization of x_{t+h}. So the expected payoff U of forecaster i making forecast $\widetilde{E}_{it}[x_{t+h}]$ is given by

$$\mathbb{E}_{it}\left(U(\widetilde{E}_{it}[x_{t+h}])\right) = \int_x \frac{\mathbb{1}(x_{t+h} = \widetilde{E}_{it}[x_{t+h}])}{\gamma(\widetilde{E}_{it}[x_{t+h}]|x_{t+h}, g_t)} f(x_{t+h}|\mathcal{I}_{it}) dx_{t+h}$$

where $\mathbb{1}(x_{t+h} = \widetilde{E}_{it}[x_{t+h}])$ is an indicator function which captures the fact that forecaster i wins the prize only in the state of the world where $x_{t+h} = \widetilde{E}_{it}[x_{t+h}]$, and otherwise the payoff is zero (winner-take-all). In addition, the expected payoff is also divided by the mass of other forecasters that make the same exact guess, $\widetilde{E}_{it}[x_{t+h}]$, which is given by their PDF $\gamma(.)$. The mass of other forecasts is a function of the underlying state x_{t+h} and the realization of the public signal g_t, and this is made explicit in the fact that the PDF $\gamma(.)$ is conditional on both of these variables. Lastly, the agent is taking an expectation over this payoff since the value of x_{t+h} is unknown (this is the only stochastic part of the payoff, given $\mathcal{I}_{it}$), hence we integrate against the unknown x_{t+h}, using the PDF of x_{t+h} conditional on the information set of agent i, $\mathcal{I}_{it}$, which we denote as $f(x_{t+h}|\mathcal{I}_{it})$.

Given that the agent only wins in one state of the world, $x_{t+h} = \widetilde{E}_{it}[x_{t+h}]$, the expected payoff simplifies down to the following ratio of PDFs evaluated at that state of the world:

$$\mathbb{E}_{it}\left(U(\widetilde{E}_{it}[x_{t+h}])\right) = \frac{f(x_{t+h}|\mathcal{I}_{it})}{\gamma(\widetilde{E}_{it}[x_{t+h}]|x_{t+h}, g_t)}\bigg|_{x_{t+h}=\widetilde{E}_{it}[x_{t+h}]}$$

We will now construct the numerator and denominator of this expression. Given the information set of forecaster i, $\mathcal{I}_{it}$, the conditional distribution of the unknown x_{t+h} is Gaussian and takes the form $x_{t+h}|\mathcal{I}_{it} \sim N(\mathbb{E}_{it}(x_{t+h}), \widehat{\Sigma}^{RE})$, where as we have detailed in the main text:

$$\mathbb{E}_{it}(x_{t+h}\ ) = \mu + G_1^{RE}(s_t^{(i)} - \mu) + G_2^{RE}(g_t - \mu)$$

with $G_1^{RE} = \frac{\tau}{\tau+\nu+\chi}$, $G_2^{RE} = \frac{\nu}{\tau+\nu+\chi}$, and $\widehat{\Sigma}^{RE} = (\chi + \tau + \nu)^{-1}$. Thus, the probability that x_{t+h} takes on exactly the value of the forecast made by agent i is thus

$$f(\widetilde{E}_{it}[x_{t+h}]|\mathcal{I}_{it}) = \frac{1}{\sqrt{2\pi\widehat{\Sigma}^{RE}}}\exp\left(-\frac{(\widetilde{E}_{it}[x_{t+h}] - \mathbb{E}_{it}[x_{t+h}])^2}{2\widehat{\Sigma}^{RE}}\right)$$

To construct the mass of other forecasters that make the same guess and thus share the prize, $\gamma(.)$, we observe that since for all agents j the forecast follows the linear structure (which we guess and verify):

$$\widetilde{E}_{jt}[x_{t+h}] = (1 - G)\mu + G_1 s_t^{(j)} + G_2 g_t \tag{56}$$

it follows that

$$\widetilde{E}_{jt}[x_{t+h}]\bigg|x_{t+h}, g_t \sim N((1 - G)\mu + G_1 x_{t+h} + G_2 g_t, \frac{G_1^2}{\tau})$$

Evaluating this PDF at the realization $x_{t+h} = \widetilde{E}_{it}[x_{t+h}]$ and for $\widetilde{E}_{jt}[x_{t+h}] = \widetilde{E}_{it}[x_{t+h}]$, we have

$$\gamma(\widetilde{E}_{it}[x_{t+h}]|x_{t+h}=\widetilde{E}_{it}[x_{t+h}],g_t)=\frac{\sqrt{\tau}}{\sqrt{2\pi G_1^2}}\exp\left(-\frac{\tau(\widetilde{E}_{it}[x_{t+h}]-(1-G)\mu-G_1\widetilde{E}_{it}[x_{t+h}]-G_2g_t)^2}{2G_1^2}\right).$$

Finally, putting it all together and taking logs of the expected payoff we have

$$\ln(U(\widetilde{E}_{it}[x_{t+h}]))=\frac{1}{2}\left(\ln(2\pi\widehat{\Sigma}^{RE})-\ln(2\pi\frac{G_1}{\tau})\right)-\frac{(\widetilde{E}_{it}[x_{t+h}]-\mathbb{E}_{it}[x_{t+h}])^2}{2\widehat{\Sigma}^{RE}}+\frac{\tau((1-G_1)\widetilde{E}_{it}[x_{t+h}]-(1-G)\mu-G_2g_t)^2}{2G_1^2}$$

Taking FOCs with respect to $\widetilde{E}_{it}[x_{t+h}]$, and solving for $\widetilde{E}_{it}[x_{t+h}]$ we can express the FOCs as:

$$\widetilde{E}_{it}[x_{t+h}]=\frac{(\widehat{\Sigma}^{RE})^{-1}+\frac{\tau(1-G_1)}{G_1}}{(\widehat{\Sigma}^{RE})^{-1}-\frac{\tau(1-G_1)^2}{G_1^2}}\mathbb{E}_{it}[x_{t+h}]-\frac{\frac{\tau(1-G_1)}{G_1^2}}{(\widehat{\Sigma}^{RE})^{-1}-\frac{\tau(1-G_1)^2}{G_1^2}}\mathbb{E}_{it}(\overline{E}[x_{t+h}]) \tag{57}$$

where on the right hand side we have used the fact that the agent's expectation of the consensus forecast is given by

$$\mathbb{E}_{it}(\overline{E}[x_{t+h}])=(1-G)\mu+G_1\mathbb{E}_{it}[x_{t+h}]+G_2g_t$$

So we can see that the optimal response of each forecaster i follows the same type of structure as in our reduced form game in Section 3 in the main text – the optimal reported forecast is a function of the true belief of the agent, $\mathbb{E}_{it}[x_{t+h}]$, but shifted away from what he beliefs everyone else is reporting $\mathbb{E}_{it}(\overline{E}[x_{t+h}])$.

Next, we solve for the equilibrium of this winner-take all game, looking for the equilibrium coefficients G_1 and G_2 that satisfy the fixed point defined by the FOCs of the agents and the linear structure of the optimal response. Solving this fixed point, we get

$$G_1=\frac{\sqrt{(G_1^{RE})^2+4G_1^{RE}(1-G_1^{RE})}-G_1^{RE}}{2(1-G_1^{RE})}$$

$$G_2 = G_2^{RE} \frac{G_1}{G_1 + G_1^{RE}(1 - G_2)}$$

While this is not pretty to look at, using the fact that $G_1^{RE} = \frac{\tau}{\tau+\chi+\nu}$ it follows that

$$G_1 > G_1^{RE}$$

$$G_2 < G_2^{RE}$$

which is the key behavioral implication of our reduced form Morris-Shin game that we also exploit and test for our main empirical results.

Thus, the formal winner-take-all game we analyze here is one micro-founded way to obtain the basic behavioral implication that our paper focuses on taking to the data. The basic intuition is that forecasters want to "stand out" from the crowd. This intuition is captured by the negative weight put on the expected consensus forecast, $\mathbb{E}_{it}(\overline{E}[x_{t+h}])$, in the FOCs in equation (57). And this has the specific behavioral implication that forecasts optimally overweight any idiosyncratic information, and underweight informative signals that are commonly seen by all agents.

The reduced form game we use in the main text is simply analytically more convenient and tractable, and also leads to some cleaner and more intuitive expressions. However, the basic behavioral implications and empirical predictions are the same.